

Absolute Values and Original-Metric Control

Full proofs, source corrections and backward receivers

Split-Zero research programme

19 September 2026

This successor reader gives the complete accepted calculations received and extended on 19 September 2026, with their original provider proofs and human-source citations. The accompanying three cumulative LaTeX successors retain every byte of their preceding editions by reversible insertion. This edition has no newly assigned deposited DOI.

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Part I

Absolute values and original-metric control

This continuation integrates the nine new notes and their complete receivers. The preceding editions remain recoverable byte for byte. Source corrections and original-minor scope are stated at their exact use.

Absolute Gamma baseline and the complete canonical action

The incoming note 07_ABSOLUTE_GAMMA_BASELINE.md identified the previously unevaluated absolute Gamma coefficient. This proof retains its complete root polynomial, original monic frames and source mass. It strengthens its remainder to $O_h(\log q)$ for every fixed multiplicity. The original four blocks are expressed using a single fixed potential and three analytic factors, and a finite convexity argument controls the remaining bounded root parameter. The previously proved arithmetic correction is then returned to the absolute baseline and the complete canonical action. The human analytic input is Borot–Guionnet [1]; it is applied after a comparison of the complete quadratic forms on both complementary regions.

1 Original objects and the determinant map

Fix $0 < \delta < 1/2$, $\gamma > 2$ and $m_0 \in \mathbb{N}$. For the arithmetic receivers these are the stipulated complete quartet data of the preceding programme, with $h(s)$ its actual quartet factor in $2\xi(s)$. The Gamma calculation itself requires only these fixed parameters. Put

$$k = 4\ell + 1, \quad e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \quad c = k/2, \quad (AG1)$$

$$\chi(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e.$$

The monic polynomial $Q(y) = i^{-q}\chi(c + iy)$ has its full roots

$$\omega_{ab} = (2b - k)\gamma - i(2a - k)\delta \quad \text{with multiplicity } e. \quad (AG2)$$

Conjugation and negation preserve the root multiset. Since k is odd, no root is real. Thus Q is real, even and strictly positive on $\mathbb{R}$. Here q is divisible by four. Retain

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad \int_{\mathbb{R}} \sigma(y) dy = \sqrt{2\pi}, \quad \alpha = \pi/2.$$

Let H_N be the monic source Gram through degree N , J_N the remainder map into $\mathbb{C}[y]/Q$, and $G_N = (J_N H_N^{-1} J_N^*)^{-1}$ for $N \geq q - 1$. Its definition is the squared minimum norm in each original remainder fibre. The monic source norms are

$$\gamma_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2\pi} \frac{\Gamma(2n + 1)}{4^n}. \quad (AG3)$$

These are the original Meixner–Pollaczek specialization; the generating function is $(1 + t^2)^{-1/4} \exp(y \arctan t)$, with coefficients $p_n(y)/n!$. The general polynomial convention is recorded in DLMF [2]. Put $R_n = \det[\int y^{i+j} |Q|^2 \sigma dy]_{0 \leq i,j < n}$ and $R_0 = 1$. The frame $(1, \dots, y^{q-1}, Q, yQ, \dots, y^{N-q}Q)$ has determinant one. Taking its Schur complement, with every source/relation cross term present, proves $\det G_N = \det H_N / R_{N-q+1}$. Consequently

$$\mathcal{B}_k[\sigma] = \log R_q + \log R_{q+1} - \log R_1 - \sum_{n=q}^{2q} c_{n-q} \log \gamma_n, \quad c_0 = c_q = 1, \quad c_r = 2 \quad (0 < r < q). \quad (AG4)$$

The affine substitution $S = c + iy$ has triangular coefficient matrix with diagonal $1, i, \dots, i^N$, of determinant modulus one. Its inverse is $y = (S - c)/i$; it intertwines the relation injections and remainder maps. Thus (AG4) computes the original S -coordinate return, without a new mass.

2 The fixed equilibrium measure and its inverse moment

Use the EIQ measure for $V(x) = \pi\sqrt{x} - 2\log x$, with support $[u^2, v^2]$. In the original endpoint convention

$$\begin{aligned} \kappa^2 &= 1 - u^2/v^2, & uK(\kappa) &= 2, & vE(\kappa) &= 4, \\ F &= -\min \mathcal{E}_{2,\pi}, & L &= \int \log x \, d\rho(x), & M &= \int x^{-1} \, d\rho(x). \end{aligned} \quad (\text{AG5})$$

All integrals use the original probability density, rather than a scaled endpoint model. Its complete proof is EIQ1–26 in `EIQ_GEL_COMPLETE.tex`. For application of the analytic theorem we recall its decisive exact formulas. For general $a = u^2, b = v^2$, $R_s = \sqrt{(a+s)(b+s)}$, and parameters $A, B > 0$, they are

$$h_{A,B}(x) = \frac{B}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x+s)R_s} \, ds > 0, \quad \rho_{A,B}(x) = \frac{\sqrt{(b-x)(x-a)}}{2\pi} h_{A,B}(x). \quad (\text{AG6})$$

The density has mass one, positive endpoint coefficients, and its effective potential derivative off the support is $\sqrt{(x-a)(x-b)}h_{A,B}(x)$, with the square root negative below a . This proves the strict exterior gap on every closed interval outside the support. The functions extend holomorphically near every compact positive interval. These are precisely the offcriticality and exterior properties used below. Variation under $x \mapsto tx$ at its minimum gives $B \int \sqrt{x} \, d\rho_{A,B} = 2(A+1)$; hence

$$\int \sqrt{x} \, d\rho = 6/\pi. \quad (\text{AG7})$$

Indeed the energy of the pushed-forward measure is $B\sqrt{t} \int \sqrt{x} \, d\rho - A \log t - A \int \log x \, d\rho - \log t - \iint \log |x-y| \, d\rho \, d\rho$; its derivative at $t=1$ is zero.

Here is an explicit derivation of the required new moment, including its sign. Set $r_0 = \sqrt{ab}$, $m = (a+b)/2$, and $R(z) = \sqrt{(z-a)(z-b)}$ with $R(z)/z \rightarrow 1$. EIQ's elementary transform is

$$H_s(z) = 1 - \frac{R_s + R(z)}{z+s}, \quad G(z) = \frac{A}{2r_0} H_0(z) - \frac{B}{4\pi} \int_0^\infty \frac{H_s(z)}{\sqrt{s}R_s} \, ds.$$

Since $R(0) = -r_0$, $R'(0) = m/r_0$, and $\int_0^\infty (\sqrt{s}R_s)^{-1} \, ds = 2\pi A/(Br_0)$, evaluation at zero, followed by an integration by parts, gives

$$\begin{aligned} \int x^{-1} \, d\rho_{A,B} &= \frac{A(a+b)}{4ab} - \frac{B}{4\pi} \int_0^\infty \frac{1 - r_0/R_s}{s^{3/2}} \, ds \\ &= \frac{A(a+b)}{4ab} - \frac{Br_0}{4\pi} \int_0^\infty \frac{a+b+2s}{\sqrt{s}[(a+s)(b+s)]^{3/2}} \, ds. \end{aligned} \quad (\text{AG8})$$

The boundary terms vanish: $1 - r_0/R_s = O(s)$ at zero and is bounded at infinity. Differentiating the absolutely convergent identity $\int (\sqrt{s}R_s)^{-1} \, ds = 2K(\kappa)/v$ with the operator $-2(\partial_a + \partial_b)$ evaluates the last integral as $2E(\kappa)/(u^2v)$. This derivative also follows directly from the defining elliptic integrals, and every differentiated integral is integrable at both ends. Therefore

$$M = \frac{u^2 + v^2 - 4uv}{2u^2v^2} > 0. \quad (\text{AG9})$$

Its positivity follows from the original positive probability measure.

3 Full-form bounds on both complementary regions

Write $r_k = k\sqrt{\delta^2 + \gamma^2}/q$, so $r_k \rightarrow 0$. Pair opposite roots before estimating. The normalized factor $E_q(x) = |Q(q\sqrt{x})|^2/q^{2q}$ satisfies for $x > r_k^2$

$$(x - r_k^2)^q \leq E_q(x) \leq (x + r_k^2)^q; \quad E_q(x) \leq (x + r_k^2)^q \quad (x \geq 0). \quad (\text{AG10})$$

Stirling's formula and positivity on compact real intervals give fixed $c_\sigma, C'_\sigma > 0$ such that, for all $y \geq 0$,

$$c_\sigma(1+y)^{-1/2}e^{-\alpha y} \leq \sigma(y) \leq C'_\sigma(1+y)^{-1/2}e^{-\alpha y}.$$

For $\varepsilon = 0, 1$ use the exact rescaled density $g_{q,\varepsilon}(x) = q^{1/2}x^{\varepsilon-1/2}E_q(x)\sigma(q\sqrt{x})$. Every relevant polynomial has degree $d \leq q/2$. Its shifted Legendre expansion on $[1, 2]$ proves the elementary pointwise bounds

$$|P(x)|^2 \leq (d+1)^2 \begin{cases} 7^{2d}, & 0 \leq x \leq 1, \\ (5x)^{2d}, & x \geq 2 \end{cases} \int_1^2 |P(t)|^2 dt. \quad (\text{AG11})$$

For completeness, the normalized basis is $\sqrt{2j+1}P_j(2x-3)$; Cauchy-Schwarz and $|P_j(t)| \leq (2|t|+1)^j$ outside $[-1, 1]$ give (AG11), since $\sum_{j=0}^d (2j+1) = (d+1)^2$. The polynomial bound follows from the integral representation of P_j or its real three-term recurrence. On $[1, 2]$, with $r_k^2 \leq 1/2$ and $q \geq 1$, the full density is at least $c_0 e^{-C_0 q}$, where one may take $C_0 = \log 2 + \alpha\sqrt{2}$ and $c_0 = c_\sigma/[\sqrt{2}\sqrt{1+\sqrt{2}}]$.

Choose fixed $0 < \epsilon < \min(1/2, u^2/2)$ with $14\epsilon e^{C_0} < e^{-2}$. Impose the finite guard $r_k^2 \leq \epsilon$. On $[0, \epsilon]$, (AG10)–(AG11) bound the integral of $|P|^2 g_{q,\varepsilon}$, relative to its $[1, 2]$ integral, by

$$\frac{C'_\sigma}{c_0}(d+1)^2(14\epsilon e^{C_0})^q \frac{\epsilon^{\varepsilon+1/4}}{\varepsilon+1/4}. \quad (\text{AG12})$$

The bound uses $q^{1/2}\sigma(q\sqrt{x}) \leq C'_\sigma x^{-1/4}e^{-\alpha q\sqrt{x}}$, valid for every $x > 0$. In particular it includes the entire region $q\sqrt{x} \leq 1$.

Choose $M_0 > \max(2, 2v^2, (8/\alpha)^2)$ such that $\log 10 + C_0 + 2\log x \leq \alpha\sqrt{x}/2$ for $x \geq M_0$. It suffices to check this at M_0 and the displayed derivative guard. For $x \geq M_0$, the same formulas bound the upper-tail ratio by

$$\frac{C'_\sigma}{c_0}(d+1)^2 \int_{M_0}^\infty x^{\varepsilon-3/4}e^{-\alpha q\sqrt{x}/2} dx. \quad (\text{AG13})$$

Here $(5x)^{2d}(2x)^q \leq (10x^2)^q$ and $d \leq q/2$. Substitution $z = \sqrt{x}$ shows that (AG13) is bounded by a fixed polynomial in q times $e^{-\alpha q\sqrt{M_0}/4}$. Thus the sum b_q of (AG12)–(AG13) is $O_h(q^2 e^{-c_1 q})$ for a fixed $c_1 > 0$. These are inequalities on every vector before minimization:

$$H_{[\epsilon, M_0]} \preceq H_{(0, \infty)} \preceq (1 + b_q)H_{[\epsilon, M_0]}, \quad 0 \leq \log \frac{\det H_{(0, \infty)}}{\det H_{[\epsilon, M_0]}} \leq n \log(1 + b_q). \quad (\text{AG14})$$

The identical argument with $E_q = x^q$ gives the comparison for the power reference. No polynomial root, minimum or complementary integral is omitted.

4 Exact fixed-potential block and a bounded-parameter lemma

On the fixed interval, the real part of Stirling's expansion gives

$$\sigma(q\sqrt{x}) = \sqrt{2} q^{-1/2} x^{-1/4} e^{-\pi q\sqrt{x}/2} (1 + O(q^{-2})), \quad (\text{AG15})$$

uniformly in x . Its inverse- y contribution has zero real part; the next term is $O(y^{-2})$. This relative bound passes through every Gram in (AG14), costing $O(nq^{-2})$ in its logarithmic determinant. The finite root sums give

$$\sum \omega_{ab}^2 = \frac{qk(k+2)}{3}(\gamma^2 - \delta^2), \quad \log \frac{E_q(x)}{x^q} = -\frac{\tau_k}{x} + r_q(x), \quad \tau_k = \frac{\eta_h k(k+2)}{q}, \quad \eta_h = \frac{\gamma^2 - \delta^2}{3}. \quad (\text{AG16})$$

All sums include multiplicity. Indeed $\sum_{a=0}^k (2a-k) = 0$ and $\sum_{a=0}^k (2a-k)^2 = (k+1)k(k+2)/3$. In a fixed complex neighborhood of $[\epsilon, M_0]$ the logarithmic series converges once $r_k^2/|x| \leq 1/2$; opposite

roots remove odd terms and the sum of the fourth and higher terms is bounded by $C_h k^4/q^3$. Hence $\|r_q\|_\infty \leq C_h k^4/q^3$. Replacing e^{r_q} in a complete determinant costs at most $n\|r_q\|_\infty = O_h(k^4/q^2) = O_h(1)$, since $q \geq (k+1)^2$.

Let $q = 2N$, $n = N + s$, and use $(s, \varepsilon) = (0, 0), (1, 0), (0, 1)$. After its exact powers of q and $\sqrt{2}$ are extracted, the block density is

$$x^{q+\varepsilon-3/4} e^{-\pi q \sqrt{x}/2 - \tau_k/x} = e^{-nV(x) + W_{s,\varepsilon,\tau_k}(x)}, \quad W_{s,\varepsilon,t} = (\varepsilon - 2s - 3/4) \log x + \pi s \sqrt{x} - t/x. \quad (\text{AG17})$$

This equality, not a limiting replacement, fixes every parity increment. The potential V is now independent of n and k .

Borot–Guionnet’s soft-edge expansion and analytic linear-statistic formula [1, Propositions 1.2 and 5.2, equations (1-3)–(1-6)], applied to (AG6), give, for fixed analytic real W on this interval,

$$\log D_n[e^{-nV+W}] = n^2 F + n \log(2\pi) + n \int W d\rho + O_W(\log n). \quad (\text{AG18})$$

Here D_n is the monic Gram determinant. Its Vandermonde integral is the partition function divided by $n!$. At ensemble index two the potential’s order- n correction vanishes. The Gaussian reference is $(2\pi)^{n/2} n^{-n^2/2} \prod_{j=0}^{n-1} j!$; Stirling summation gives $-3n^2/4 + n \log(2\pi) + O(\log n)$. This fixes the stated mass and factorial. For the ratio with analytic factor the error after its order- n term is $O_W(1)$ by the cited linear-statistic formula. The support lies strictly inside the actual integration interval, the potential is analytic there, and (AG6) verifies its positive density and exterior gap; analyticity at $x = 0$ is not used.

We need no unproved uniformity in the varying τ_k . Choose $T > 1 + |\eta_h|$ and fix one of the three $W_{s,\varepsilon,0}$. The function

$$f_n(t) = \log \frac{D_n[e^{-nV+W_{s,\varepsilon,t}}]}{D_n[e^{-nV}]} - n \int W_{s,\varepsilon,t} d\rho$$

is convex in t , by differentiating the positive squared-Vandermonde integral twice (its second derivative is a variance). Formula (AG18)’s ratio statement bounds $|f_n(-2T)|, |f_n(0)|, |f_n(2T)|$ by a common constant C . Convexity bounds $f_n(t) \leq C$ on $[-T, T]$. For $0 \leq t \leq T$, write $0 = \lambda t + (1 - \lambda)(-2T)$, $\lambda = 2T/(t + 2T) \geq 2/3$; then $f_n(t) \geq -2C$. The negative half is identical. This proves the uniform ratio error $O_h(1)$ for every actual τ_k , since $|\tau_k| \leq |\eta_h|$. It includes fixed higher multiplicities without a fractional-power potential expansion or a differentiated asymptotic remainder.

5 Evaluate every block, source norm and root term

The even block consists of $1, y^2, \dots, y^{2n-2}$, and the odd block of $y, y^3, \dots, y^{2n-1}$. The substitution $y = q\sqrt{x}$, including both halflines, contributes $qx^{-1/2} dx$. Factoring the row and column powers and using (AG7), (AG14)–(AG18) proves

$$\begin{aligned} \log H_n^{q+\varepsilon} &= [2(q + \varepsilon)n + 2n^2 - \tfrac{3}{2}n] \log q + \tfrac{n}{2} \log 2 + n^2 F + n \log(2\pi) \\ &\quad + n[(\varepsilon - 2s - \tfrac{3}{4})L + 6s - \tau_k M] + O_h(\log q). \end{aligned} \quad (\text{AG19})$$

The complete parity identity is

$$\log R_q + \log R_{q+1} = \log H_N^q + \log H_{N+1}^q + 2 \log H_N^{q+1}. \quad (\text{AG20})$$

Their dimensions sum to $2q + 1$. Their squared dimensions sum to $q^2 + q + 1$; their s -weighted dimensions sum to $q/2 + 1$; and their ε -weighted dimensions sum to q . Keeping these exact sums until the end gives

$$\log R_q + \log R_{q+1} = (6q^2 + 3q + \tfrac{1}{2}) \log q + Fq^2 \quad (\text{AG21})$$

$$+ q[F + 3 - \frac{3}{2}L + 3 \log 2 + 2 \log \pi] - 2\eta_h M k(k+2) + O_h(\log q).$$

The extra $-\tau_k M$ from $2q+1$ is bounded and included in the displayed error. Every constant dropped here is bounded independently of k .

For R_1 the reference integral is exactly $2\sqrt{2}\Gamma(2q+1/2)/\alpha^{2q+1/2}$. On $[\epsilon, M_0]$ the full root multiplier has logarithm bounded by $|\tau_k|/\epsilon + \|r_q\|_\infty$, and (AG15) has relative error $O(q^{-2})$. Both references satisfy (AG14), so their full integral ratio has bounded logarithm. Stirling now gives

$$\log R_1 = 2q \log q + (4 \log 2 - 2 \log \pi - 2)q + O_h(\log q). \quad (\text{AG22})$$

For the source norms, (AG3) yields uniformly for $q \leq n \leq 2q$ $\log \gamma_n = 2n \log n - 2n + \frac{1}{2} \log n + \frac{3}{2} \log 2 + \log \pi + O(n^{-1})$. The coefficients in (AG4) are twice the trapezoidal weights. Integrating $2x \log x - 2x$ and $\frac{1}{2} \log x$ from q to $2q$, and bounding the trapezoidal remainder by its second derivative integral, gives

$$\sum_{n=q}^{2q} c_{n-q} \log \gamma_n = 6q^2 \log q + (8 \log 2 - 9)q^2 + q \log q + (5 \log 2 + 2 \log \pi - 1)q + O(\log q). \quad (\text{AG23})$$

For example $2 \int_q^{2q} (2x \log x - 2x) dx = 6q^2 \log q + (8 \log 2 - 9)q^2$ and $\int_q^{2q} \log x dx = q \log q + (2 \log 2 - 1)q$; the constant part contributes $(3 \log 2 + 2 \log \pi)q$.

Theorem 5.1 (Absolute Gamma value on every fixed multiplicity). *With all original definitions retained, put*

$$C_B = 9 - 8 \log 2 + F, \quad C_\sigma = F + 6 - \frac{3}{2}L - 6 \log 2 + 2 \log \pi. \quad (\text{AG24})$$

Then

$$\boxed{\mathcal{B}_k[\sigma] = C_B q^2 + C_\sigma q - 2\eta_h M k(k+2) + O_h(\log q).} \quad (\text{AG25})$$

Proof. Substitute (AG21)–(AG23) into the exact identity (AG4). The $q \log q$ coefficient is $3-2-1=0$. The remaining order- q coefficient is $F+3-\frac{3}{2}L+3 \log 2+2 \log \pi-(4 \log 2-2 \log \pi-2)-(5 \log 2+2 \log \pi-1)$, which is exactly C_σ . All preceding bounds are uniform in the bounded τ_k and use $k^4/q^2 \leq 1$; they apply to every fixed m_0 . $\square$

6 Backwards and forwards through the original action

Retain $w_h = |(\xi/h)(1/2 + iy)|^2/(2\pi)$, $m_k = w_h^{*k}$, $I_h = \int_0^{\pi/2} [(\log M_h)'(\theta)]^2 d\theta$, $d = 4m_0 - 2$, $c_\zeta = 2\gamma_E - \log(2\pi)$ and $C_\Gamma = 2L - 8 \log 2 + 4$. The already proved SR1–48 receiver gives

$$\mathcal{B}_k[m_k] - \mathcal{B}_k[\sigma] = -dC_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + R_{h,k}, \quad R_{h,k} = o_h(q/\log q). \quad (\text{AG26})$$

Its full finite remainder, annulus guards, inner and outer integrals and the eight positive Grams are retained in `SLOW_TAIL_AND_NONLINEAR_RECEIVER.tex`; its result is used without differentiating $R_{h,k}$ or assigning its constituent signed pieces. Adding (AG25) to that exact scalar difference proves

$$\begin{aligned} \mathcal{B}_k[m_k] &= C_B q^2 + (C_\sigma - dC_\Gamma)q + \frac{\log 2}{\pi} I_h k^2 - 2\eta_h M k(k+2) \\ &\quad + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + R_{h,k} + O_h(\log q). \end{aligned} \quad (\text{AG27})$$

This evaluates the absolute baseline through the $q/\log q$ scale. It strengthens the incoming $o(q)$ statement; no new arithmetic estimate is presumed. The simple-packet root term contributes at order q . At higher fixed multiplicity its displayed k^2 value is retained even though the arithmetic error does not yet resolve all terms of that size.

The original CAI15 identity is $J_k^{\text{action}} = \mathcal{B}_k[m_k] + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+$. Its finite guards and penalties give (CAI29–35)

$$\begin{aligned} -E_{\text{hi}} &\leq \mathcal{W}_k^{\text{ar}} - \mathcal{W}_q^\sigma \leq E_{\text{lo}}, \quad 0 \leq \mathcal{P}_- + \mathcal{P}_+ \leq \Pi_{h,k}, \\ \mathcal{W}_q^\sigma &= 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}, \quad E_{\text{hi}} + E_{\text{lo}} + \Pi_{h,k} = O_h(k + \log q). \end{aligned} \quad (\text{AG28})$$

In particular its exact factorial window is $4q \log q + (8 \log 2 - 4)q - \log 2 + O(q^{-1})$. Returning (AG27) through (AG28) proves

$$\boxed{\begin{aligned} J_k^{\text{action}} &= C_B q^2 + 4q \log q + [C_\sigma + 8 \log 2 - 4 - dC_\Gamma]q \\ &\quad + \frac{\log 2}{\pi} I_h k^2 - 2\eta_h M k(k+2) + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + o_h(q/\log q). \end{aligned}} \quad (\text{AG29})$$

Indeed $(k + \log q)/(q/\log q) \rightarrow 0$ on the original sequence. At finite k the exact interval (AG28) and the complete SR error remain the bounds, rather than an assigned coefficient for any separate penalty or native loss. The constant identity

$$C_\sigma = C_B - \frac{3}{4}C_\Gamma + 2 \log(\pi/4) \quad (\text{AG30})$$

follows by substituting the original definitions; no mass is changed. For the original benchmark $L_{h,k} = \delta q(k+1)/2$, subtracting its actual logarithm gives the further receiver

$$\begin{aligned} J_k^{\text{action}} - 4q \log L_{h,k} &= C_B q^2 - 4q \log(k+1) + [C_\sigma + 8 \log 2 - 4 - dC_\Gamma - 4 \log(\delta/2)]q \\ &\quad + \frac{\log 2}{\pi} I_h k^2 - 2\eta_h M k(k+2) + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + o_h(q/\log q). \end{aligned} \quad (\text{AG31})$$

The full observation kernel still enters through its original inclusion I and quotient Λ : $H_{K,N} = I^* G_N I$ and $Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}$. For any fixed section S_* of Λ the exact map $[I, G_N^{-1} \Lambda^* Q_{B,N}]$ has determinant equal to $[I, S_*]$ and orthogonal columns between its two blocks. Thus

$$\det H_{K,N} \det Q_{B,N} = |\det[I, S_*]|^2 \det G_N, \quad F_K + F_B = \mathcal{B}_k[m_k]. \quad (\text{AG32})$$

Equation (AG27) replaces the previously unevaluated total in this equality; it does not allocate that total to either original block. The independent proper-source return and the original projected-current response ratios retain their own exact maps and remain subsequent calculations.

Full-primary positive-trace minimum and fixed-metric control

Abstract

For the unchanged arithmetic multiplication packet, the minimum of the positive Hermitian weight trace over all positive metrics is evaluated, attained, and its entire minimizing set is parametrized by finite Lyapunov sums. All primary jets and all same-half cross terms are retained. A rank-one observable forcing on each complete spectral half constructs an attaining metric whose weight has precisely two nonzero eigenvalues. The canonical attained polynomial metric is then evaluated through its actual boundary rows: its exact rank-two scalar exceeds the minimum by an explicit positive spectral-projection expression. The positive-primary determinant line has exactly the same growth exponent for every metric. Finally an original finite polynomial section realizes every minimizing metric up to a specified scalar, and its complete residual in each smaller polynomial source quotient is computed.

7 Original packet, source and coordinate maps

The incoming note [3] proposes the minimum and its complete Lyapunov parametrization. The present text proves them, adds the rank-two attaining construction, and evaluates their precise receivers. The fixed source and cutoff conventions are GMF1–3 of [5]; the earlier exact metric and source maps are TC1–29d of [4]. These source references record provenance. Every new finite assertion below is proved here.

Keep the original hypothetical actual quartet with full multiplicity:

$$\begin{aligned} \rho &= \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m_0 \geq 1, \\ h(s) &= \prod_{\zeta \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \zeta)^{m_0}, \quad v_h = 2\xi/h, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ m_k &= w_h^{*k}, \quad k \equiv 1 \pmod{4}, \quad k \geq 5, \quad e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \quad c = k/2, \quad (\text{PT1}) \\ \lambda_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad 0 \leq a, b \leq k, \\ \chi(S) &= \prod_{a,b=0}^k (S - \lambda_{ab})^e, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S : E \rightarrow E, \quad B = A - cI. \end{aligned}$$

The original basis of E is $1, S, \dots, S^{q-1}$. The numbers λ_{ab} are pairwise distinct: equality first forces the real indices a , and then the imaginary indices b , to coincide. Thus χ has exactly the displayed complete primary factors. None is divided out in this paper.

For $G > 0$ in this original basis define

$$W(G) = A^*G + GA - kG = B^*G + GB, \quad H_G = G^{-1/2}W(G)G^{-1/2}, \quad T_+(G) = \text{Tr}(H_G)_+. \quad (\text{PT2})$$

The symbol X_+ means the positive part of a Hermitian matrix, obtained by replacing each negative eigenvalue by zero; write $X_- = (-X)_+$. Positive square roots always refer to the fixed standard coordinate Hermitian form. The physical coordinate in the source remains $S = c + iy$; its measure is exactly $m_k(y) dy$.

Here is the complete confluent coordinate map and its inverse. Define

$$C : E \longrightarrow \bigoplus_{a,b=0}^k \mathbb{C}^e, \quad (CP)_{(\lambda,r)} = P^{(r)}(\lambda)/r!, \quad C_{(\lambda,r),j} = \begin{cases} \binom{j}{r} \lambda^{j-r}, & j \geq r, \\ 0, & j < r. \end{cases} \quad (\text{PT3})$$

For each λ put $h_\lambda = \chi/(S - \lambda)^e$ and write the actual Taylor series $h_\lambda(\lambda + z)^{-1} = \sum_{n \geq 0} a_{\lambda,n} z^n$. Its constant denominator $h_\lambda(\lambda)$ is the full product of the other root differences, each with exponent e , and is nonzero. Set

$$p_{\lambda,r}(S) = h_\lambda(S)(S - \lambda)^r \sum_{n=0}^{e-r-1} a_{\lambda,n}(S - \lambda)^n, \quad 0 \leq r < e. \quad (\text{PT4})$$

This polynomial has degree at most $q - 1$. At λ its Taylor class modulo $(S - \lambda)^e$ is $(S - \lambda)^r$; at every other root its first e Taylor coefficients vanish. Consequently these polynomials are the columns of C^{-1} . This proves both inverse laws of (PT3), without replacing any primary factor by a simple root. Multiplication by S now gives

$$CAC^{-1} = \bigoplus_{\lambda} (\lambda I_e + J_e), \quad (J_e)_{r+1,r} = 1 \quad (0 \leq r < e - 1), \quad J_e^e = 0. \quad (\text{PT5})$$

The direct sum and the subsequent maps are coordinate morphisms; the recorded operator remains the original A .

Let E_+ be the sum of the complete primary spaces with $a \geq (k + 1)/2$, and E_- its complementary sum. Both have dimension $r = q/2$. Formulae (PT3)–(PT4) give explicit column frames for their inclusions. For a column frame $X_+ : \mathbb{C}^r \rightarrow E_+$ define the actual Euclidean orthogonal projection

$$P_G = G^{1/2} X_+ (X_+^* G X_+)^{-1} X_+^* G^{1/2} : \mathbb{C}^q \longrightarrow G^{1/2} E_+. \quad (\text{PT6})$$

It is self-adjoint and idempotent by matrix multiplication, and its image is precisely the stated subspace. It is independent of the choice of frame within E_+ .

8 The trace bound, exact defect and equality

The finite projection argument below is the positive-part form of Ky Fan's variational trace principle [6]. We give the full argument rather than use it as a black box.

Theorem 8.1. *For every original-coordinate $G > 0$,*

$$L := \frac{\delta q(k + 1)}{2} = \text{Tr}(P_G H_G), \quad T_+(G) - L = \text{Tr}((I - P_G)(H_G)_+) + \text{Tr}(P_G(H_G)_-) \geq 0. \quad (\text{PT7})$$

The two terms in the defect are exactly

$$T_+(G) - L = \left\| (H_G)_+^{1/2} (I - P_G) \right\|_F^2 + \left\| (H_G)_-^{1/2} P_G \right\|_F^2. \quad (\text{PT8})$$

Equality holds precisely when E_+ and E_- are G -orthogonal and their actual restricted metrics satisfy

$$B_+^* G_+ + G_+ B_+ \succeq 0, \quad C_-^* G_- + G_- C_- \succeq 0, \quad B_+ = A|_{E_+} - cI, \quad C_- = cI - A|_{E_-}. \quad (\text{PT9})$$

Proof. Put $K = G^{1/2} B G^{-1/2}$, so $H_G = K + K^*$. The subspace $G^{1/2} E_+$ is K -invariant, and the compression of K to it is similar to B_+ . Therefore

$$\text{Tr}(P_G H_G) = 2\Re \text{Tr } B_+ = e(k + 1)2\delta \sum_{a=(k+1)/2}^k (2a - k).$$

The summands in the last sum are the odd integers $1, 3, \dots, k$; their sum is $((k + 1)/2)^2$. This proves the first equality in (PT7). Substitute $H_G = (H_G)_+ - (H_G)_-$ and use linearity of trace to obtain the second equality. For a positive matrix X and an orthogonal projection P , $\text{Tr}(PX) = \text{Tr}(P X^{1/2} X^{1/2} P) = \|X^{1/2} P\|_F^2$. Apply this twice to get (PT8) and nonnegativity, without assuming that either projection commutes with either positive part.

The defect vanishes exactly when $(H_G)_+^{1/2} (I - P_G) = 0$ and $(H_G)_-^{1/2} P_G = 0$. These equalities say that every positive eigenvector of H_G belongs to $\text{im } P_G$, and every negative eigenvector belongs to its orthogonal complement. The zero eigenspace may be split in any way. In particular $P_G H_G = H_G P_G$. In an orthonormal frame adapted to $\text{im } P_G$, invariance gives

$$K = \begin{pmatrix} K_+ & D \\ 0 & K_- \end{pmatrix}, \quad H_G = \begin{pmatrix} K_+ + K_+^* & D \\ D^* & K_- + K_-^* \end{pmatrix}.$$

Commutation forces $D = 0$. Thus the orthogonal complement is also K -invariant; its spectrum is precisely the negative-real part of the original spectrum. To identify this complement exactly, apply the complete spectral idempotents supplied by (PT4) with $r = 0$. A vector in this invariant complement is killed by the product of the negative primary factors, so all positive idempotent components vanish. Dimension r then proves that the complement is $G^{1/2}E_-$. The restrictions of H_G on the two halves are respectively nonnegative and nonpositive, which is exactly (PT9) by congruence with their positive square roots. Conversely, orthogonality and (PT9) give those two signs on the two invariant halves. The positive trace is then their positive half trace $2\Re \operatorname{Tr} B_+ = L$. This proves both directions, including all possible zero eigenvalues within either half. $\square$

9 An explicit full-primary attainer

Put $t_0 = \delta/2$, $D_e = \operatorname{diag}(1, t_0, \dots, t_0^{e-1})$, and define in the original frame

$$T = \left(\bigoplus_{\lambda} D_e \right) C, \quad G_* = T^* T. \quad (\text{PT10})$$

The map $T : (E, G_*) \rightarrow (\mathbb{C}^q, I)$ is an exact isometry. Indeed $U = TG_*^{-1/2}$ is unitary, and directly

$$UH_{G_*}U^* = TAT^{-1} + (TAT^{-1})^* - kI.$$

Since $D_e J_e D_e^{-1} = t_0 J_e$, each block on the right is $2(2a - k)\delta I_e + t_0(J_e + J_e^*)$. The vector with r -th entry $\sin((r+1)j\pi/(e+1))$, $0 \leq r < e$, is an eigenvector of $J_e + J_e^*$ with eigenvalue $2\cos(j\pi/(e+1))$: the first and last missing entries are zero, and the interior recurrence is the sine addition identity. These e eigenvalues are distinct, giving a full eigenbasis. Consequently

$$\operatorname{spec} H_{G_*} = \left\{ 2(2a - k)\delta + \delta \cos \frac{j\pi}{e+1} : 0 \leq a, b \leq k, 1 \leq j \leq e \right\}. \quad (\text{PT11})$$

Every one of these values has the sign of $2a - k$, since $|2a - k| \geq 1$ and $|\cos| < 1$. The cosine terms sum to zero in each block by pairing j with $e + 1 - j$. Thus

$$\min_{G>0} T_+(G) = T_+(G_*) = L, \quad \operatorname{rank}(H_{G_*})_+ = \operatorname{rank}(H_{G_*})_- = q/2, \quad \operatorname{Tr} H_{G_*} = 0, \quad \operatorname{Tr} |H_{G_*}| = 2L. \quad (\text{PT12})$$

In particular $\|H_{G_*}\| = 2k\delta$ when $e = 1$, and $2k\delta + \delta \cos(\pi/(e+1))$ when $e > 1$. These are norms of this displayed metric. They do not replace the positive trace of any other original metric.

10 Every minimizing metric and every nilpotent term

The method here is the classical Lyapunov matrix equation method, named for A. M. Lyapunov [7]. All convergence, positivity and inverse statements needed here are proved below. Use the fixed primary bases (PT3) on each half and put

$$D_+ = \bigoplus_{a>(k/2), b} ((\lambda_{ab} - c)I_e + J_e), \quad D_- = \bigoplus_{a<(k/2), b} ((c - \lambda_{ab})I_e - J_e). \quad (\text{PT13})$$

Every eigenvalue of D_+ and D_- has strictly positive real part. Notice the minus sign on every negative-half nilpotent.

Theorem 10.1. *All minimizing metrics, with no duplicates suppressed, are obtained by choosing $Q_+, Q_- \succeq 0$ such that*

$$\bigcap_{j=0}^{r-1} \ker(Q_{\pm}^{1/2} D_{\pm}^j) = \{0\}, \quad r = q/2, \quad (\text{PT14})$$

and setting

$$\Gamma_{\pm} = \int_0^{\infty} e^{-tD_{\pm}^*} Q_{\pm} e^{-tD_{\pm}} dt, \quad G = C^* \text{diag}(\Gamma_+, \Gamma_-) C. \quad (\text{PT15})$$

In this formula C is ordered first by the positive half and then by the negative half, with the same primary coordinates within each half. Every integral is a finite rational block sum. For blocks $D_{\alpha} = b_{\alpha}I + N_{\alpha}$ its exact value is

$$\Gamma_{\alpha\beta} = \sum_{i,j=0}^{e-1} \frac{(-1)^{i+j}(i+j)!}{i!j!(\bar{b}_{\alpha} + b_{\beta})^{i+j+1}} (N_{\alpha}^*)^i Q_{\alpha\beta} N_{\beta}^j. \quad (\text{PT16})$$

All blocks within each half, including off-diagonal blocks, occur.

Proof. The full block exponential is the finite sum $e^{-tb_{\alpha}} \sum_{j=0}^{e-1} (-t)^j N_{\alpha}^j / j!$. Every entry of the integrand in (PT15), and its derivative, is therefore a polynomial times an exponential of strictly negative real exponent. The integrals converge absolutely and the endpoint at infinity is zero. Differentiation and integration give

$$D_{\pm}^* \Gamma_{\pm} + \Gamma_{\pm} D_{\pm} = Q_{\pm}.$$

The quadratic form of $\Gamma_{\pm}$ on v is the integral of $\|Q_{\pm}^{1/2} e^{-tD_{\pm}} v\|^2$. It vanishes exactly when this continuous nonnegative integrand is identically zero. Its entire power series is zero exactly when all $Q_{\pm}^{1/2} D_{\pm}^j v$ vanish. The characteristic polynomial of $D_{\pm}$ has degree r ; division by it expresses every power $j \geq r$ as a linear combination of lower powers. This proves the finite test (PT14) and strict positivity.

Conversely let $\Gamma > 0$ on one half satisfy $Q = D^* \Gamma + \Gamma D \succeq 0$. Differentiating $e^{-tD^*} \Gamma e^{-tD}$, and integrating between zero and infinity, recovers (PT15) exactly. The preceding positivity argument proves (PT14). The same argument shows that the homogeneous Lyapunov equation has only the zero solution. The characterization (PT9) now proves that (PT15) gives every minimizer and only minimizers.

For (PT16), expand both finite exponentials. The term with powers i, j has scalar integral $\int_0^{\infty} t^{i+j} e^{-zt} dt = (i+j)! / z^{i+j+1}$, where $z = \bar{b}_{\alpha} + b_{\beta}$ has positive real part. The integral identity starts with $1/z$ and follows by successive integration by parts, whose endpoints vanish by the same exponential decay. This proves every coefficient and sign. On the negative half $N_{\alpha} = -J_e$, so its two powers supply an additional $(-1)^{i+j}$; these signs remain explicit in (PT13)–(PT16). No inverse of a nilpotent is used. $\square$

11 A minimizing metric with rank-two weight

There is an attaining family with many zero weight directions, even when $e > 1$. On each half define the single literal row

$$\ell_{\pm}(v) = \sum_{\lambda \text{ in that half}} v_{\lambda, e-1}, \quad Q_{\pm} = \ell_{\pm}^* \ell_{\pm}. \quad (\text{PT17})$$

The row extracts the final divided-derivative coefficient in every complete primary block, with coefficient exactly one in each.

Proposition 11.1. *The two choices (PT17) satisfy (PT14). The metric G_{two} obtained from (PT15)–(PT16) has*

$$\text{spec } H_{G_{\text{two}}} = \{L, -L, 0, \dots, 0\}, \quad T_+(G_{\text{two}}) = \|H_{G_{\text{two}}}\| = L. \quad (\text{PT18})$$

Hence the positive-trace minimum is attained within the rank-two weight stratum as well as within the full-rank weight stratum.

Proof. Write the blocks of either half as $b_\lambda I + sJ_e$, with the fixed common sign $s = 1$ on the positive half and $s = -1$ on the negative half. Then

$$\ell e^{-tD}v = \sum_{\lambda} e^{-tb_{\lambda}} \sum_{j=0}^{e-1} \frac{(-st)^j}{j!} v_{\lambda, e-1-j}.$$

Distinct λ 's give distinct b_{λ} 's on either half. The functions $e^{-tb_{\lambda}}p_{\lambda}(t)$, with $\deg p_{\lambda} < e$, are linearly independent. Here is an explicit proof: apply $\prod_{\mu \neq \lambda} (\partial_t + b_{\mu})^e$ to a vanishing sum. It kills every summand except the λ summand, where it gives $e^{-tb_{\lambda}}$ times $\prod_{\mu \neq \lambda} (\partial_t + b_{\mu} - b_{\lambda})^e p_{\lambda}$. On the polynomials of degree less than e , each operator $\partial_t + a$ with $a \neq 0$ is invertible: its matrix is triangular with diagonal a , or its inverse is the finite geometric sum in $-\partial_t/a$. Therefore every p_{λ} is zero. Their displayed coefficients force every coordinate of v to be zero. Thus $\ell e^{-tD}v = 0$ for all t implies $v = 0$, and (PT14) follows as in the preceding theorem.

In each half the Lyapunov weight is $Q_{\pm}$, of rank one. Whitening by the positive $\Gamma_{\pm}$ preserves this rank and gives a nonnegative rank-one matrix on the positive half and a nonpositive rank-one matrix on the negative half. Its nonzero eigenvalue is its trace. Cyclicity of trace gives

$$\text{Tr}(\Gamma_+^{-1}Q_+) = 2\Re \text{Tr } D_+ = L, \quad \text{Tr}(\Gamma_-^{-1}Q_-) = 2\Re \text{Tr } D_- = L.$$

The two complete halves are orthogonal in the constructed metric, so these are exactly the eigenvalues in (PT18). All vectors of all primary blocks remain present; zero Hermitian weight does not remove those vectors or their original operator action. $\square$

The rank-two metric has fully evaluated entries as well. On either half put $\sigma = 1$ for D_+ , $\sigma = -1$ for D_- , and, for indices $0 \leq i, j < e$, put $u = e - 1 - i$, $v = e - 1 - j$. In (PT16), $Q_{\alpha\beta}$ is the matrix with its only nonzero entry, equal to one, at $(e - 1, e - 1)$. Thus

$$(\Gamma_{\pm})_{\alpha i, \beta j} = \frac{(-\sigma)^{u+v}(u+v)!}{u!v!(\bar{b}_{\alpha} + b_{\beta})^{u+v+1}}. \quad (\text{PT18a})$$

For $u + v > 0$, substituting these entries into the Lyapunov equation reduces its numerator to

$$\frac{(u+v)!}{u!v!} - \mathbf{1}_{u>0} \frac{(u+v-1)!}{(u-1)!v!} - \mathbf{1}_{v>0} \frac{(u+v-1)!}{u!(v-1)!} = 0.$$

For $u = v = 0$ the single term gives one. This directly verifies every row and both nilpotent signs independently of the integral.

Every positive/negative weight-rank pair $1 \leq r_+, r_- \leq q/2$ occurs among minimizers. In each half choose a positive semidefinite matrix of the prescribed rank whose range contains $\ell_{\pm}^*$. Its kernel is contained in $\ker \ell_{\pm}$, so its finite observability kernel is zero by the proof of (PT17). Formula (PT15) gives a positive metric, and invertible congruence preserves the prescribed weight rank. This proves the claim, including all intermediate zero-weight dimensions, with every original primary vector present.

12 The original positive-primary exterior line

Equip every exterior power with the Gram-determinant metric induced by its stated G . The inclusion of the positive-primary line is the actual map $\Lambda^r X_+ : \Lambda^r \mathbb{C}^r \rightarrow \Lambda^r E$, not a choice of a positive eigenspace of H_G . On this line the generator induced by B is multiplication by $\text{Tr } B_+$. Its imaginary part vanishes because for each fixed a the sum of $2b - k$ over $0 \leq b \leq k$ is zero. Its real part is $L/2$ by (PT7). Thus for any nonzero element $\omega_+ \in \Lambda^r E_+$ and every real t ,

$$\|\Lambda^r e^{tB} \omega_+\|_{\Lambda^r G}^2 = e^{Lt} \|\omega_+\|_{\Lambda^r G}^2, \quad \|\Lambda^r e^{tA} \omega_+\|_{\Lambda^r G}^2 = e^{(kr+L)t} \|\omega_+\|_{\Lambda^r G}^2. \quad (\text{PT19})$$

To verify the claimed action directly, triangularize each full primary block by its fixed basis (PT5). The determinant of e^{tB_+} is the product of its diagonal entries, $e^{t \operatorname{Tr} B_+}$; all nilpotent exponential diagonals are one. Squaring the scalar modulus gives (PT19). In the original column frame the derivative has the equally exact expression

$$\left. \frac{d}{dt} \right|_{t=0} \log \det(X_+^* e^{tB^*} G e^{tB} X_+) = \operatorname{Tr}((X_+^* G X_+)^{-1} X_+^* W(G) X_+) = \operatorname{Tr}(P_G H_G) = L. \quad (\text{PT20})$$

The derivative of $\log \det M(t)$ follows by multilinear expansion of $\det(I + tM(0)^{-1}M'(0))$. Substituting (PT6) and moving finite factors around the trace proves the second equality. This supplies the exact receiver to the original exterior benchmark for every G , including the fixed attained metric.

13 The canonical attained metric and its exact rank-two gap

Use the original source $m_k(y) dy$ and physical coordinate $S = c + iy$ from (PT1). Its positivity and moments are the original source data of [5]; in particular it is positive on a set of positive measure and all polynomial moments are finite. For $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $N \geq q - 1$, put

$$\begin{aligned} \langle f, g \rangle &= \int_{\mathbb{R}} \overline{f(c + iy)} g(c + iy) m_k(y) dy, \quad H_N = (\langle S^i, S^j \rangle)_{0 \leq i, j \leq N}, \\ J_N : \mathcal{P}_N &\rightarrow E, \quad f \mapsto [f]_{\chi}, \quad G_N = (J_N H_N^{-1} J_N^*)^{-1}, \quad R_N = H_N^{-1} J_N^* G_N. \end{aligned} \quad (\text{PT21})$$

A nonzero polynomial has only finitely many zeros on the vertical line; thus $H_N > 0$. Surjectivity of J_N gives $G_N > 0$. Matrix multiplication gives $J_N R_N = I$ and $R_N^* H_N R_N = G_N$. For $z \in \ker J_N$, $z^* H_N R_N = z^* J_N^* G_N = 0$. Therefore $R_N u$ is the actual attained minimum representative of u ; every other representative has squared norm $u^* G_N u + \|z\|^2$.

Define the following original rows and polynomial, with no omitted leading coefficient:

$$\ell_N(u) = [S^N] R_N u, \quad r_N(S) = \chi(S) S^{N+1-q}, \quad s_N(u) = \langle r_N, R_N u \rangle. \quad (\text{PT22})$$

The exponent $N + 1 - q$ is nonnegative even at $N = q - 1$. Set $T_N f = S f - [S^N] f r_N$. As χ is monic, the degree- $N + 1$ coefficient cancels, so $T_N : \mathcal{P}_N \rightarrow \mathcal{P}_N$. It satisfies the exact quotient intertwining relation $J_N T_N = A J_N$. Thus $R_N A$ is the orthogonal projection of $T_N R_N$ onto $(\ker J_N)^\perp$. The physical identity $\bar{S} + S = k$ on the vertical line gives $\langle S f, g \rangle + \langle f, S g \rangle = k \langle f, g \rangle$. Taking $f = R_N u$, $g = R_N v$, and subtracting the two displayed leading-coefficient corrections, proves

$$W(G_N) = -(\ell_N^* s_N + s_N^* \ell_N), \quad \operatorname{rank} W(G_N) \leq 2, \quad \operatorname{Tr} H_{G_N} = 2\Re \operatorname{Tr} A - kq = 0. \quad (\text{PT23})$$

The last equality follows by summing $2a - k$ over all a, b . Since (PT7) gives a strictly positive trace lower bound, the rank is exactly two and its two nonzero eigenvalues are $+\epsilon_N, -\epsilon_N$, with $\epsilon_N > 0$.

Define, in the original packet coordinates,

$$u_N = s_N G_N^{-1} s_N^*, \quad v_N = \ell_N G_N^{-1} \ell_N^*, \quad z_N = s_N G_N^{-1} \ell_N^*. \quad (\text{PT24})$$

The trace identity in (PT23) gives $\Re z_N = 0$. Put $a = G_N^{-1/2} \ell_N^*$ and $b = G_N^{-1/2} s_N^*$, so $H_{G_N} = -(ab^* + ba^*)$. Direct multiplication yields

$$\operatorname{Tr}(H_{G_N}^2) = 2u_N v_N + 2\Re(z_N^2) = 2u_N v_N - 2(\Im z_N)^2.$$

The eigenvalues in (PT23) give the exact certificate and bound

$$\epsilon_N^2 = u_N v_N - (\Im z_N)^2 = u_N v_N - |z_N|^2 \geq L^2. \quad (\text{PT25})$$

No coefficient of R_N , no root difference in χ , and no term in either original source region was changed to obtain this identity.

Choose orthonormal eigenvectors p_N, n_N of the whitened canonical matrix with eigenvalues $\epsilon_N, -\epsilon_N$. Then the full exact receiver of (PT7) is

$$\epsilon_N - L = \epsilon_N \left(1 - \|P_{G_N} p_N\|^2 + \|P_{G_N} n_N\|^2 \right). \quad (\text{PT26})$$

This follows by substituting $(H_{G_N})_+ = \epsilon_N p_N p_N^*$ and $(H_{G_N})_- = \epsilon_N n_N n_N^*$ in (PT7). In particular equality means precisely that p_N lies in $G_N^{1/2} E_+$ and n_N lies in its orthogonal complement; equivalently (PT9) holds for the fixed G_N . Equations (PT22)–(PT26) evaluate the comparison through its actual minimum map. They do not assert that this minimum sequence happens to lie in the minimizing set (PT15).

14 Exact metric realization in the original finite source

Use $N = 2q - 1$, the unmodified H_N, J_N, R_N, G_N in (PT21), and the complete relation map

$$K_\chi : \mathcal{P}_{q-1} \hookrightarrow \mathcal{P}_{2q-1}, \quad f \mapsto \chi f, \quad Q_\chi = K_\chi^* H_N K_\chi > 0, \quad B_\chi = K_\chi Q_\chi^{-1/2}. \quad (\text{PT27})$$

Injectivity follows from the integral domain $\mathbb{C}[S]$. Polynomial division gives $\text{im } K_\chi = \ker J_N$, exactly, and this space has dimension q . Hence $J_N B_\chi = 0$, $B_\chi^* H_N B_\chi = I$, and $B_\chi^* H_N R_N = 0$.

For every displayed minimizing metric G_0 from (PT15), including G_* and G_{two} , set

$$\alpha = 1 + \left\| G_0^{-1/2} G_N G_0^{-1/2} \right\|, \quad \Delta = \alpha G_0 - G_N, \quad F = Q_\chi^{-1/2} \Delta^{1/2} : E \longrightarrow \mathcal{P}_{q-1}, \quad (\text{PT28})$$

$$R = R_N + K_\chi F.$$

The norm is the ordinary largest eigenvalue of the positive matrix shown. Consequently $G_0^{-1/2} \Delta G_0^{-1/2} \succeq I$, so $\Delta > 0$ and F is invertible. The orthogonal cross terms vanish and give

$$J_N R = I_E, \quad R^* H_N R = G_N + F^* Q_\chi F = \alpha G_0, \quad H_{\alpha G_0} = H_{G_0}. \quad (\text{PT29})$$

The last equality follows directly from (PT2), because a positive scalar commutes with the specified positive square root. This is an actual finite polynomial section realizing the trace minimum. It retains the full relation χ , not a root-value ideal.

Its residual at every smaller polynomial source is also exact. For $W \subseteq \mathcal{P}_{q-1}$, write

$$Q_W = \mathcal{P}_{2q-1} / K_\chi W, \quad \eta_W : \mathcal{P}_{q-1} / W \hookrightarrow Q_W, \quad [f] \mapsto [K_\chi f], \quad \pi_W : Q_W \twoheadrightarrow E.$$

The first map is well-defined and injective because K_χ is injective, and the second is induced by J_N . Their image and kernel agree by $\ker J_N = \text{im } K_\chi$. Therefore this is the explicit exact sequence and section comparison

$$0 \longrightarrow \mathcal{P}_{q-1} / W \xrightarrow{\eta_W} Q_W \xrightarrow{\pi_W} E \longrightarrow 0,$$

$$[R(\cdot)]_W - [R_N(\cdot)]_W = \eta_W([F(\cdot)]_W), \quad \ker([F]_W) = F^{-1}W, \quad \text{rank}([F]_W) = q - \dim W. \quad (\text{PT30})$$

Surjectivity of $[F]_W$ follows from invertibility of F . Thus every proper finite polynomial source retains a nonzero residual map of the computed rank. The arithmetic quotient class is nevertheless identical, because $\pi_W[R]_W = I_E$.

For completeness, the entire change in the multiplication boundary is a relation as well, with its original degree recorded. Interpret the identity below in $\mathcal{P}_{2q}$, and define $\widehat{F} : E \rightarrow \mathcal{P}_q$ by $\widehat{F} = SF - FA$. Then direct substitution gives

$$(SR - RA) - (SR_N - R_N A) = K_\chi^{(q)} \widehat{F}, \quad K_\chi^{(q)} : \mathcal{P}_q \hookrightarrow \mathcal{P}_{2q}, \quad f \mapsto \chi f. \quad (\text{PT31})$$

For a target source $W' \subseteq \mathcal{P}_q$ this difference has precisely the class $[\widehat{F}]_{W'}$ under the injective map induced by $K_\chi^{(q)}$. Thus a full-quotient equality has not erased the smaller-source generator residual.

15 Backward receivers for the earlier theta source

The original TC source has weight one and a packet of actual zeros; the present cyclic sum packet has weight k . We give the exact receiving operations at those two types. For any positive Gram and operator pair (G, A_0) of weight w , the operation is $(G, A_0, w) \mapsto B_0 = A_0 - wI/2$, followed by the trace projection proof (PT7)–(PT9) on its complete positive-primary sum. The polynomial coordinate change to (PT1), when used, is the original full-primary sum realization, whose explicit CRT map is (PT3). This operation does not replace an ordered tensor space by its cyclic image: TC21 remains on its original ordered tensor space.

In the original TC22 coordinates, write $W_m = -z_m b_m - b_m^* z_m^*$, where z_m is its actual column and b_m its actual row. The receiver to our row form is exactly

$$\ell = b_m, qqads = z_m^*, \quad sG_m^{-1}s^* = u_m, quadlG_m^{-1}\ell^* = v_m, quadsG_m^{-1}\ell^* = \bar{c}_m. \quad (\text{PT32})$$

Complex conjugation changes the sign of the imaginary part but not its square; (PT25) is therefore precisely the original scalar certificate TC26 and lower bound TC28 in these coordinates. The new projection formula (PT26) strengthens its description by evaluating the excess relative to the unchanged positive-primary exterior line. The known rank-one Gram update TC27 remains exactly as stated; it is not assigned to the distinct sequence (PT21). TC18 and TC21 are operator-norm infima. Equations (PT12) and (PT18) add the positive-trace minimum and a minimizing rank-two stratum, without changing those previously proved infima.

Here is the exact smaller-source map for the original theta realization TC29a–d [4], including its finite correction. Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x, \quad \Theta\phi = 2 \sum_{n \geq 1} \phi(nx),$$

and the original space $\mathcal{B}$ of smooth functions rapidly decreasing at both ends on $\mathbb{R}_{>0}$. The map $\Theta : V \rightarrow \mathcal{B}$ is injective. Indeed on $\Re s > 1$ absolute convergence and the substitution $u = nx$ give $\mathcal{M}\Theta\phi(s) = 2\zeta(s)\mathcal{M}\phi(s)$. The Euler product is nonzero there, and uniqueness of the Mellin transform on a vertical line is Fourier uniqueness after $x = e^y$. Thus $\Theta\phi = 0$ forces $\phi = 0$. For each $W \subseteq V$ define the actual maps

$$\eta_W : V/W \longrightarrow \mathcal{B}/\Theta W, quad[\phi] \longmapsto [\Theta\phi], \quad \pi_W : \mathcal{B}/\Theta W \longrightarrow \mathcal{B}/\Theta V.$$

Injectivity of Θ proves injectivity of η_W , and its image is exactly $\ker \pi_W$. For the original reference section s_h and its finite primitive correction $T = \Phi_L C : E_h \rightarrow V$, put $R = s_h + \Theta T$. Subtraction of actual functions gives

$$[R]_W - [s_h]_W = \eta_W([T]_W), \quad \pi_W[R]_W = [s_h]_V, qqquad \ker([R - s_h]_W) = T^{-1}W. \quad (\text{PT33})$$

The last equality uses the proved injectivity. This is the precise map from a full-source correction to its smaller-source class, for the original theta packet E_h .

The original reference relation is $Ds_h - s_h A_h = \Theta(\phi_* \ell_h)$. Since termwise differentiation gives $D\Theta = \Theta D$, substitution proves

$$DR - RA_h = \Theta K_R, \quad K_R = \phi_* \ell_h + DT - TA_h : E_h \longrightarrow V. \quad (\text{PT34})$$

For $DW \subseteq W'$, the induced map $\bar{D} : \mathcal{B}/\Theta W \rightarrow \mathcal{B}/\Theta W'$ is well-defined because $D\Theta W \subseteq \Theta W'$. Its defect on (PT33) is exactly $\eta_{W'}([K_R]_{W'})$, which vanishes on a vector precisely when that original primitive belongs to W' . This proves the typed generator receiver.

For $a > 0$ let $\mathcal{R}_a F(x) = F(x/a)$ and let $A_{a,h} = e^{(\log a)A_h}$ be the original arithmetic action. Retain the reference primitive $\Psi_{a,h}$ defined by $\mathcal{R}_a s_h - s_h A_{a,h} = \Theta \Psi_{a,h}$. The equality $\mathcal{R}_a \Theta = \Theta \mathcal{R}_a$ and substitution give

$$\begin{aligned} c_{R,a} &= \Psi_{a,h} + \mathcal{R}_a T - TA_{a,h}, & \mathcal{R}_a R - RA_{a,h} &= \Theta c_{R,a}, \\ \bar{\mathcal{R}}_a^{W \rightarrow \mathcal{R}_a W} [R]_W - [R]_{\mathcal{R}_a W} A_{a,h} &= \eta_{\mathcal{R}_a W}([c_{R,a}]_{\mathcal{R}_a W}), \\ c_{R,ab} &= \mathcal{R}_a c_{R,b} + c_{R,a} A_{b,h}. \end{aligned} \quad (\text{PT35})$$

For the last identity, expand the preceding two definitions and cancel the terms $\mathcal{R}_a T A_{b,h}$. The reference primitives satisfy the same identity because the difference has zero Θ -image and Θ is injective. The target source remains $\mathcal{R}_a W$, so no source invariance is silently inserted. Equations (PT30)–(PT31) give the complete polynomial counterpart on the unchanged sum packet, and (PT33)–(PT35) are the actual TC proper-source receiver.

16 Quantitative excess and the original cross metric

The trace defect also measures the original metric interaction between the positive and negative primary halves. In the fixed CRT ordering let

$$\begin{aligned} \mathcal{G} &= (C^{-1})^* G C^{-1} = \begin{pmatrix} P & X \\ X^* & Y \end{pmatrix}, \quad K = P^{-1}X, \quad Z = Y - X^*P^{-1}X > 0, \\ F &= \begin{pmatrix} P^{-1/2} & -KZ^{-1/2} \\ 0 & Z^{-1/2} \end{pmatrix}, \quad \mathcal{F} = C^{-1}F, \quad \mathcal{F}^*G\mathcal{F} = I. \end{aligned} \quad (\text{PT36})$$

Positivity of P is restriction positivity; positivity of Z follows by testing $\mathcal{G}$ on $(-Kv, v)$. Multiplication verifies the displayed isometry. Write the original CRT operator as $CBC^{-1} = \text{diag}(B_+, B_-)$, with $B_- = -D_-$. Its matrix in this isometry is

$$\begin{aligned} \mathcal{F}^{-1}B\mathcal{F} &= \begin{pmatrix} \widehat{B}_+ & D \\ 0 & \widehat{B}_- \end{pmatrix}, \quad \widehat{B}_+ = P^{1/2}B_+P^{-1/2}, \quad \widehat{B}_- = Z^{1/2}B_-Z^{-1/2}, \\ D &= P^{1/2}(KB_- - B_+K)Z^{-1/2}, \quad \mathcal{H} = \mathcal{F}^*W(G)\mathcal{F} = \begin{pmatrix} D_+^H & D \\ D_-^* & D_-^H \end{pmatrix}, \\ D_\pm^H &= \widehat{B}_\pm + \widehat{B}_\pm^*, \quad \text{Tr } D_+^H = L, \quad \text{Tr } D_-^H = -L. \end{aligned} \quad (\text{PT37})$$

The matrix $G^{1/2}\mathcal{F}$ is unitary, so $\mathcal{H}$ and H_G have exactly the same eigenvalues. Let $\mathcal{E} = T_+(G) - L$, and let $\|D\|_1$ denote the sum of the singular values of D . Then

$$\|D\|_1^2 \leq \mathcal{E}(2L + \mathcal{E}), \quad \mathcal{E} \geq \sqrt{L^2 + \|D\|_1^2} - L. \quad (\text{PT38})$$

To prove this, take a singular-value decomposition $D = U\Sigma V^*$, extending the singular frames to full unitary bases, and put $W = UV^*$. The matrices

$$J_0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad T_0 = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}$$

square to I and anticommute. Hence $Z_\theta = \cos \theta J_0 + \sin \theta T_0$ is a Hermitian contraction for every real θ . Diagonalizing $\mathcal{H}$, every diagonal entry of Z_θ lies between -1 and 1 , so $\text{Tr}(Z_\theta \mathcal{H}) \leq \text{Tr} |\mathcal{H}|$. The two sides are respectively $2L \cos \theta + 2\|D\|_1 \sin \theta$ and $2(L + \mathcal{E})$, since $\text{Tr } \mathcal{H} = 0$. Maximizing over θ proves (PT38).

There is a complete finite inverse from this measured cross block to the original X . For positive-half block parameters $b_\alpha = \lambda_\alpha - c$ and negative-half parameters $\beta_\eta = \lambda_\eta - c$, the original operator blocks are $b_\alpha I + J_e$ and $\beta_\eta I + J_e$. Set $R = P^{-1/2}DZ^{1/2}$. Then

$$K_{\alpha\eta} = - \sum_{u,v=0}^{e-1} \frac{(-1)^u (u+v)!}{u!v! (b_\alpha - \beta_\eta)^{u+v+1}} J_e^u R_{\alpha\eta} J_e^v, \quad X = PK. \quad (\text{PT39})$$

Indeed $R = KB_- - B_+K$ by (PT37). The absolutely convergent integral $-\int_0^\infty e^{-tB_+} R e^{tB_-} dt$ solves this equation by integration of its derivative. Each real exponent is at most $-2\delta t$, multiplied by a finite polynomial. Expansion of its two full nilpotent exponentials gives (PT39), with sign $(-1)^u$ because the right exponential is e^{+tB_-} . Uniqueness follows from the homogeneous equation:

$e^{-tB_+} K e^{tB_-} = K$ then holds for all t , while its left side tends to zero. Thus the integral and the finite sum invert the original cross map exactly.

For an explicit bound in the same full primary coordinates set

$$\kappa = \max_{\alpha, \eta} \sum_{u, v=0}^{e-1} \frac{(u+v)!}{u!v!|b_\alpha - \beta_\eta|^{u+v+1}}, \quad (\text{PT40})$$

$$\|P^{-1}X\|_F^2 \leq \kappa^2 \|P^{-1/2}\|^2 \|Z^{1/2}\|^2 \mathcal{E}(2L + \mathcal{E}).$$

Each power of J_e has norm at most one. Applying the triangle inequality to every block in (PT39), then summing squared block Frobenius norms, gives $\|K\|_F \leq \kappa \|R\|_F$. The multiplication bound for $R = P^{-1/2} D Z^{1/2}$, followed by $\|D\|_F \leq \|D\|_1$ and (PT38), proves (PT40). All original metric factors are displayed; no uniform bound on their condition numbers has been inserted. For the canonical metric (PT21) one substitutes exactly $G = G_N$ and $\mathcal{E} = \epsilon_N - L$. Thus the same scalar which occurs in (PT25)–(PT26) controls the actual positive/negative primary metric cross block through (PT36)–(PT40). In particular the original scalar certificate itself strengthens to

$$u_N v_N - (\Im z_N)^2 \geq L^2 + \|D_N\|_1^2, \quad D_N = P_N^{1/2} (K_N B_- - B_+ K_N) Z_N^{-1/2}, \quad (\text{PT41})$$

where P_N, X_N, Y_N, K_N, Z_N are obtained by substituting the unchanged G_N in (PT36). Indeed $\mathcal{E}(2L + \mathcal{E}) = (\epsilon_N - L)(\epsilon_N + L) = \epsilon_N^2 - L^2$, so (PT38) and (PT25) give precisely (PT41). This is an estimate on the actual attained metric; every factor in its cross block has its original coordinate map and its finite inverse in (PT36)–(PT39).

17 Scope of the finite result

The positive-trace minimum (PT12), every minimizing metric (PT14)–(PT16), the rank-two attainer (PT18), the exterior growth (PT19)–(PT20), the canonical scalar and defect (PT25)–(PT26), the finite source and proper-source residuals (PT28)–(PT35), and the complete cross-metric receiver (PT36)–(PT41) are established by the displayed proofs. The fixed attained metrics remain the matrices in (PT21). The variational minimum does not evaluate their separate projected kernel and boundary allocations; their exact trace gap is (PT26). No value for an unevaluated projection is assigned in this result.

Every original coefficient section and native–arithmetic metric return

18 The prescribed conductor minor and the exact arithmetic image

This derives the coefficient-image continuation supplied in [8], retaining its exact residue map. All parameters and full polynomial factors are those of CAL1. Let $\mathbb{V} = (\beta_\alpha^n)_{0 \leq n < q, \alpha}$ be the original upper Vandermonde, with the original root labels, and let P_v insert zeros in its first v derivative coordinates. Write $q' = (a - 7)^2$ and retain the original conductor

$$E_A(z) = \sum_{r,t=0}^8 a_{rt} e^{\beta_{8,r,t} z}, \quad \mu_j = E_A^{(j)}(0), \quad \mu_0 = \cdots = \mu_{v-1} = 0, \quad \mu_v \neq 0. \quad (\text{COI1})$$

The coefficient map $C_a^\top : \mathbb{C}^{q'} \rightarrow \mathbb{C}^q$ is exactly the unscaled convolution by E_A :

$$\mathcal{N}_{C_a^\top \eta}(z) = E_A(z) \sum_{\lambda \in \text{lower}} \eta_\lambda e^{\lambda z}. \quad (\text{COI2})$$

The original DNE2–3, NSO12 and NCT6–7 specify $J_v = P_v^\top \mathbb{V} C_a^\top$ and $Z_I = \mathbb{V}^{-1} P_v E_{J_I}$, where I is its selected invertible q' -row minor and J_I is the complementary set. These sources are pinned in [11, 12].

18.1 The first consecutive block is evaluated completely

Leibniz differentiation of COI2 gives, for $r \geq 0$,

$$(J_v)_{r,\lambda} = \sum_{j=v}^{v+r} \binom{v+r}{j} \mu_j \lambda^{v+r-j}. \quad (\text{COI3})$$

This is a degree- r polynomial in the actual lower root λ , with leading coefficient $\binom{v+r}{v} \mu_v$. Let $I_0 = (0, 1, \dots, q' - 1)$ in the increasing derivative order, and let V_- be the original increasing-power lower Vandermonde. Then $(J_v)_{I_0} = H V_-$, where the complete lower-triangular factor is

$$H_{rt} = \binom{v+r}{v+r-t} \mu_{v+r-t} \quad (0 \leq t \leq r < q'), \quad H_{rt} = 0 \quad (t > r). \quad (\text{COI4})$$

It follows that

$$\det(J_v)_{I_0} = \mu_v^{q'} \prod_{r=0}^{q'-1} \binom{v+r}{v} \prod_{\lambda < \lambda'} (\lambda' - \lambda) \neq 0. \quad (\text{COI5})$$

All signs are with the original ordered lower-root columns and the increasing rows. The Vandermonde determinant follows by alternation, divisibility by every root difference, and its leading monomial of coefficient one; no absolute value or ordering change is made. The inverse is equally explicit: solve $H_{rr} u_r = b_r - \sum_{t < r} H_{rt} u_t$ successively and then apply V_-^{-1} . Thus every original conductor derivative in this block remains in its inverse.

The exact consequence for a first-minor rule needs its actual order. In increasing lexicographic enumeration of the increasing q' -tuples of row indices, I_0 is the very first tuple, so COI5 evaluates that rule before any search for a later nonzero determinant. The original frozen DNE3, however, states a “fixed ordering of all subsets”; NSO12 and NCT7 refer to the first ordered nonzero minor. Their printed lexicographic convention is for upper-root coefficient columns. It does not specify an enumeration of the row subsets. Accordingly COI5 proves the consecutive determinant but does not alter an independently retained original choice I . The exact calculation for that original choice is COI9–17 below. This records a source-specification distinction and supplies the full morphism; it does not discard the relation between the two charts.

18.2 Every residue column and its full inverse

Write $\chi(S) = \sum_{n=0}^q c_n S^n$, $c_q = 1$, and define the original residue map

$$\mathcal{L}(\nu) = \sum_{\alpha} \nu_{\alpha} \frac{\chi(S)}{S - \beta_{\alpha}}, \quad \mathfrak{p}_n(S) = \sum_{t=n+1}^q c_t S^{t-n-1}. \quad (\text{COI6})$$

The exact partial-fraction identity $\mathcal{L}(\nu)/\chi = \sum_{\alpha} \nu_{\alpha}/(S - \beta_{\alpha})$ has Laurent coefficients $[S^{-n-1}]\mathcal{L}(\nu)/\chi = \sum_{\alpha} \nu_{\alpha} \beta_{\alpha}^n$. The first q coefficients determine its degree-below- q numerator: after multiplication by monic χ , any remaining tail is strictly negative in S and has no polynomial part. This proves $\mathcal{L}\mathbb{V}^{-1}e_n = \mathfrak{p}_n$ for every $0 \leq n < q$, including all coefficients and phases. For the consecutive block, $J_0 = \{q', \dots, q - v - 1\}$, whence

$$L_0 = \mathcal{L}\mathbb{V}^{-1}P_v E_{J_0}, \quad L_0 e_j = \mathfrak{p}_{q-m+j} = \sum_{t=q-m+j+1}^q c_t S^{t-q+m-j-1}, \quad 0 \leq j < m, \quad (\text{COI7})$$

$$\text{im } L_0 = \mathcal{P}_{m-1}, \quad (L_0^{-1}P)_j = [S^{-q+m-j-1}] \frac{P(S)}{\chi(S)}.$$

The first equality has one polynomial of every degree $m-1, m-2, \dots, 0$, each monic. Its coefficient matrix in reversed degree order is triangular with diagonal one, proving the asserted image and invertibility. The displayed Laurent extraction is its inverse by the coefficient identity just proved. If $\deg P < m$, all earlier coefficients $[S^{-n-1}]P/\chi$, $n < q-m$, are zero because the rational function starts at order S^{m-1-q} . For an arbitrary original I the exact image instead is

$$L_I = \mathcal{L}\mathbb{V}^{-1}P_v E_{J_I}, \quad \text{im } L_I = \text{span}\{\mathfrak{p}_{v+j} : j \in J_I\}. \quad (\text{COI8})$$

These are original injections, not isometric identifications of their arithmetic pullbacks with native minimum metrics. In fact COI8 equals $\mathcal{P}_{m-1}$ exactly for $I = I_0$. If a selected index $v+j$ is less than $q-m$, its individual column has degree $q-v-j-1 \geq m$ and cannot belong to that low-degree space. Otherwise all m selected indices lie in the m -element set $\{q-m, \dots, q-1\}$ and hence equal it; their complement is precisely I_0 .

18.3 The complete transition for every original minor

Retain the actually selected I and define, with ordered coordinate extractions in $\mathbb{C}^{q-v}$,

$$D_{0I} = (J_v)_{I_0}^{-1} E_{I_0}^T E_{J_I}, \quad M_{0I} = E_{J_0}^T E_{J_I} - (J_v)_{J_0} D_{0I}. \quad (\text{COI9})$$

Extraction in the I_0 rows and the J_0 rows verifies exactly

$$E_{J_I} = J_v D_{0I} + E_{J_0} M_{0I}. \quad (\text{COI10})$$

The induced M_{0I} is invertible. Indeed both $[J_v, E_{J_I}]$ and $[J_v, E_{J_0}]$ are square isomorphisms; their transition is the block triangular matrix $\begin{pmatrix} I & D_{0I} \\ 0 & M_{0I} \end{pmatrix}$. To see the first isomorphism directly, extraction in I of a relation between its columns annihilates its first coefficient, and then its complementary coefficient vanishes; the second uses COI5. Consequently no determinant or inverse is merely presumed. Using the literal convolution identity gives the full exact arithmetic transition

$$Z_I = C_a^T D_{0I} + Z_0 M_{0I}, \quad \boxed{L_I = F_C D_{0I} + L_0 M_{0I}}, \quad F_C = \mathcal{L} C_a^T. \quad (\text{COI11})$$

The term $F_C D_{0I}$ is an actual arithmetic vector. It remains under the injective full-unit theta realization and cannot be removed there as a relation. On the native side, dividing its numerator by E_A gives exactly the lower exponential $\sum_{\lambda} (D_{0I} z)_{\lambda} e^{\lambda z}$. That vector is among the very same lower evaluations minimized in the original native metric. Translation of the full affine fibre by it

is a bijection of that fibre. Therefore, at every original cutoff and source order, the exact native congruence is

$$D_{I,a,N-v}^{(s)} = M_{0I}^* D_{0,a,N-v}^{(s)} M_{0I}, \quad s = 1, a. \quad (\text{COI12})$$

This proves the native comparison by its actual minimum, rather than imposing it on the different arithmetic metric.

For the full arithmetic quotient forms put $G_{00,N} = L_0^* G_N L_0$, $G_{CC,N} = F_C^* G_N F_C$ and $G_{0C,N} = L_0^* G_N F_C$, $G_{C0,N} = G_{0C,N}^*$. Multiplication of COI11 gives the complete receiver

$$\boxed{L_I^* G_N L_I = M_{0I}^* G_{00,N} M_{0I} + D_{0I}^* G_{CC,N} D_{0I} + M_{0I}^* G_{0C,N} D_{0I} + D_{0I}^* G_{C0,N} M_{0I}.} \quad (\text{COI13})$$

The identical expansion holds with G_N replaced by the full centered action form $A^* G_N + G_N A - a G_N$, and with every boundary difference $G_{N_l} - G_{N_h}$. Thus no complete-action or boundary cross block is lost in transport.

For a more quantitative bound on a genuine original chart, put $\Delta G_N = G_{q-1} - G_N \succeq 0$. For $u = L_0 M_{0I} z$ and $w = F_C D_{0I} z$, CAL13 gives $u^* \Delta G_N u \leq \varepsilon_a \|u\|_{G_{q-1}}^2$. The complete minimum also gives $w^* \Delta G_N w \leq \|w\|_{G_{q-1}}^2$. The Cauchy–Schwarz inequality for the positive form ΔG_N follows by minimizing $(u+tw)^* \Delta G_N (u+tw)$ over complex t ; it remains true at its zero directions. Expanding the square proves

$$0 \leq z^* L_I^* \Delta G_N L_I z \leq (\sqrt{\varepsilon_a} \|L_0 M_{0I} z\|_{G_{q-1}} + \|F_C D_{0I} z\|_{G_{q-1}})^2. \quad (\text{COI14})$$

The exact cross block remains COI13; COI14 is not substituted for the last conductor norm. For example applying $2xy \leq x^2 + y^2$ to the displayed nonnegative terms gives the valid whole-space inequality

$$L_I^* \Delta G_N L_I \preceq 2\varepsilon_a M_{0I}^* G_{00,q-1} M_{0I} + 2D_{0I}^* G_{CC,q-1} D_{0I}. \quad (\text{COI15})$$

The sharper unexpanded form COI14 can be used on each actual direction. For four copies use $I_4 \otimes M_{0I}$ and $I_4 \otimes D_{0I}$ in all these expressions. A native coefficient flag X passes to $(I_4 \otimes M_{0I})X$ in the consecutive coordinates, while its arithmetic image still carries the entire conductor term in COI11. These are the exact typed maps needed to apply a metric estimate to the original flags.

In the specified lexicographic row-subset enumeration, $I = I_0$ has $D_{0I} = 0$ and $M_{0I} = I_m$ by COI9. This is the exact specialization of all preceding formulas; COI13 then becomes the low-degree form controlled in CAL13–18. For any other retained enumeration, COI9 computes its actual conductor matrix explicitly, and COI13–15 are its full receiving formulas.

18.4 Primary intersections and the actual invariant closure

For a monic divisor g of χ of degree J , polynomial cancellation in the integral domain $\mathbb{C}[S]$ proves

$$\ker g(A) = (\chi/g) \mathcal{P}_{J-1}, \quad \mathcal{P}_{m-1} \cap \ker g(A) = (\chi/g) \mathcal{P}_{m-q+J-1}. \quad (\text{COI16})$$

Indeed $\chi \mid gP$ is equivalent to $\chi/g \mid P$; the degree-below- q representative then gives the first degree bound, and membership in $\mathcal{P}_{m-1}$ gives the second. Multiplication by χ/g is injective on that literal polynomial space. The intersection dimension is therefore $\max(0, m+J-q)$, with negative-degree polynomial spaces equal to zero. This proves terminal ($J=1$) and reflected-pair ($J=2$) exclusion on the original $m \leq q-2$ domain. Since a is odd, exactly half of the grid has positive real displacement; its corresponding divisor has $J = q/2$. For $a \geq 29$, the elementary inequality $16a - 48 - v \leq (a+1)^2/2$ shows that the intersection is zero. The original full-unit arithmetic injection preserves these intersection equalities because it is injective.

The exact increasing invariant hull is

$$\sum_{r=0}^d A^r \mathcal{P}_{m-1} = \mathcal{P}_{\min(q-1, m+d-1)}, \quad d \geq 0. \quad (\text{COI17})$$

Before the right side reaches degree $q - 1$, multiplication by S is literal and successively produces every next monomial. Once the full quotient basis has appeared, the left side equals E and remains there. This proves both inclusions and every rank, including the four-copy rank $4 \min(q, m+d)$. The corresponding full-space action and its original metric cross terms are retained in the companion action calculation.

19 Complete-source localization before any relation minimum

This supplies the complete proof of the arithmetic concentration calculation in [9], with a stronger finite factorial tail and an explicit original-chart receiver. Keep the original actual simple quartet, its entire unit $v_h = 2\xi/h$, the original conductor and branch, and the original degree $a = k + 4$, $k = 4\ell + 1 \geq 9$. Set

$$\begin{aligned} c &= a/2, \quad q = (a+1)^2, \quad q' = (a-7)^2, \quad m = q - q' - v = 16a - 48 - v, \\ \beta_{rt} &= c + (2r - a)\delta + i(2t - a)\gamma, \quad \chi(S) = \prod_{r,t=0}^a (S - \beta_{rt}), \\ 0 &< \delta < 1/2, \quad \gamma > 2, \quad 0 \leq v \leq 80, \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), \quad d\mu_a(y) = w_h^{*a}(y) dy, \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \end{aligned} \tag{CAL1}$$

The full masses are $(\int w_h)^a$ and $\sqrt{2\pi}$. Every polynomial norm in this section is evaluated at the literal physical coordinate $S = c + iy$, on both real half-lines. The arithmetic source order is a . The two native Gamma orders $s = 1, a$ enter the separate native comparison, and are not substituted into $d\mu_a$.

19.1 The original source envelope and a sharper finite polynomial bound

The complete TW/BT bounds, retained in RTM2 and GRA3-4 [13, 14], use the original positive constants c_h, C_h^{tilt} and, on the simple branch, are

$$\begin{aligned} \alpha_* &= \pi/2, \quad \vartheta_h = \int_{-1}^1 w_h(y) dy > 0, \quad M_{\alpha_*} = \int e^{\alpha_* y} w_h(y) dy \in (0, \infty), \\ w_h^{*3}(y) &\geq c_h e^{-\alpha_* |y|} (1 + |y|)^{-50}, \quad e^{\alpha_* y} w_h(y) \leq C_h^{\text{tilt}} (1 + |y|)^{-7/2}, \\ c_\Gamma e^{-\alpha_* |y|} (1 + |y|)^{-1/2} &\leq \sigma(y) \leq C_\Gamma e^{-\alpha_* |y|} (1 + |y|)^{-1/2}, \\ c_\Gamma &= \sqrt{2} e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5} e^{2/3}}. \end{aligned} \tag{CAL2}$$

Here c_h is a three-factor convolution constant, not a pointwise positive lower bound for the one-factor zeta numerator. For clarity, the passage to the all-factor envelope is reproduced. In the last $a - 3$ convolution variables, restrict each to $[-1, 1]$. Their sum z has $|z| \leq a - 3$ and their retained product mass is ϑ_h^{a-3} . Positivity and CAL2 therefore give

$$w_h^{*a}(y) \geq c_h \vartheta_h^{a-3} e^{-\alpha_* (|y| + a - 3)} (a - 2 + |y|)^{-50}.$$

For the upper envelope put $f(y) = e^{\alpha_* y} w_h(y)$. On a convolution fibre whose coordinates sum to $y \geq 0$, at least one coordinate is at least y/a . Cover by these a choices, apply the bound on that coordinate, and integrate every other coordinate with its full mass M_{α_*} . Evenness supplies the negative half-line. This proves

$$w_h^{*a}(y) \leq a C_h^{\text{tilt}} M_{\alpha_*}^{a-1} e^{-\alpha_* |y|} (1 + |y|/a)^{-7/2}.$$

Consequently the following complete constants satisfy the literal pointwise inequalities:

$$b_a = \frac{c_h \vartheta_h^{a-3} e^{-\alpha_*(a-3)}}{C_\Gamma 2^{25} (a-2)^{50}}, \quad B_a = \frac{C_h^{\text{tilt}}}{C_\Gamma} a^{9/2} M_{\alpha_*}^{a-1}, \quad (\text{CAL3})$$

$$b_a (1+y^2)^{-25} \sigma(y) \leq w_h^{*a}(y) \leq B_a \sigma(y).$$

For the lower bound use $a-2+|y| \leq (a-2)(1+|y|)$ and $(1+|y|)^{50} \leq 2^{25}(1+y^2)^{25}$, dropping only the positive factor $(1+|y|)^{1/2} \geq 1$. For the upper bound use $1+|y|/a \geq (1+|y|)/a$; the displayed $a^{9/2}$ is exactly $a \cdot a^{7/2}$.

The original Gamma orthonormal recurrence is

$$ye_n = \sqrt{(n+1)(n+1/2)} e_{n+1} + \sqrt{n(n-1/2)} e_{n-1}.$$

Its Meixner–Pollaczek constants are those of [15, 18.19.8–9, 18.22.8]. The raising and lowering shifts on degree at most D have norms at most $D+1$ and D , respectively. The triangle inequality therefore proves $\|yp\|_\sigma \leq 2(D+1)\|p\|_\sigma$. There is a sharper way to use this one-step estimate than iterating twenty-five times. Set $z_0 = \|yp\|_\sigma^2 / \|p\|_\sigma^2$ for $p \neq 0$. The second derivative of $f(z) = (1+z)^{-25}$ is positive on $z \geq 0$, so $f(z) \geq f(z_0) + f'(z_0)(z - z_0)$ by integrating its increasing first derivative. Multiply this inequality at $z = y^2$ by $|p(y)|^2 d\sigma(y)$ and integrate. The linear term cancels exactly. Using $z_0 \leq 4(D+1)^2$ gives, also trivially for $p = 0$,

$$\frac{b_a}{[1+4(D+1)^2]^{25}} \|P\|_\sigma^2 \leq \|P\|_{\mu_a}^2 \leq B_a \|P\|_\sigma^2, \quad \deg P \leq D, \quad K_{a,D} = \frac{B_a}{b_a} [1+4(D+1)^2]^{25}. \quad (\text{CAL4})$$

Here $p(y) = P(c+iy)$ is the actual polynomial. No source measure or polynomial vector was replaced. Taking logarithms of the explicit positive constants proves $\log K_{a,D} = O_h(a + \log(D+2))$ uniformly in a, D .

19.2 Every full relation is concentrated outside the inner interval

Keep precisely the original finite quantities and guards

$$R = a\sqrt{\delta^2 + \gamma^2}, \quad Y = 2^{-16}q, \quad d = \lfloor Y/16 \rfloor, \quad R \leq Y/4, \quad m+1 \leq Y/16, \quad d \geq 1. \quad (\text{CAL5})$$

The physical roots of $i^{-q}\chi(c+iy)$ have modulus at most R . Thus $|\chi(c+iy)| \leq (Y+R)^q$ when $|y| \leq Y$, whereas $|\chi(c+iy)| \geq (q-R)^q$ on $[q, 2q]$. Every factor of the full polynomial is used in these triangle inequalities.

We give the complete finite polynomial estimate used between these two regions. Let L_n denote the Legendre polynomial on $[-1, 1]$. Rodrigues' formula $L_n(x) = (2^n n!)^{-1} (d/dx)^n (x^2 - 1)^n$ gives orthogonality by n integrations by parts, with no endpoint terms. Its leading coefficient is $(2n)!/(2^n (n!)^2)$. Moreover

$$\int_{-1}^1 x^n L_n(x) dx = 2^{-n} \int_{-1}^1 (1-x^2)^n dx = \frac{2^{n+1} (n!)^2}{(2n+1)!}.$$

The last elementary integral follows inductively by integration by parts, starting with 2 at $n = 0$. Subtracting the lower-degree part of L_n in the first factor now gives $\int L_n^2 = 2/(2n+1)$. These are the standard classical formulas [15, 18.3, 18.5.5], with their required proof included. Leibniz applied to $(x-1)^n (x+1)^n$ also gives

$$L_n(x) = 2^{-n} \sum_{j=0}^n \binom{n}{j}^2 (x-1)^{n-j} (x+1)^j.$$

For $|x| \leq 4$ it follows that $|L_n(x)| \leq (5/2)^n \binom{2n}{n} \leq 10^n$; the sum of squared binomial coefficients is the coefficient of z^n in $(1+z)^n (1+z)^n$, and $\binom{2n}{n} \leq 4^n$.

On $[q, 2q]$ the functions $\sqrt{(2n+1)/q} L_n(2y/q - 3)$ are therefore an orthonormal polynomial basis. For $|y| \leq Y$ their argument has modulus less than 4. Expanding any polynomial u of degree at most q in that basis and applying complex Cauchy–Schwarz proves

$$|u(y)|^2 \leq \frac{(q+1)^2}{q} 100^q \int_q^{2q} |u(t)|^2 dt \quad (|y| \leq Y). \quad (\text{CAL6})$$

The exact sum of the coefficients $2n+1$ is $(q+1)^2$. Apply CAL6 to $u(y) = Q(c+iy)$ for an arbitrary $F = \chi Q \in \chi \mathcal{P}_q$. The full Gamma mass bounds its inner integral; on $[q, 2q]$, CAL2 gives $\sigma(y) \geq c_\Gamma e^{-\pi q} / \sqrt{1+2q}$. Consequently

$$\begin{aligned} \int_{|y| \leq Y} |F(c+iy)|^2 d\mu_a &\leq \eta_a \|F\|_{\mu_a}^2, \\ \eta_a &= K_{a,2q} \frac{\sqrt{2\pi}}{c_\Gamma} \frac{(q+1)^2 \sqrt{1+2q}}{q} e^{\pi q} 100^q \left(\frac{Y+R}{q-R} \right)^{2q}. \end{aligned} \quad (\text{CAL7})$$

In the final transfer the degree of F is at most $2q$, so CAL4 applies with exactly that degree. Since $(Y+R)/(q-R) \leq 2^{-15}$ by CAL5, this proves

$$\log \eta_a \leq -c_\eta q + O_h(a + \log q), \quad c_\eta = 30 \log 2 - \pi - \log 100 > 0. \quad (\text{CAL8})$$

Both the central interval and the exterior source remain in the complete norms; no relation row or cross pairing was dropped.

19.3 Every actual low-degree polynomial is concentrated inside

For $P \in \mathcal{P}_m$, successive applications of the Gamma multiplication bound at the actual intermediate degrees give

$$\|y^d P(c+iy)\|_\sigma \leq 2^d \frac{(m+d)!}{m!} \|P\|_\sigma.$$

On $|y| > Y$, the factor $|y|^{2d}$ is at least Y^{2d} . Using the upper and lower inequalities of CAL4 on this same P therefore proves the sharpened finite bound

$$\begin{aligned} \int_{|y| > Y} |P(c+iy)|^2 d\mu_a &\leq \zeta_a^* \|P\|_{\mu_a}^2, \\ \zeta_a^* &= K_{a,m} \left(\frac{2^d (m+d)!}{m! Y^d} \right)^2 \leq K_{a,m} 4^{-2d} \leq K_{a,m} 3^{-2d}. \end{aligned} \quad (\text{CAL9})$$

For the first comparison, every factor $2(m+j)/Y$, $1 \leq j \leq d$, is at most $1/4$, since $m+1+d \leq Y/8$ by CAL5. This improves the supplied note's 3^{-2d} without changing its auxiliary integer or any original space. Since $d \geq 2^{-20}q - 1$,

$$\log \zeta_a^* \leq -c_\zeta q + O_h(a + \log q), \quad c_\zeta = 2^{-19} \log 4 > 0. \quad (\text{CAL10})$$

The factorial expression in CAL9, rather than its weaker displayed upper estimates, can be used in every subsequent finite receiver.

Let $\mathcal{R} = \chi \mathcal{P}_q$ with its original arithmetic inner product, and let $\Pi_{\mathcal{R}}$ be its attained orthogonal projector. For every unit $F \in \mathcal{R}$, split $\langle F, P \rangle$ at $|y| = Y$. Apply Cauchy–Schwarz separately to the two full regions, using CAL7 on the first and CAL9 on the second. Taking the supremum over unit F gives the exact whole-relation estimate

$$\|\Pi_{\mathcal{R}} P\|_{\mu_a}^2 \leq \varepsilon_a \|P\|_{\mu_a}^2, \quad \varepsilon_a = (\sqrt{\eta_a} + \sqrt{\zeta_a^*})^2, \quad P \in \mathcal{P}_m. \quad (\text{CAL11})$$

The supremum equals the projection norm: Cauchy–Schwarz bounds it above, and its nonzero projected vector divided by its own norm attains the bound. This argument incurs no row count or unestimated relation cross block. The literal finite values CAL7 and CAL9 give

$$\varepsilon_a \leq \exp\{-c_0 q + O_h(a + \log q)\}, \quad c_0 = \min\{c_\eta, c_\zeta\} > 0. \quad (\text{CAL12})$$

This is an upper estimate. It is not an assertion of equality with that exponential for every smaller choice of decay constant. The condition $\varepsilon_a < 1$, when needed for a logarithm, is a direct finite check on CAL7 and CAL9 and holds eventually because $q = (a+1)^2$ and all original data in their constants are fixed.

19.4 Attained metrics, all fixed flags, and both boundary bands

Let $E = \mathbb{C}[S]/(\chi)$ in its original increasing-power representatives, and let G_N be the minimum arithmetic form over $\mathcal{P}_N \rightarrow E$. For $q-1 \leq N \leq 2q$, the fibre above a literal $P \in \mathcal{P}_m$ is precisely $P + \chi\mathcal{P}_{N-q}$, with a zero relation space at $N = q-1$. Write Π_N for the projector onto that entire relation space. Expansion into the orthogonal sum gives

$$G_N[P] = \|P - \Pi_N P\|_{\mu_a}^2 = \|P\|_{\mu_a}^2 - \|\Pi_N P\|_{\mu_a}^2, \quad (1 - \varepsilon_a)G_{q-1}|_{\mathcal{P}_m} \preceq G_N|_{\mathcal{P}_m} \preceq G_{q-1}|_{\mathcal{P}_m}. \quad (\text{CAL13})$$

The smaller relation space is contained in $\mathcal{R}$, so its projection norm is no larger. This proves CAL13 before choosing any coefficient flag or taking any further minimum.

For the consecutive coefficient map L_0 whose image is proved to be $\mathcal{P}_{m-1}$, put $M_0 = I_4 \otimes L_0$ and $\mathcal{G}_N^0 = M_0^* \text{diag}(G_N, 4)M_0$. The map M_0 is precisely specified; the following statement does not silently replace another actual minor by this one. CAL13 gives

$$(1 - \varepsilon_a)\mathcal{G}_{q-1}^0 \preceq \mathcal{G}_N^0 \preceq \mathcal{G}_{q-1}^0. \quad (\text{CAL14})$$

Every fixed restriction, and every successive attained quotient of such a restriction, satisfies the same two bounds. For a restriction substitute its original columns in the form inequality. For a quotient, every representative of one value obeys the inequality; minimizing both sides in the identical full affine fibre preserves both bounds. Successive minima range over the union of those fibres, so this proof also covers every original nested flag, including $B, S, W, J, I, K_c, K_i, T, W_i$, after its actual coefficient map has been specified. Empty value spaces have determinant one.

For any such attained form $H_{X,N}$ of dimension r , at the four original cutoffs its signed arithmetic return satisfies

$$0 \leq F_X^{\text{ar}} := \log \det H_{X,q-1} + \log \det H_{X,q} - \log \det H_{X,2q-1} - \log \det H_{X,2q} \leq -2r \log(1 - \varepsilon_a). \quad (\text{CAL15})$$

Indeed set $l_N = \log \det H_{X,N} - \log \det H_{X,q-1}$. Nested source fibres make these forms and logarithms decrease, so $0 \geq l_q \geq l_{2q-1} \geq l_{2q}$. The return is $l_q - l_{2q-1} - l_{2q} \geq 0$; CAL14 gives each high logarithm at least $r \log(1 - \varepsilon_a)$, while $l_q \leq 0$. This proves the stated coefficient $2r$. For example the full four-copy form has $r = 4m$, giving $8m$. For $\varepsilon_a \leq 1/2$, the identity $-\log(1-x) = \int_0^x (1-t)^{-1} dt \leq 2x$ proves $F_X^{\text{ar}} \leq 4r\varepsilon_a$. Since $r \leq 4m = O(a)$, CAL12 gives exponential smallness, in particular zero at every fixed polynomial scale, on this precisely specified image.

The two exact original boundary differences pulled through M_0 are

$$\Gamma_0^0 = M_0^* \text{diag}(G_{q-1} - G_{2q-1}, 4)M_0, \quad \Gamma_1^0 = M_0^* \text{diag}(G_q - G_{2q}, 4)M_0. \quad (\text{CAL16})$$

Positivity of successive projection losses and CAL13 give

$$0 \preceq \Gamma_i^0 \preceq \varepsilon_a \mathcal{G}_{q-1}^0, \quad i = 0, 1. \quad (\text{CAL17})$$

Here the strict sign can also be proved directly on the stated simple odd-degree domain, rather than imported. The physical polynomial $\chi(c + iy)$ is strictly positive: its roots in the physical coordinate have nonzero imaginary parts and occur in conjugate pairs, and $q = (a+1)^2$ is divisible by four. For a nonzero P of degree below $m < q$, the first band contains χP , and $\langle \chi P, P \rangle = \int \chi(c + iy)|P|^2 d\mu_a > 0$. Thus the first projection loss is positive. For the second band, let $e = \chi$, let $b = \langle e, P \rangle / \langle e, e \rangle$ and put $P_1 = P - be$. This is the attained representative at q . The polynomial χP_1 has degree at most $2q$ and lies in the high relation space. Since $P_1 \perp e$, subtracting the e

component from χP_1 preserves its pairing with P_1 . That pairing equals $\int \chi |P_1|^2 d\mu_a > 0$; $P_1 \neq 0$ because a polynomial of degree below q cannot be a nonzero multiple of χ . The second projection loss is therefore positive as well. Applied to the nonzero components of four copies, this proves $\Gamma_i^0 \succ 0$. Consequently the exact inverse comparison is

$$\lambda_{\min} \left((\Gamma_i^0)^{-1/2} \mathcal{G}_{q-1}^0 (\Gamma_i^0)^{-1/2} \right) \geq \varepsilon_a^{-1}. \quad (\text{CAL18})$$

This is a statement about the displayed original boundary comparison. It does not compare Γ_0^0 with Γ_1^0 , does not replace the native Gamma metric, and does not evaluate any individual native allocation. The exact original-minor return, including every conductor correction when that minor differs from the consecutive one, is calculated in the companion coefficient-image section.

20 Physical compression, retained metric defects, and invariant enlargement

20.1 The consecutive comparison image and the original source metric

Retain the original simple-quartet domain

$$\begin{aligned} a &= k + 4, & k &= 4\ell + 1 \geq 9, & 0 < \delta < \frac{1}{2}, & \gamma > 2, \\ q &= (a + 1)^2, & q' &= (a - 7)^2, & 0 \leq v \leq 80, & m = q - q' - v > 0, & c = \frac{a}{2}. \end{aligned}$$

In particular a is odd and $m < q$. Put

$$\beta_{rt} = c + (2r - a)\delta + i(2t - a)\gamma, \quad \chi(S) = \prod_{r,t=0}^a (S - \beta_{rt}) = \sum_{j=0}^q c_j S^j, \quad c_q = 1. \quad (\text{CJA1})$$

Let $E = \mathbb{C}[S]/(\chi)$, with the literal increasing-power representative of degree below q . Its action A is multiplication by the original S , followed by reduction modulo the complete χ . The exact minor correction COI3–7 evaluates the consecutive comparison injection L_0 , retaining the supplied note's provenance [8]. Throughout CJA2–19a, L denotes L_0 , and $M = I_4 \otimes L_0$. The originally selected injection L_I is retained separately in CJA31–37. Its full conductor transition is used there; no row-subset ordering is inferred from the order of root labels. The consecutive injection is

$$L : \mathbb{C}^m \longrightarrow E, \quad Le_j = \ell_j(S) = \sum_{t=q-m+j+1}^q c_t S^{t-q+m-j-1}, \quad 0 \leq j < m. \quad (\text{CJA2})$$

The polynomial ℓ_j is monic of degree $m - 1 - j$. Thus these columns are a basis of the literal subspace $\mathcal{P}_{m-1} \subset E$: elimination in descending degree proves both spanning and independence. The Jacobi calculation uses this consecutive map, including every coefficient of χ . It applies directly to the original map when the retained row-subset rule selects this consecutive block.

The arithmetic measure, with its complete mass, is

$$d\mu(y) = w_h^{*a}(y) dy, \quad w_h(y) = \frac{|v_h(1/2 + iy)|^2}{2\pi}, \quad v_h = 2\xi/h, \quad S = c + iy. \quad (\text{CJA3})$$

It is the original source of NCRP8 [16]. Its density is even, positive almost everywhere, and has all polynomial moments finite. Evenness follows from conjugation of the complete quartet and of v_h , then from convolution; the complete quartet and all convolution factors are retained. Every nonzero polynomial has positive squared integral. Write $G = G_{q-1}$ for the q -by- q Gram in the literal $1, S, \dots, S^{q-1}$ coordinates. At this degree the remainder map is the identity and its fibre is a singleton. Consequently

$$\|P\|_G^2 = \int_{\mathbb{R}} |P(c + iy)|^2 d\mu(y), \quad \deg P < q. \quad (\text{CJA4})$$

This is precisely the original attained arithmetic metric at $q - 1$. The coefficient metric used below is $G_L = L^*GL$. The native conductor minimum and the original boundary forms Γ_i keep their own values; no equality of those metrics with G_L is used.

20.2 The exact physical Jacobi compression and its nonzero residual

Apply orthogonal elimination to $1, y, y^2, \dots$ in the measure (CJA3). Let $\pi_r(y)$ be the resulting monic real polynomials, $\omega_r = \int |\pi_r|^2 d\mu > 0$, and $\varphi_r = \pi_r / \sqrt{\omega_r}$. The full source mass is retained: in particular $\varphi_0 = 1 / \sqrt{\int d\mu}$. Uniqueness of monic orthogonal polynomials follows because the difference of two candidates has lower degree and is orthogonal to itself. Evenness of the measure and uniqueness give $\pi_r(-y) = (-1)^r \pi_r(y)$. Set

$$a_r = \sqrt{\omega_r / \omega_{r-1}} > 0 \quad (r \geq 1), \quad a_0 = 0, \quad \varphi_{-1} = 0. \quad (\text{CJA5})$$

These original norm ratios are not assigned external values.

The exact recurrence is

$$y\varphi_r = a_{r+1}\varphi_{r+1} + a_r\varphi_{r-1}. \quad (\text{CJA6})$$

Here is the complete argument in this measure. For $s \leq r-2$, $\langle \varphi_s, y\varphi_r \rangle = \langle y\varphi_s, \varphi_r \rangle = 0$ by degree. The coefficient along φ_r vanishes by parity. Comparison of leading coefficients gives a_{r+1} on φ_{r+1} . Symmetry of multiplication by the real y , and the same leading-coefficient calculation at $r-1$, give a_r on φ_{r-1} . This proves every coefficient. The general recurrence and its orthogonal-polynomial provenance are recorded in NIST DLMF, section 18.2(iv) [15]; the proof here supplies its specialization without changing the arithmetic measure.

Define the unique matrix $T \in \mathbb{C}^{m \times m}$ by

$$(Lz)(c + iy) = \sum_{r=0}^{m-1} (Tz)_r \varphi_r(y), \quad T_{rj} = \int_{\mathbb{R}} \varphi_r(y) \ell_j(c + iy) d\mu(y). \quad (\text{CJA7})$$

Both systems in this identity are bases of $\mathcal{P}_{m-1}$; therefore T is invertible. Integrating the identity against itself proves the exact pullback

$$T^*T = L^*GL = G_L. \quad (\text{CJA8})$$

Thus T is an isometry from the coefficient space with its specified G_L metric into orthonormal coordinates. It is not a replacement of the metric on that coefficient space.

For four copies set

$$\begin{aligned} H &= (\mathbb{C}^m)^4, & \mathbf{E} &= E^4, & M &= I_4 \otimes L, & \mathbf{G} &= I_4 \otimes G, \\ \mathbf{A} &= I_4 \otimes A, & \mathcal{T} &= I_4 \otimes T, & G_H &= M^* \mathbf{G} M = \mathcal{T}^* \mathcal{T}. \end{aligned}$$

Let J_m have zero diagonal and off-diagonals $a_1, \dots, a_{m-1}$, and put $\mathbf{J} = I_4 \otimes J_m$. The source-metric compression of this consecutive map and its residual are

$$A_H = G_H^{-1} M^* \mathbf{G} A M, \quad R = \mathbf{A} M - M A_H. \quad (\text{CJA9})$$

By definition $M^* \mathbf{G} R = 0$. Since $m < q$, multiplication of a polynomial of degree below m by S has degree at most $m < q$. No reduction by χ occurs. Applying (CJA6) to that multiplication and dropping only its orthogonal degree- m component proves

$$\boxed{\mathcal{T} A_H \mathcal{T}^{-1} = cI + i\mathbf{J}, \quad R \mathcal{T}^{-1} = i a_m \mathcal{U}_m.} \quad (\text{CJA10})$$

Here $\mathcal{U}_m : \mathbb{C}^{4m} \rightarrow \mathbf{E}$ is completely specified: in each copy it sends a coordinate column to its last entry times the polynomial $\varphi_m((S - c)/i)$, and has no other component. Those four target polynomials are orthonormal in the direct-sum metric $\mathbf{G}$. Hence R has rank exactly four, and its four nonzero singular values from (H, G_H) to $(\mathbf{E}, \mathbf{G})$ are exactly a_m . More explicitly, if P_{last} projects onto the last coordinate of each of the four copies, then

$$R^* \mathbf{G} R = a_m^2 \mathcal{T}^* P_{\text{last}} \mathcal{T}. \quad (\text{CJA11})$$

This retains the domain and codomain norms of the residual.

The determinant $D_r(y) = \det(yI_r - J_r)$ obeys $D_0 = 1$, $D_1 = y$, and $D_{r+1} = yD_r - a_r^2 D_{r-1}$, by expansion on the last row. Multiplying (CJA6) by $\sqrt{\omega_r}$ gives exactly the same monic recurrence for π_r . Induction proves $D_r = \pi_r$. The real symmetric J_m has an orthonormal eigenbasis and real eigenvalues: maximizing its Rayleigh quotient gives an eigenvector, its orthogonal complement is invariant, and induction diagonalizes it. An eigenvector is determined by its first entry, because its successive equations have nonzero off-diagonal coefficients. If that entry vanishes, all entries vanish. Each eigenspace thus has dimension at most one; the orthonormal diagonalization proves all eigenvalues are simple. Writing these eigenvalues as $t_1, \dots, t_m$, we have proved

$$\text{spec } A_H = \{c + it_j : 1 \leq j \leq m\}, \quad \text{each repeated four times}, \quad A_H^* G_H + G_H A_H - a G_H = 0. \quad (\text{CJA12})$$

The latter identity also follows immediately by substituting (CJA10) and $2c = a$. It is accompanied by the exact nonzero residual (CJA10).

20.3 Every boundary-metric and action cross term

We first give the literal COB endpoint maps [17]. For $n \geq q - 1$, let $J_n^\chi : \mathcal{P}_n \rightarrow E$ be remainder, let H_n be the original moment Gram, and let

$$\mathcal{R}_n = H_n^{-1} (J_n^\chi)^* (J_n^\chi H_n^{-1} (J_n^\chi)^*)^{-1}.$$

The identity $J_n^\chi \mathcal{R}_n = I$, together with $\mathcal{R}_n^* H_n \ker J_n^\chi = 0$, proves its exact attained minimum property by the Pythagorean identity on each affine fibre. Define the two distinct original windows

$$\Delta_0 = \mathcal{R}_{2q-1} - \mathcal{R}_{q-1}, \quad \Delta_1 = \mathcal{R}_{2q} - \mathcal{R}_q.$$

All these polynomial-valued maps are embedded in the same original polynomial space before they are subtracted. On the physical line put $Q(y) = \chi(c + iy)$. Since a is odd, the real displacements $2r - a$ are nonzero. Pairing r with $a - r$, and using that q is divisible by four, writes Q as the product of positive factors $((y - (2t - a)\gamma)^2 + ((2r - a)\delta)^2)$, one for each pair. Thus Q is real, even, and strictly positive.

Let π_r^ν be the real monic orthogonal polynomials for $d\nu = Q^2 d\mu$, with squared norms $\nu_r > 0$, and put

$$e_r(S) = \frac{\chi(S) \pi_r^\nu((S - c)/i)}{\sqrt{\nu_r}}, \quad c_r(x) = -\langle e_r, x \rangle_\mu, \quad C_0 = (c_0, \dots, c_{q-1})^\top, \quad C_1 = (c_1, \dots, c_q)^\top.$$

The argument used in (CJA6) applies to the even positive measure $d\nu$. These e_r are orthonormal in the original measure $d\mu$, and they span the successive original relation spaces. Projecting the literal representative $x \in \mathcal{P}_{q-1}$ against those spaces proves

$$\Delta_0 x = \sum_{r=0}^{q-1} e_r c_r(x), \quad \Delta_1 x = \sum_{r=1}^q e_r c_r(x), \quad M_i^{\text{one}} = C_i^* C_i. \quad (\text{CJA12a})$$

For the second equality, the common $e_0 c_0(x)$ term cancels between the two minima at q and $2q$. Both maps are injective. For $\Delta_0 x = 0$, the common minimum is $P = x$, and its orthogonality to the relation χP at degree $2q - 1$ contradicts $\int Q|P|^2 d\mu > 0$ unless $x = 0$. For $\Delta_1 x = 0$, use the common minimum $P = \mathcal{R}_q x$ of degree at most q , and its relation χP of degree at most $2q$, to obtain the same conclusion. Therefore C_i is invertible and $M_i^{\text{one}} > 0$.

Write $\Delta_i = I_4 \otimes \Delta_i$, $M_i = I_4 \otimes M_i^{\text{one}}$. For each original endpoint retain the pullback of its actual COB form along the consecutive map:

$$\Gamma_i = M^* M_i M, \quad Q_i = \mathcal{T}^{-*} \Gamma_i \mathcal{T}^{-1}.$$

Here M_i is the full original form on $\mathbf{E}$ just defined by both original attained minima. Direct substitution of (CJA10), with no commutation assumption, gives

$$\boxed{\mathcal{T}^{-*}(A_H^* \Gamma_i + \Gamma_i A_H - a \Gamma_i) \mathcal{T}^{-1} = i(Q_i \mathbf{J} - \mathbf{J} Q_i).} \quad (\text{CJA13})$$

Thus this compressed defect vanishes exactly when $[Q_i, \mathbf{J}] = 0$. Its arithmetic-relative trace is

$$\text{tr}\{G_H^{-1}(A_H^* \Gamma_i + \Gamma_i A_H - a \Gamma_i)\} = i \text{tr}(Q_i \mathbf{J} - \mathbf{J} Q_i) = 0.$$

When $\Gamma_i > 0$, its own relative trace is also zero: $\text{tr}(Q_i^{-1} i[Q_i, \mathbf{J}]) = 0$. The first assertion does not require inverting Γ_i .

The full pulled-back ambient action contains both additional terms:

$$\begin{aligned} M^*(\mathbf{A}^* M_i + M_i \mathbf{A} - a M_i) M &= A_H^* \Gamma_i + \Gamma_i A_H - a \Gamma_i \\ &\quad + R^* M_i M + M^* M_i R. \end{aligned} \quad (\text{CJA14})$$

To prove it, substitute $\mathbf{A} M = M A_H + R$ into the two action terms and multiply out. Orthogonality $M^* \mathbf{G} R = 0$ applies to $\mathbf{G}$; it does not remove either of the displayed M_i cross terms. In the physical orthonormal coordinates, put $M_0 = M \mathcal{T}^{-1}$, $R_0 = R \mathcal{T}^{-1} = i a_m \mathcal{U}_m$. The last two terms become, exactly,

$$R_0^* M_i M_0 + M_0^* M_i R_0 = i a_m (M_0^* M_i \mathcal{U}_m - \mathcal{U}_m^* M_i M_0). \quad (\text{CJA15})$$

All four degree- m boundary directions remain in this expression.

For any specified transported coefficient cone subspace $X = B, S, W$ in these consecutive coordinates, take a full independent column frame I_X and its original arithmetic projector

$$P_X^{G_H} = I_X (I_X^* G_H I_X)^{-1} I_X^* G_H.$$

The inverse is the complete inverse on that frame. Equation (CJA8) gives $\mathcal{T} P_X^{G_H} \mathcal{T}^{-1} = P_{\mathcal{T}X}$, the Euclidean orthogonal projector onto the actual transformed subspace. Hence

$$\mathcal{T}(I - P_X^{G_H}) A_H P_X^{G_H} \mathcal{T}^{-1} = i(I - P_{\mathcal{T}X}) \mathbf{J} P_{\mathcal{T}X}. \quad (\text{CJA16})$$

This is an exact preservation test for the compression. The complete failure of the specified inclusion M to intertwine $\mathbf{A}$ is still R , as given in (CJA10).

The proper-source action receives the same residual. Put $\mathbf{a}_r = \sqrt{\nu_r / \nu_{r-1}} > 0$ for $r \geq 1$. Multiplying the relation recurrence by χ gives

$$S e_r = c e_r + i \mathbf{a}_{r+1} e_{r+1} + i \mathbf{a}_r e_{r-1}.$$

Let J_0^{rel} be its q -by- q Jacobi matrix on indices $0, \dots, q-1$, and J_1^{rel} the one on indices $1, \dots, q$. Their off-diagonals are respectively $\mathbf{a}_1, \dots, \mathbf{a}_{q-1}$ and $\mathbf{a}_2, \dots, \mathbf{a}_q$. Define the exact maps

$$\begin{aligned} A_i^{\text{tan}} &= C_i^{-1} (c I + i J_i^{\text{rel}}) C_i, & C_i^{\text{act}} &= A_i^{\text{tan}} - A, \\ N_0 &= i \mathbf{a}_q e_q c_{q-1}, \\ N_1 &= i \mathbf{a}_1 e_0 c_1 + i \mathbf{a}_{q+1} e_{q+1} c_q. \end{aligned}$$

Retain their four-copy maps $\mathbf{C}_i = I_4 \otimes C_i^{\text{act}}$ and $\mathbf{N}_i = I_4 \otimes N_i$. The displayed recurrence, projected onto the stated band and its orthogonal complement, proves

$$S \Delta_i - \Delta_i \mathbf{A} = \Delta_i \mathbf{C}_i + \mathbf{N}_i, \quad \Delta_i^* \mathbf{H} \mathbf{N}_i = 0. \quad (\text{CJA16a})$$

Here $\mathbf{H} = I_4 \otimes H_{2q+1}$ is the original moment metric on the polynomial representatives through degree $2q+1$. Both Δ_i and $\mathbf{N}_i$ are embedded in that same coefficient space. Their degrees are at most $2q$ and $2q+1$, respectively. They and their action edges inject into $(\mathbb{C}[S]/\chi^{a+1})^4$, because $(a+1)q > 2q+1$. Thus (CJA16a) is an identity of the original proper-source maps, with

a specified norm on the displayed polynomial bands and edge. No norm on the entire quotient is being introduced.

In this notation,

$$\beta_i = \Delta_i M, \quad M_i = \Delta_i^* H \Delta_i.$$

Composing (CJA16a) with M , and using (CJA9), proves

$$K_i := S\beta_i - \beta_i A_H = \Delta_i R + \Delta_i C_i M + N_i M. \quad (\text{CJA17})$$

For clarity, the three terms here are maps from H to the original proper-source value space. Writing $U_i = \Delta_i R$, $V_i = \Delta_i C_i M$, and $W_i = N_i M$, its original squared residual form is

$$\begin{aligned} K_i^* H K_i &= U_i^* H U_i + V_i^* H V_i + W_i^* H W_i \\ &\quad + U_i^* H V_i + V_i^* H U_i \\ &\quad + U_i^* H W_i + W_i^* H U_i \\ &\quad + V_i^* H W_i + W_i^* H V_i. \end{aligned} \quad (\text{CJA18})$$

All nine directed terms have now been displayed. The last four cross terms are exactly zero by (CJA16a), with its proved band orthogonality. The remaining original form is

$$\begin{aligned} K_i^* H K_i &= R^* M_i R + R^* M_i C_i M + M^* C_i^* M_i R \\ &\quad + M^* C_i^* M_i C_i M + M^* N_i^* H N_i M. \end{aligned} \quad (\text{CJA18a})$$

In particular the two Gamma-metric cross terms involving R remain. The normal energies are exactly

$$\begin{aligned} N_0^* H N_0 &= I_4 \otimes \mathfrak{a}_q^2 c_{q-1}^* c_{q-1}, \\ N_1^* H N_1 &= I_4 \otimes (\mathfrak{a}_1^2 c_1^* c_1 + \mathfrak{a}_{q+1}^2 c_q^* c_q). \end{aligned} \quad (\text{CJA18b})$$

These follow from orthonormality of the different e_r ; neither original boundary edge is discarded. Similarly the full proper-source centered form is

$$\begin{aligned} (S\beta_i)^* H \beta_i + \beta_i^* H (S\beta_i) - a\Gamma_i &= A_H^* \Gamma_i + \Gamma_i A_H - a\Gamma_i \\ &\quad + R^* M_i M + M^* M_i R \\ &\quad + M^* C_i^* M_i M + M^* M_i C_i M \\ &\quad + M^* N_i^* H \Delta_i M + M^* \Delta_i^* H N_i M. \end{aligned} \quad (\text{CJA19})$$

This follows by first substituting $S\beta_i = \beta_i A_H + K_i$, then inserting (CJA17). On the actual line $\bar{S} + S = a$, so the left side is exactly zero, pointwise before integration. The last two terms are zero by (CJA16a). Put $B_i = R + C_i M$. The complete original weight identity is consequently

$$\begin{aligned} A_H^* \Gamma_i + \Gamma_i A_H - a\Gamma_i &= -M^* M_i B_i - B_i^* M_i M, \\ i[Q_i, \mathbf{J}] &= -M_0^* M_i (B_i \mathcal{T}^{-1}) - (B_i \mathcal{T}^{-1})^* M_i M_0. \end{aligned} \quad (\text{CJA19a})$$

The centered scalar is the same retained a throughout. The surviving normal leakage is measured by (CJA18a)–(CJA18b), while the weight identity retains both the arithmetic compression residual and the boundary tangential correction.

20.4 The actual cyclic enlargement and its positive trace

The one-copy COB proper-source packet is $E_2 = \mathbb{C}[S]/(\chi^{a+1})$. For its original nonzero boundary vector, with the originally selected L_I , COB7–9 give the literal form

$$p = \Delta_i L_I z = \chi R_p \pmod{\chi^{a+1}}, \quad 0 \neq p, \quad \deg R_p \leq q. \quad (\text{CJA20})$$

The subscript prevents confusion with the action residual R . The monic greatest common divisor $g_p = \gcd(\chi^a, R_p)$ gives the exact annihilator and exact cyclic-module map

$$f_p = \frac{\chi^a}{g_p}, \quad \mathbb{C}[S]/(f_p) \longrightarrow \mathbb{C}[S]p \subset E_2, \quad [F] \longmapsto Fp. \quad (\text{CJA21})$$

This map is an isomorphism of $\mathbb{C}[S]$ -modules. Indeed, $Fp = 0$ is equivalent in the original quotient to $\chi^{a+1} \mid F\chi R_p$, and cancellation of the nonzero polynomial χ gives $\chi^a \mid FR_p$. At each of the distinct roots β of χ this requires $\text{ord}_\beta F \geq a - \min(a, \text{ord}_\beta R_p)$, and requires no other factor. These are exactly the factors of f_p . The map is onto by the definition of the cyclic module, and the calculated kernel proves injectivity. Writing $d_\beta = \min(a, \text{ord}_\beta R_p)$, we obtain

$$\dim \mathbb{C}[S]p = aq - \sum_{\beta} d_\beta \geq (a-1)q, \quad \sum_{\beta} d_\beta = \deg g_p \leq \deg R_p \leq q. \quad (\text{CJA22})$$

Thus every invariant subspace containing this actual nonzero vector contains its explicitly given cyclic module. The original full-unit proper-source map in COB.8 is

$$\Psi_2 : E_2 \longrightarrow Q_h^{\otimes a}, \quad [P] \longmapsto [P(D_1 + \cdots + D_a)F_h^{\otimes a}], \quad Q_h = \mathcal{B}/\Theta h(D)W_*.$$

It is injective and intertwines S with the sum of the original Euler actions. Here is the algebra behind that retained assertion. The original identities are $h(D)F_h = \Theta\phi_*$, $\mathcal{M}F_h = v_h$, $g = 2\xi = hv_h$, and $W_* = \mathbb{C}[D]\phi_*$. If $P(D)F_h = \Theta h(D)T(D)\phi_*$, its original Mellin identity is $Pv_h = ghT = h^2v_hT$, so $P = h^2T$, since v_h is not the zero entire function. The converse follows by applying $h(D)T(D)$ to $h(D)F_h = \Theta\phi_*$, using the same original Euler intertwining. The single-factor cyclic annihilator is exactly h^2 . Tensoring the resulting injections over $\mathbb{C}$ preserves injectivity. In every ordered root tuple the local nilpotents $\epsilon_1, \dots, \epsilon_a$ satisfy $\epsilon_j^2 = 0$; their sum has a -th power $a!\epsilon_1 \cdots \epsilon_a \neq 0$ and $(a+1)$ -st power zero. Every original grid sum occurs, so the cyclic sum has precisely the annihilator χ^{a+1} . The actual Taylor observation multiplies by the full unit $[v_h]_{h^2}^{\otimes a}$, which is invertible at every original simple root and is retained. These facts prove the stated map and injectivity with the original source labels. Now $F\Psi_2(p) = \Psi_2(Fp)$; injectivity proves equality of the two annihilators, without changing or discarding that unit.

Here is a direct metric proof of the needed positive-trace estimate. Let B_p denote multiplication by S on the cyclic module (CJA21), let $H_p > 0$ be any specified positive metric on that module, and define

$$D_p = B_p^*H_p + H_pB_p - aH_p, \quad K_p = H_p^{-1/2}D_pH_p^{-1/2}.$$

The polynomial factorization of f_p gives its primary subspaces with dimensions $a - d_\beta$. In the local basis $1, S - \beta, \dots, (S - \beta)^{a-d_\beta-1}$, multiplication is βI plus a nilpotent shift. This proves its eigenvalue and trace, retaining all multiplicities. Let I_+ be any full basis of the direct sum of the primary subspaces with $\text{Re } \beta > c$, and write $B_pI_+ = I_+B_+$, $H_+ = I_+^*H_pI_+$. The orthogonal projector in Euclidean coordinates onto $H_p^{1/2} \text{im } I_+$ is

$$P_+ = H_p^{1/2}I_+H_+^{-1}I_+^*H_p^{1/2}.$$

Cyclicity of matrix trace and the invariant-subspace identity give

$$\text{tr}(P_+K_p) = \text{tr}\{H_+^{-1}I_+^*D_pI_+\} = 2 \text{Re } \text{tr } B_+ - a \dim \text{im } I_+.$$

For every Hermitian K and orthogonal projector P , $\text{tr}(PK) \leq \text{tr } K_+$. To verify this last elementary inequality, diagonalize K in an orthonormal basis: the diagonal entries of P lie in $[0, 1]$, so each positive eigenvalue contributes at most itself, and each negative eigenvalue contributes a nonpositive number. We have therefore proved, for the original metric,

$$\text{tr}(K_p)_+ \geq \sum_{\beta} (a - d_\beta) \max(2 \text{Re } \beta - a, 0). \quad (\text{CJA23})$$

The notation on the left means the trace of the positive spectral part of K_p , rather than the positive part of its scalar trace.

For odd a , the full simple-packet positive sum is, explicitly,

$$L_{h,a} = 2\delta(a+1) \sum_{r=(a+1)/2}^a (2r-a) = \frac{\delta(a+1)^3}{2} = \frac{\delta q(a+1)}{2}.$$

Since the largest positive displacement is $2a\delta$, (CJA22) and (CJA23) yield the required finite bound

$$\mathrm{tr}(K_p)_+ \geq aL_{h,a} - 2a\delta q = \frac{\delta a q(a-3)}{2}. \quad (\text{CJA24})$$

The exact preceding spectral sum remains available with the actual d_β ; the inequality does not assign them generic values.

20.5 The consecutive invariant hull and exact Sylvester inverse

There is no nonzero $\mathbf{A}$ -invariant subspace contained in $\mathrm{im} M = (\mathcal{P}_{m-1})^4$. Indeed, in a nonzero such subspace choose a vector whose largest component degree d is maximal among all its vectors. Then $d \leq m-1$. Multiplication by S raises that degree to $d+1 \leq m < q$; no reduction modulo χ is available to change the leading coefficient. The resulting vector has larger degree, contradicting invariance. This proves that the invariant core is exactly zero. On the other hand, the invariant enlargement is exact:

$$\sum_{r=0}^d \mathbf{A}^r \mathrm{im} M = (\mathcal{P}_{\min(q-1, m+d-1)})^4 \quad (d \geq 0). \quad (\text{CJA25})$$

For $m+d-1 < q$, ordinary polynomial multiplication proves both containments, since every power up to $m+d-1$ occurs. At $d = q-m$ this is all of $\mathbf{E}$, and it stays all of $\mathbf{E}$ for every larger d .

Let $\mathcal{V} = I_4 \otimes V$ be the original root-evaluation map, $V_{\beta n} = \beta^n$, $0 \leq n < q$. Distinctness of the complete grid makes V invertible: a degree- $(q-1)$ polynomial vanishing at all q roots is zero. Let O be a real orthogonal eigenvector matrix for J_m , with $O^* J_m O = \mathrm{diag}(t_1, \dots, t_m)$, and put

$$\mathcal{O} = I_4 \otimes O, \quad \mathcal{W} = \mathcal{O}^* \mathcal{T}, \quad K = \mathcal{V}^{-*} \mathbf{G} \mathcal{V}^{-1}.$$

Thus $\mathcal{W}^* \mathcal{W} = G_H$. The original target metric in root-value coordinates is the full positive matrix K , not the identity.

For a map $Y : H \rightarrow \mathbf{E}$ the equation $\mathbf{A}X - XA_H = Y$ has the exact solution

$$Z = \mathcal{V} Y \mathcal{W}^{-1}, \quad \hat{X}_{(b,\beta),(e,j)} = \frac{Z_{(b,\beta),(e,j)}}{\beta - c - it_j}, \quad (\text{CJA26})$$

$$X = \mathcal{V}^{-1} \hat{X} \mathcal{W}.$$

The indices b, e range independently over the four copies; this formula includes all sixteen copy-to-copy blocks. Indeed, conjugation by these exact maps turns the Sylvester equation into those scalar equations, because the target action is the diagonal matrix of all original β , and the source action is the diagonal matrix of all $c + it_j$. The real part of each denominator is $(2r-a)\delta$. Since a is odd, it is nonzero and has absolute value at least δ . Thus every entry is uniquely determined, proving invertibility on the full original space $\mathrm{Hom}(H, \mathbf{E})$.

The entrywise bound does not by itself give a δ^{-1} bound in the original metrics. The exact metric formula can be stated without that inference. Use the original Hilbert–Schmidt norm

$$\|X\|_{\mathrm{HS}(G_H, \mathbf{G})}^2 = \mathrm{tr}(G_H^{-1} X^* \mathbf{G} X).$$

Since $\mathcal{W}$ is the proved source isometry, transport gives

$$\|X\|_{\mathrm{HS}(G_H, \mathbf{G})}^2 = \mathrm{tr}(\hat{X}^* K \hat{X}). \quad (\text{CJA27})$$

For source index j , let $D_j = \text{diag}_{(b,\beta)}(\beta - c - it_j)$. Different source columns in (CJA27) contribute separately. For each column the exact norm of multiplication by D_j^{-1} in the target K metric is $\|K^{1/2}D_j^{-1}K^{-1/2}\|_2$. Taking the largest column norm, and choosing a maximizing vector in that column for the reverse inequality, proves

$$\|\mathcal{S}^{-1}\|_{\text{HS}(G_H, \mathbf{G})} = \max_j \|K^{1/2}D_j^{-1}K^{-1/2}\|_2 \leq \frac{\sqrt{\lambda_{\max}(K)/\lambda_{\min}(K)}}{\delta}. \quad (\text{CJA28})$$

Here $\mathcal{S}(X) = \mathbf{A}X - XA_H$. Both target coordinate factors remain. No bound on their condition ratio has been assumed.

For completeness, a different specified positive source form Γ , including the boundary pullback for the indicated map, is retained by setting $Q_\Gamma = \mathcal{W}^{-*}\Gamma\mathcal{W}^{-1}$. If columns are vectorized in their actual order, put

$$\mathcal{H}_\Gamma = \overline{Q_\Gamma^{-1}} \otimes K, \quad \mathcal{D} = \text{diag}_{((e,j),(b,\beta))}(\beta - c - it_j).$$

The identity $\text{tr}(Q_\Gamma^{-1}\widehat{X}^*K\widehat{X}) = \text{vec}(\widehat{X})^*\mathcal{H}_\Gamma\text{vec}(\widehat{X})$ is obtained by expanding all row and column indices. Thus the corresponding exact original-metric inverse norm is

$$\|\mathcal{S}^{-1}\|_{\text{HS}(\Gamma, \mathbf{G})} = \|\mathcal{H}_\Gamma^{1/2}\mathcal{D}^{-1}\mathcal{H}_\Gamma^{-1/2}\|_2. \quad (\text{CJA29})$$

It retains both the source and target Gram changes, including their cross entries. No native metric is replaced by G_H .

Finally the computed residual satisfies $\mathcal{S}(M) = R$. The complete correction equation $\mathbf{A}(M + X) = (M + X)A_H$ is therefore $\mathcal{S}(X) = -R$. Invertibility just proved gives its unique solution

$$X = -M. \quad (\text{CJA30})$$

This computes the correction for the consecutive source action A_H , the original target action $\mathbf{A}$, and their full map space. Its relation to enlargement is already explicit in (CJA25) and (CJA20)–(CJA24): the small compressed image has zero invariant core, while the corresponding invariant closure retains the original action and grows by the stated exact module maps.

20.6 The original selected minor: its core and complete action receiver

We now retain the originally selected row set I , its complement J_I , and its actual injection L_I , without a consecutive-row assumption. Put

$$\mathcal{K}_I = \{v + j : j \in J_I\} \subset \{0, \dots, q-1\}, \quad \mathbf{p}_n(S) = \sum_{r=n+1}^q c_r S^{r-n-1}.$$

The residue calculation COI6–8 proves $\text{im } L_I = \text{span}\{\mathbf{p}_n : n \in \mathcal{K}_I\}$; the full family $\mathbf{p}_0, \dots, \mathbf{p}_{q-1}$ is a basis because its degrees are respectively $q-1, \dots, 0$ with leading coefficient one. For each original root β , the original Lagrange class is

$$v_\beta = \frac{\chi(S)}{(S - \beta)\chi'(\beta)} = \sum_{n=0}^{q-1} \frac{\beta^n}{\chi'(\beta)} \mathbf{p}_n(S).$$

To prove the last identity, expand $\chi(S)/(S - \beta) = \sum_{n=0}^{q-1} \beta^n \mathbf{p}_n(S)$; the coefficient of S^r on the right is $\sum_{n=0}^{q-r-1} c_{n+r+1}\beta^n$, exactly the quotient coefficient in division by $S - \beta$. All displayed coordinates of v_β are nonzero: $\chi'(\beta) \neq 0$ by the simple grid, and $\text{Re } \beta \geq a(1/2 - \delta) > 0$, so $\beta \neq 0$. Since $|\mathcal{K}_I| = m < q$, at least one nonzero coordinate is omitted. Therefore

$$v_\beta \notin \text{im } L_I \quad \text{for every original root and every retained minor.} \quad (\text{CJA31})$$

Let $\mathcal{U}$ be a nonzero invariant subspace of $\mathbf{E}$ contained in $\text{im } M_I$, where $M_I = I_4 \otimes L_I$. For any nonzero $u \in \mathcal{U}$, at least one original Lagrange spectral projector $P_\beta = v_\beta(\mathbf{A})$ has $P_\beta u \neq 0$, since their sum is the identity. Here $v_\beta(\mathbf{A})$ means evaluation of its literal Lagrange polynomial at $\mathbf{A}$, on each of the four copies. Each P_β is a polynomial in $\mathbf{A}$, so this vector still belongs to $\mathcal{U}$. It has the form $(z_1 v_\beta, z_2 v_\beta, z_3 v_\beta, z_4 v_\beta)$, with some $z_b \neq 0$. Its membership in the four-copy image would imply $v_\beta \in \text{im } L_I$, contradicting (CJA31). This proves the actual all-minor strengthening

$$\boxed{\text{core}_{\mathbf{A}}(\text{im } M_I) = \{0\} \quad (\text{every retained } I).} \quad (\text{CJA32})$$

Its exact invariant hull is also given by original evaluations: put $\mathcal{B}_I = \{\beta : \mathfrak{p}_n(\beta) \neq 0 \text{ for some } n \in \mathcal{K}_I\}$. The same spectral projectors prove

$$\mathbb{C}[\mathbf{A}] \text{im } M_I = \left(\bigoplus_{\beta \in \mathcal{B}_I} \mathbb{C} v_\beta \right)^4.$$

For inclusion from left to right, every missing root evaluation stays zero under multiplication by β . For the reverse inclusion apply P_β to a column with nonzero evaluation. Thus this hull formula retains the actual selected image rather than assigning it the consecutive hull without a map.

The exact original-to-consecutive transition COI9–11 is

$$\begin{aligned} L_I &= F_C D_{0I} + L_0 M_{0I}, & M_I &= Y + Z, \\ Y &= (I_4 \otimes L_0)(I_4 \otimes M_{0I}), & Z &= (I_4 \otimes F_C)(I_4 \otimes D_{0I}). \end{aligned} \quad (\text{CJA33})$$

Its matrices are evaluated using the full original conductor and the selected invertible minor in COI9; M_{0I} is invertible there by the full block transition. In particular Z is a retained actual arithmetic vector, not a discarded relation. For every full Hermitian form Q on $\mathbf{E}$, direct multiplication gives the exact receiver

$$M_I^* Q M_I = Y^* Q Y + Z^* Q Z + Y^* Q Z + Z^* Q Y. \quad (\text{CJA34})$$

This applies separately to $\mathbf{G}$, to each $\mathbf{M}_i$, and to the full centered action $\mathbf{D} = \mathbf{A}^* \mathbf{G} + \mathbf{G} \mathbf{A} - a \mathbf{G}$. Define the actual quantities

$$G_I = M_I^* \mathbf{G} M_I, \quad A_I = G_I^{-1} M_I^* \mathbf{G} \mathbf{A} M_I, \quad R_I = \mathbf{A} M_I - M_I A_I, \quad \Gamma_{i,I} = M_I^* \mathbf{M}_i M_I.$$

They obey

$$\begin{aligned} M_I^* \mathbf{G} R_I &= 0, \\ A_I^* G_I + G_I A_I - a G_I &= M_I^* \mathbf{D} M_I. \end{aligned} \quad (\text{CJA35})$$

The right-hand side retains all four terms of (CJA34), with $Q = \mathbf{D}$. The proper-source identities (CJA17)–(CJA19) and the first line of (CJA19a), including both Gamma-metric cross terms and both normal edges, hold verbatim with M, A_H, R, Γ_i replaced by $M_I, A_I, R_I, \Gamma_{i,I}$: their proofs used the actual compression equation and the proved COB band orthogonality, not consecutive degrees. In particular

$$\begin{aligned} A_I^* \Gamma_{i,I} + \Gamma_{i,I} A_I - a \Gamma_{i,I} &= -M_I^* \mathbf{M}_i (R_I + \mathbf{C}_i M_I) \\ &\quad - (R_I + \mathbf{C}_i M_I)^* \mathbf{M}_i M_I. \end{aligned}$$

There is a further exact original-chart domain where the critical-line and Sylvester conclusions follow without a consecutive selector:

$$\mathcal{K}_I \subset \{1, \dots, q-1\}. \quad (\text{CJA36})$$

This holds for every original minor when $v \geq 1$; for $v = 0$ it holds exactly when $0 \notin J_I$. On this explicitly identified domain the degree of every $\mathfrak{p}_n$ in the image is at most $q-2$. Multiplication by S

on that image is therefore literal through degree $q-1$, without polynomial reduction. The physical identity $\bar{S} + S = a$, applied to each pair of original image polynomials, proves $M_I^* \mathbf{D} M_I = 0$. By (CJA35), take $T_I = G_I^{1/2}$ and put

$$J_I^{\text{phys}} = -i(T_I A_I T_I^{-1} - cI). \quad \text{Then} \quad (J_I^{\text{phys}})^* = J_I^{\text{phys}}. \quad (\text{CJA37})$$

The proof is to conjugate the zero identity in (CJA35) by T_I^{-1} ; the resulting centered action is skew-Hermitian. This Hermitian compression need not be the tridiagonal J_m ; no rank-four assertion is transferred to R_I . Diagonalize this actual finite Hermitian matrix by a unitary O_I . Replacing W in (CJA26) by $O_I^* T_I$, and t_j by its actual real eigenvalues, gives the exact Sylvester inverse for $\mathbf{A}X - XA_I$. Every denominator still has real part $(2r-a)\delta \neq 0$. The original-metric norm is exactly (CJA28) with these new eigenvalues and the same retained target Gram K ; the unique action-compatible correction is $X = -M_I$. For each $\Gamma_{i,I}$, its compressed defect is the exact commutator

$$T_I^{-*}(A_I^* \Gamma_{i,I} + \Gamma_{i,I} A_I - a \Gamma_{i,I}) T_I^{-1} = i[T_I^{-*} \Gamma_{i,I} T_I^{-1}, J_I^{\text{phys}}],$$

and its own original-metric Sylvester norm is transported by (CJA29) with $O_I^* T_I$.

On the remaining possible original chart $v = 0$, $0 \in J_I$, the degree- $(q-1)$ direction is present. The actual action and all its centered terms are (CJA33)–(CJA35). The physical critical-line argument above has an explicit degree- q polynomial-reduction term there, so it does not assign that chart a critical-line compression or the consecutive Jacobi spectrum. The all-minor zero-core theorem (CJA32), the actual cyclic enlargement (CJA20)–(CJA24), and all complete proper-source cross terms remain established on that chart as well.

21 The full native–arithmetic relative return on the original coefficients

21.1 Original forms, source orders, and orientations

Retain the actual simple quartet, its period, branch, full conductor E_A , and all original root guards. Write

$$\begin{aligned} a = k + 4 &\equiv 1 \pmod{4}, \quad q = (a+1)^2, \quad Q = (a-7)^2, \quad \Delta = q - Q = 16a - 48, \\ v &= \text{ord}_0 E_A, \quad 0 \leq v \leq 80, \quad m = \Delta - v > 0, \\ s &\in \{1, a\}, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \\ (N_0, N_1, N_2, N_3) &= (q-1, q, 2q-1, 2q), \quad D_i = N_i - v, \\ \mathcal{R}f &= -f(N_0) - f(N_1) + f(N_2) + f(N_3). \end{aligned} \quad (\text{NMR1})$$

The lower grid keeps source order s , not order $a-8$. The original Gamma measure, mass, and polynomial norms are

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = (2\pi)^{s/2} n! (s/2)_n. \quad (\text{NMR2})$$

Use its original orthonormal polynomials $\phi_{n,s,c}(S) = p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}$, where $p_{0,s} = 1$, $p_{1,s} = y$, and $p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}$. These are the mass-retaining Meixner–Pollaczek formulas with $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$ [15].

Let $Z = \mathbb{V}^{-1} P_v E_{\mathcal{J}}$ be the actual DNE numerator section selected by its actual admitted lower-conductor minor, and let $L = \mathcal{L}Z : \mathbb{C}^m \rightarrow E_a = \mathbb{C}[S]/(\chi_a)$ be its residue map, retaining $\chi'_a(\beta)$ in $\mathcal{L}$. The arithmetic form G_N^{ar} is the attained polynomial remainder metric for the complete source $d\mu_a(y) = w_h^{*a}(y) dy$ on the physical line $S = c_0 + iy$. Set

$$\begin{aligned} M &= I_4 \otimes L, \quad \mathcal{G}_N = M^* (G_N^{\text{ar}})^{\oplus 4} M, \quad \mathcal{D}_N = (D_{a,N-v}^{(s)})^{\oplus 4}, \\ T_N &= \mathcal{G}_N^{-1/2} \mathcal{D}_N \mathcal{G}_N^{-1/2}, \quad F_{\text{ar}} = -\mathcal{R} \log \det \mathcal{G}_N, \quad \mathcal{T}_{a,s} = \mathcal{R} \log \det D_{a,N-v}^{(s)}. \end{aligned} \quad (\text{NMR3})$$

All are literal original coefficient forms. The positive square root only represents the identity map between the two original Hilbert metrics; it does not rescale either source or discard a coordinate.

21.2 The native metric and its monotonicity before any determinant

The native basis germs are $F_b(z) = \sum_{\beta} Z_{\beta b} e^{\beta z} / E_A(z)$, analytic at zero. At every $D \geq q - 1 - v$, let

$$H_D[n, \beta'] = \phi_{n,s,c_1}(\beta'), \quad J_D[n, b] = \Lambda_{F_b}(\phi_{n,s,c_1}), \quad 0 \leq n \leq D.$$

Every derivative in this expression is retained. Explicitly, with $n_{jb} = \sum_{\beta} Z_{\beta b} \beta^j$ and $f_{rb} = F_b^{(r)}(0)$,

$$f_{rb} = \frac{n_{v+r,b} - \sum_{j=v+1}^{v+r} \binom{v+r}{j} \mu_j f_{v+r-j,b}}{\binom{v+r}{v} \mu_v}.$$

It follows directly by differentiating the original identity $E_A F_b = \sum_{\beta} Z_{\beta b} e^{\beta z}$. The exact original metric is

$$x^* D_{a,D}^{(s)} x = \min_{\eta \in \mathbb{C}^Q} \|J_D x + H_D \eta\|_2^2, \quad D_{a,D}^{(s)} = J_D^* J_D - J_D^* H_D (H_D^* H_D)^{-1} H_D^* J_D. \quad (\text{NMR4})$$

The lower evaluation columns are independent. The combined columns $[H_D, J_D]$ are independent: multiplying a germ by E_A and taking its first q numerator moments identifies them with the independent columns $[C_a^T, Z]$ through the upper Vandermonde. Thus the minimum is positive for $x \neq 0$ and its unique minimizer is $-(H_D^* H_D)^{-1} H_D^* J_D x$.

At $D+1$ the objective in (NMR4), for each fixed x, η , equals its old objective plus the squared modulus of the new observation row. Taking the minimum over the identical complete space $\mathbb{C}^Q$ therefore proves

$$D_{a,D+1}^{(s)} \succeq D_{a,D}^{(s)} > 0, \quad G_{N+1}^{\text{ar}} \preceq G_N^{\text{ar}}, \quad \mathcal{G}_{N+1} \preceq \mathcal{G}_N. \quad (\text{NMR5})$$

For the arithmetic inequalities, the full affine fibre of a remainder p grows from $p + \chi_a \mathbb{C}[S]_{\leq N-q}$ to $p + \chi_a \mathbb{C}[S]_{\leq N+1-q}$, in the same original integral norm. Minimizing over the larger fibre can only decrease its squared norm. Pullback along the fixed M preserves this inequality. In particular $\mathcal{T}_{a,s} \geq 0$ and $F_{\text{ar}} \geq 0$.

21.3 Reconstruction of the native total and its signs

Here the Gamma numerator metric is denoted $G_N^{(s)}$, to retain its distinction from the arithmetic source. Let $\mathbf{E}_{+,N}[n, \beta] = \phi_{n,s,c_0}(\beta)$ and $K_N = \mathbf{E}_{+,N}^* \mathbf{E}_{+,N}$. An exponential coefficient ν acts on a remainder coefficient vector x by $(\mathbb{V}\nu)^T x$. Its squared dual norm is $(\mathbb{V}\nu)^* (G_N^{(s)})^{-1} (\mathbb{V}\nu)$, by solving for its Hermitian representing vector. The same functional in the orthonormal polynomial coordinates has squared norm $\nu^* K_N \nu$. Equality for every ν proves

$$K_N = \overline{\mathbb{V}^* (G_N^{(s)})^{-1} \mathbb{V}}, \quad \det K_N = \frac{|\det \mathbb{V}|^2}{\det G_N^{(s)}}. \quad (\text{NMR6})$$

The conjugation and ordinary transpose are both retained.

Let $Z_v = \mathbb{V}^{-1} P_{<v}$ and $\mathbf{F} = [C_a^T, Z, Z_v]$. This frame is invertible: the first v moment coordinates remove a putative Z_v coefficient, and the selected lower-conductor minor then proves independence of the other columns. Define the three successive factors of $\mathbf{F}^* K_N \mathbf{F}$:

$$K_{C,N} = \overline{C}_a K_N C_a^T, \quad \hat{D}_N = Z^* \{K_N - K_N C_a^T K_{C,N}^{-1} \overline{C}_a K_N\} Z,$$

and let $Q_{v,N}$ be the attained last-block Gram after minimizing over every column of $[C_a^T, Z]$. Subtracting the first-block minimizer and its conjugate row operation has determinant one and

splits off $K_{C,N}$. Applying the same elimination to the second block splits off $\widehat{D}_N$. The remaining minimum is exactly $Q_{v,N}$. Consequently

$$\det K_{C,N} \det \widehat{D}_N \det Q_{v,N} = |\det F|^2 \det K_N = \frac{|\det F|^2 |\det \mathbb{V}|^2}{\det G_N^{(s)}}. \quad (\text{NMR7})$$

The two fixed frame factors are not set equal to one.

The lower original covariance is $K_{-,D} = H_D^* H_D$. The literal polynomial conductor matrix $A_{D,s}$ in the original Gamma frames has ordinary dual $B_{D,s} = A_{D,s}^\top$. The exact product-rule square is

$$B_{D,s} H_D = \mathbf{E}_{+,N}^{\text{red}} C_a^\top, \quad B_{D,s} J_D = \mathbf{E}_{+,N}^{\text{red}} Z,$$

where the omitted first v upper rows vanish on both numerator subspaces. Thus $\widehat{D}_N$ is the minimum of $\|B_{D,s}(J_D x + H_D \eta)\|^2$ over that same η , with no scalar v/μ_v inserted. Put

$$\begin{aligned} e_N^C &= \log \det K_{C,N} - \log \det K_{-,N-v}, & \epsilon_N &= \log \det \widehat{D}_N - \log \det D_{a,N-v}^{(s)}, \\ \mathcal{B}_a^{(s)} &= -\mathcal{R} \log \det G_N^{(s)}, & \mathcal{C}_{a,s}^- &= \mathcal{R} \log \det K_{-,N-v}. \end{aligned} \quad (\text{NMR8})$$

The last quotient factor is $Q_{v,N} = (P_{<v}^* G_N^{(s)} P_{<v})^{-1}$. Indeed its value map on dual coordinates is $P_{<v}^\top \mathbb{V}$; its kernel is the full vanishing-moment space, and the inverse of its covariance, using (NMR6), is precisely the displayed conjugate inverse. Hence $\mathcal{R} \log \det Q_{v,N} = F_{L_v}^{(s)}$, the original low-polynomial return, with signs $(+, +, -, -)$ on its primal Gram. Applying $\mathcal{R}$ to (NMR7) cancels its fixed frame factors and proves the literal native identity

$$\mathcal{T}_{a,s} = \mathcal{B}_a^{(s)} - \mathcal{C}_{a,s}^- - \mathcal{R} e^C - F_{L_v}^{(s)} - \mathcal{R} \epsilon. \quad (\text{NMR9})$$

This independently verifies the signs and the one-copy meaning of the LRC24 total.

For clarity its error control is on the same finite maps. The complete LRC exterior bounds give $-2I_{D,s,Q,1} \leq e_N^C \leq 2F_{D,s,Q,1}$. For the native quotient the induced map and inverse are $P_{BY \cap (BK)^\perp} B$ and $P_{Y \cap K^\perp} B^{-1}$, where $Y = \text{im}[H_D, J_D]$, $K = \text{im} H_D$. Each projection removes exactly the relevant denominator. Taking the top exterior power in dimension m gives $-2I_{D,s,m,1} \leq \epsilon_N \leq 2F_{D,s,m,1}$. These are the original all-rank bounds, each $O_{\text{actual}}(q)$; no additional denominator-rank factor is charged.

The low return is also $O_{\text{actual}}(q)$ with its fixed dimension v . Explicitly retain the original full-remainder estimate $K_{q-1} \preceq K_N \preceq Z_{a,s} K_{q-1}$, where

$$\begin{aligned} t_q &= \pi/2 - 1/q, & A_q &= 2 \log 3/t_q, & S_q &= \sum_{j=0}^{q-1} A_q^j, \\ B_{s,q} &= [(9/25) \sin(1/q)]^{-s/2} \exp\{2a\sqrt{\delta^2 + \gamma^2}(\log 3 + t_q/2)\}, \\ Z_{a,s} &= 1 + B_{s,q} S_q^2 \sum_{n=q}^{2q} \frac{n!}{(s/2)_n} (25/16)^n. \end{aligned} \quad (\text{NMR10})$$

This is the original AKS whole-remainder bound, as proved in the primitive-endpoint integration. Inverting (NMR6), restricting to the fixed low subspace, and using its actual dimension v gives $0 \leq F_{L_v}^{(s)} \leq 2v \log Z_{a,s}$. Here $\log Z_{a,s} = O_{\text{actual}}(q)$: one has $n!/(s/2)_n \leq 2n+1$, $\log S_q = O(q)$ and $\log B_{s,q} = O_{\text{actual}}(a \log q) = o(q)$. Thus every error on the right of (NMR9) is $O(q)$. At $v=0$ the low factor is the empty determinant one.

To check the evaluated coefficients rather than merely repeat LRC24, write $\ell_s = (s-1)/4$ and use the proved literal covariance expansions in their exact LRS21–22 source orders:

$$\begin{aligned} \mathcal{B}_a^{(s)} &= C_B q^2 + \ell_s q (2C_B - 4a_1) + O_{\text{actual}}(q), \\ \mathcal{C}_{a,1}^- &= C_B q^2 - 16C_\partial a q + 128q \log a + O_{\text{actual}}(q), \\ \mathcal{C}_{a,a}^- &= C_B q^2 + (-16C_\partial + C_B/2 - a_1) a q + 128q \log a + O_{\text{actual}}(q). \end{aligned} \quad (\text{NMR11})$$

These use the lower interval starting at m with width q and cutoff $N - v$, not a new natural lower window. The asymptotic statements retain the original finite domains $a \geq 257$, $a\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q$, and, for $b = a, a - 8$, $n_b = (b + 1)^2 + \ell_s \geq 100$ and $\max\{b\sqrt{\delta^2 + \gamma^2}, 2\ell_s + 1/2\} \leq 2^{-17}n_b$. All hold eventually for the fixed actual quartet.

For $s = 1$ the upper ℓ_s term is zero. For $s = a$ it is exactly $(a - 1)q(C_B/2 - a_1)$, whose aq coefficient cancels the $(C_B/2 - a_1)aq$ term in the lower covariance; the remaining $-(C_B/2 - a_1)q$ is within the stated $O(q)$. Subtraction in (NMR9) therefore proves

$$\mathcal{T}_{a,s} = 16C_\partial aq - 128q \log a + O_{\text{actual}}(q), \quad s = 1, a. \quad (\text{NMR12})$$

The sign -128 is forced by subtracting the actual lower covariance. The positive coefficient is the original $C_\partial = 2 \int_0^1 \log((1 + \rho(t))/(1 - \rho(t))) dt > 0$: the proved profile-covariance dictionary has $0 < \rho(t) < 1$ and an integrable logarithmic endpoint at zero. Thus no new outer coefficient has been substituted.

21.4 Exact arithmetic correction for the actual section

This step applies to the actual L , without assigning its image a lower polynomial degree than its original minor proves. Let $g_0 = L^* G_{q-1}^{\text{ar}} L$ in one copy. Represent its columns by their literal remainders $l_1, \dots, l_m$. At cutoff N write the complete relation frame $r_j = \chi_a S^j$, $0 \leq j \leq N - q$. For $N = q - 1$ it is empty. Define full integral matrices

$$(H_N^{\text{ar}})_{ij} = \langle r_i, r_j \rangle_{\mu_a}, \quad (C_N^{\text{ar}})_{ib} = \langle r_i, l_b \rangle_{\mu_a},$$

with conjugate-linearity in the first argument. Completing the square over every relation coefficient gives

$$g_N = L^* G_N^{\text{ar}} L = g_0 - (C_N^{\text{ar}})^* (H_N^{\text{ar}})^{-1} C_N^{\text{ar}}. \quad (\text{NMR13})$$

No relation cross block is deleted. Independence of the original remainders from the relation ideal implies $g_N > 0$. Define the finite, actual projection matrices in four copies:

$$S_i = \left[g_0^{-1/2} (C_{N_i}^{\text{ar}})^* (H_{N_i}^{\text{ar}})^{-1} C_{N_i}^{\text{ar}} g_0^{-1/2} \right]^{\oplus 4}. \quad (\text{NMR14})$$

The empty case is $S_0 = 0$. The nested relation projections and (NMR13) give $0 = S_0 \preceq S_1 \preceq S_2 \preceq S_3 < I_{4m}$. Taking determinants, the fixed $\det(g_0^{\oplus 4})$ cancels and proves

$$\begin{aligned} F_{\text{ar}} &= \log \det(I - S_1) - \log \det(I - S_2) - \log \det(I - S_3), \\ 0 \leq F_{\text{ar}} &\leq -2 \log \det(I - S_3) \leq -8m \log(1 - \beta_{\text{orig}}), \quad \beta_{\text{orig}} = \|S_3\| < 1. \end{aligned} \quad (\text{NMR15})$$

For the lower bound, $\log \det(I - S_1) \geq \log \det(I - S_2)$ and $-\log \det(I - S_3) \geq 0$. For the first upper bound, discard the nonpositive first term and use $\log \det(I - S_2) \geq \log \det(I - S_3)$. For the last upper bound, all $4m$ eigenvalues of S_3 are at most β_{orig} . This is a finite original-matrix value and bound; it asserts no unproved asymptotic rate for β_{orig} .

An explicit coarse bound also applies uniformly to every original section. The unchanged simple-quartet source estimates are

$$b_a(1 + y^2)^{-25} d\sigma(y) \leq d\mu_a(y) \leq B_a d\sigma(y), \quad d\sigma = dm_1,$$

where the complete original constants are

$$\begin{aligned} b_a &= \frac{c_h \vartheta_h^{a-3} e^{-(\pi/2)(a-3)}}{C_\Gamma 2^{25} (a-2)^{50}}, & B_a &= \frac{C_h^{\text{tilt}}}{c_\Gamma} a^{9/2} M_{\pi/2}^{a-1}, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy, & M_{\pi/2} &= \int_{\mathbb{R}} e^{(\pi/2)y} w_h(y) dy, & c_\Gamma &= \sqrt{2} e^{-7/6}, & C_\Gamma &= \sqrt{2\sqrt{5}} e^{2/3}. \end{aligned} \quad (\text{NMR16})$$

The constants c_h, C_h^{tilt} retain their original convolution lower-envelope and tilted-envelope definitions. For a polynomial of degree at most D , the Gamma recurrence gives $\|yP\|_\sigma \leq 2(D+1)\|P\|_\sigma$. For nonzero P put $I_0 = \int |P|^2 d\sigma$, $I_2 = \int y^2 |P|^2 d\sigma$ and $u = I_2/I_0$. The convex function $\varphi(x) = (1+x)^{-25}$ satisfies its exact tangent inequality $\varphi(y^2) \geq \varphi(u) + \varphi'(u)(y^2 - u)$. Multiply by $|P|^2$ and integrate in the unchanged $d\sigma$. The linear term is $\varphi'(u)(I_2 - uI_0) = 0$, so the result is at least $\varphi(u)I_0 \geq [1 + 4(D+1)^2]^{-25}I_0$. The zero polynomial obeys the same inequality directly. Consequently

$$b_a[1 + 4(D+1)^2]^{-25}\|P\|_\sigma^2 \leq \|P\|_{\mu_a}^2 \leq B_a\|P\|_\sigma^2.$$

The physical translate $P(c_0 + iy)$ remains its actual polynomial in y ; no Euclidean coefficient norm or source mass is changed. Put $K_{a,2q} = (B_a/b_a)[1 + 4(2q+1)^2]^{25}$. Taking these inequalities on every original affine fibre and then using (NMR10) at order $s = 1$ gives

$$\begin{aligned} \Omega_a &= K_{a,2q}Z_{a,1}, & \Omega_a^{-1}\mathcal{G}_{q-1} &\preceq \mathcal{G}_N \preceq \mathcal{G}_{q-1}, \\ 0 &\leq F_{\text{ar}} \leq 8m \log \Omega_a = O_{\text{actual,h}}(aq). \end{aligned} \tag{NMR17}$$

Indeed $G_N^{\text{ar}} \succeq b_a[1 + 4(2q+1)^2]^{-25}G_N^{(1)} \succeq b_a[1 + 4(2q+1)^2]^{-25}Z_{a,1}^{-1}G_{q-1}^{(1)}$, whereas $G_{q-1}^{\text{ar}} \preceq B_aG_{q-1}^{(1)}$. Pulling back proves the first line. The second uses the same two high-endpoint determinant bound as (NMR15). Here $\log K_{a,2q} = O_h(a + \log q)$ and $\log Z_{a,1} = O_{\text{actual}}(q)$.

21.5 The exact relative determinant and its mandatory spectral cost

Taking determinants in (NMR3), with dimension exactly $4m$, yields at each endpoint $\log \det T_N = 4 \log \det D_{a,N-v}^{(s)} - \log \det \mathcal{G}_N$. Therefore

$$\mathcal{R} \log \det T_N = 4\mathcal{T}_{a,s} + F_{\text{ar}}. \tag{NMR18}$$

Combining the exact finite bounds above gives

$$4\mathcal{T}_{a,s} \leq \mathcal{R} \log \det T_N \leq 4\mathcal{T}_{a,s} + \min\{-2 \log \det(I - S_3), 8m \log \Omega_a\}. \tag{NMR19}$$

On the original LRC finite domains, its correct evaluated expression for the actual section is

$$\mathcal{R} \log \det T_N = 64C_{\partial}aq - 512q \log a + F_{\text{ar}} + O_{\text{actual}}(q). \tag{NMR20}$$

The explicit arithmetic term cannot be absorbed into $O(q)$ merely from the coarse bound (NMR17).

Let $\mathcal{L}_{a,s}$ be the larger maximum generalized eigenvalue on the two original native links D_2 relative to D_0 and D_3 relative to D_1 , with $D_i = D_{a,N_i-v}^{(s)}$. Each determinant ratio is at most $\mathcal{L}_{a,s}^m$, so

$$\log \mathcal{L}_{a,s} \geq \frac{\mathcal{T}_{a,s}}{2m}, \quad \liminf_{a \rightarrow \infty} \frac{\log \mathcal{L}_{a,s}}{q} \geq \frac{C_{\partial}}{2} > 0. \tag{NMR21}$$

The determinant assertion follows by taking the eigenvalues of $D_{\text{low}}^{-1/2}D_{\text{high}}D_{\text{low}}^{-1/2}$. For the limit, substitute (NMR12) and $m = 16a - 48 - v$, keeping $s = 1$ or $s = a$ on its original sequence.

There is a stronger finite result for every uniform comparison of the two actual metrics at all four endpoints. Such a comparison has $\alpha_a \mathcal{G}_N \preceq \mathcal{D}_N \preceq \beta'_a \mathcal{G}_N$ with $0 < \alpha_a \leq \beta'_a$. Every eigenvalue of T_N then lies in $[\alpha_a, \beta'_a]$, so its four-endpoint determinant return is at most $8m \log(\beta'_a/\alpha_a)$. By (NMR18),

$$\log \frac{\beta'_a}{\alpha_a} \geq \frac{4\mathcal{T}_{a,s} + F_{\text{ar}}}{8m} = \frac{\mathcal{T}_{a,s}}{2m} + \frac{F_{\text{ar}}}{8m}, \quad \liminf_{a \rightarrow \infty} \frac{1}{q} \log \frac{\beta'_a}{\alpha_a} \geq \frac{C_{\partial}}{2}. \tag{NMR22}$$

This applies in particular to the actual optimal constants $\alpha_a = \min_i \lambda_{\min}(T_{N_i}) > 0$ and $\beta'_a = \max_i \lambda_{\max}(T_{N_i}) < \infty$. It thus measures a finite, existing spectral comparison, not an assumed additional estimate. It strengthens the submitted N9 by retaining the nonnegative exact arithmetic

correction. Alternatively (NMR5) gives on either link $\mathcal{D}_{\text{high}} \preceq \beta'_a \mathcal{G}_{\text{high}} \preceq \beta'_a \mathcal{G}_{\text{low}} \preceq (\beta'_a/\alpha_a) \mathcal{D}_{\text{low}}$, recovering its original weaker bound (NMR21). Any single fixed invertible coefficient change applied to both sequences preserves these inequalities and determinant returns by congruence. No claim is made that every individual eigenvalue has this growth.

For each original fixed positive boundary form Γ_i , $i = 0, 1$, the exact identity is independently

$$\mathcal{R} \log \det(\Gamma_i^{-1/2} \mathcal{D}_N \Gamma_i^{-1/2}) = 4\mathcal{T}_{a,s}. \quad (\text{NMR23})$$

The fixed $\log \det \Gamma_i$ has coefficient $-1 - 1 + 1 + 1 = 0$; its individual endpoint value is retained.

21.6 The consecutive-chart value and the retained original-section correction

Let $I_0 = \{0, \dots, Q-1\}$ be the proved nonzero consecutive lower-conductor minor. Its section L_0 has the proved image $\mathbb{C}[S]_{\leq m-1}$. Denote its arithmetic pullback by $\mathcal{G}_N^0$ and its arithmetic return by F_{ar}^0 . Use $T_N^0 = (\mathcal{G}_N^0)^{-1/2} (D_{0,N}^{(s)})^{\oplus 4} (\mathcal{G}_N^0)^{-1/2}$ for its native form in that same chart. The full low-degree localization proved in CAL1–18 gives on its stated eventual guards

$$(1 - \varepsilon_a) \mathcal{G}_{q-1}^0 \preceq \mathcal{G}_N^0 \preceq \mathcal{G}_{q-1}^0, \quad \varepsilon_a = (\sqrt{\eta_a} + \sqrt{\zeta_a^*})^2 < 1.$$

Here η_a is CAL7 and CAL9 retains the sharper literal bound $\zeta_a^* = K_{a,m} [2^d (m+d)! / (m! Y^d)]^2 \leq K_{a,m} 4^{-2d}$, where $Y = 2^{-16}q$, $d = \lfloor Y/16 \rfloor$. These are the proved CAL constants, not an estimate for an unspecified original minor. It follows by precisely the determinant argument of (NMR15) that

$$0 \leq F_{\text{ar}}^0 \leq -8m \log(1 - \varepsilon_a), \quad \mathcal{R} \log \det T_N^0 = 64C_{\partial} a q - 512q \log a + O_{\text{actual,h}}(q). \quad (\text{NMR24})$$

Here the exponential estimate for ε_a is a proved low-degree estimate on this actual section.

For the original admitted minor I , keep the exact chart matrices $U = M_{I_0 \leftarrow I}$ and $D = D_{I_0 \leftarrow I}$ from NCB16–28. They satisfy $L_I = L_0 U + L_C D$ with $L_C = \mathcal{L} C_a^T$ and U invertible. Division of the numerator by E_A turns $L_C D$ into the actual lower exponential evaluation germ. Its whole contribution is already inside the unchanged native minimization (NMR4). Therefore, at every cutoff,

$$D_{I,N} = U^* D_{0,N} U, \quad L_I U^{-1} = L_0 + L_C W, \quad W = D U^{-1}. \quad (\text{NMR25})$$

The native identity follows by the bijective shift of the minimizing lower coefficient η by Dx , so it does not assume an isometry of the arithmetic section.

Since $\det U$ is fixed, the native four-endpoint determinant return is the same in the original and consecutive charts. The arithmetic section instead retains the full $L_C W$ term in (NMR25). Its exact arithmetic graph metric, with both mixed conductor–native blocks, is proved in the companion original-coefficient image calculation. The direct original-section formula (NMR13) already retains those full polynomial representatives and provides its finite correction F_{ar} without any chart identification.

Thus the exact identities, native monotonicity, and exponential lower uniform comparison cost in the submitted Note09 are retained. Its exponentially small arithmetic correction and the expression with that correction absorbed into $O(q)$ are established for the proved consecutive section. For the literal original section the correct complete formula is (NMR20), with its explicit finite arithmetic matrices. None of these determinant identities allocates a scalar to an individual native flag, replaces the independent degree- k proper-source metric, or adds a term to the canonical action.

Negative reflected trace, both boundary energies and uniform period recovery

22 An explicit reflected trace and both original boundary metrics

Keep the original fixed hypothetical simple quartet, its full arithmetic unit, actual period, logarithm branch and conductor. Write

$$\begin{aligned} a &\equiv 1 \pmod{4}, \quad a \geq 17, \quad c = a/2, \\ q &= (a+1)^2, \quad q' = (a-7)^2, \quad m = 16a - 48 - v, \quad 0 \leq v \leq 80, \\ \lambda_{u,b} &= c + (2u-a)\delta + i(2b-a)\gamma, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad 0 \leq u, b \leq a, \end{aligned} \quad (\text{RBV1})$$

$$\chi(S) = \prod_{u,b=0}^a (S - \lambda_{u,b}), \quad E = \mathbb{C}[S]/(\chi), \quad \mathcal{P}_{q-1} \xrightarrow{\sim} E.$$

The last map sends an actual polynomial to its residue class, with the unique degree-below- q remainder as inverse. Roots are distinct by equality of real and imaginary parts. No packet factors or coordinate phases are discarded.

22.1 The explicit polynomial and its exact trace

Put

$$t_0 = \frac{\pi}{2\gamma}, \quad r_0 = t_0\delta, \quad c_0 = 8 \log 4 - 10 > 0, \quad P_a(S) = \sum_{j=0}^{4a} \frac{t_0^{2j} (S-c)^{2j}}{(2j)!}. \quad (\text{RBV2})$$

Its degree is $8a \leq m-1$ since $8a \geq 129 \geq 49+v$. Also $8a < q$. Define the original finite reflected form

$$\mathcal{W}(f, g) = \sum_{\lambda} \overline{f(a - \bar{\lambda})} g(\lambda). \quad (\text{RBV3})$$

The reflection permutes the complete grid and is an involution. Changing its index in the conjugate of $\mathcal{W}(g, f)$ proves that this form is Hermitian. The polynomial P_a has real even coefficients in $S-c$, so $\overline{P_a(a - \bar{\lambda})} = P_a(\lambda)$. Hence $\mathcal{W}(P_a, P_a) = \sum_{\lambda} P_a(\lambda)^2$, a real number because conjugation also permutes the complete grid.

For $z = \lambda_{u,b} - c$, the actual comparison value is

$$\cosh(t_0 z) = i \sin\left(\frac{\pi(2b-a)}{2}\right) \sinh(r_0(2u-a)). \quad (\text{RBV4})$$

The sine is $+1$ or -1 , and $2u-a$ is a nonzero odd integer. Moreover $t_0|z| \leq at_0\sqrt{\delta^2 + \gamma^2} < 2a$: $\delta/\gamma < 1/4$ and $\pi < 22/7$ give the final strict bound. For $x \geq 0$ and an integer $n \geq 1$, the exponential series gives $\sum_{j \geq n} x^j/j! \leq x^n e^x/n!$, because $(n+j)! \geq n!j!$. The elementary integral comparison $\log n! = \sum_{j=1}^n \log j \geq \int_1^n \log x \, dx = n \log n - n + 1$ gives $n! \geq (n/e)^n$. Therefore

$$\begin{aligned} |P_a(c+z) - \cosh(t_0 z)| &\leq \frac{e^{2a}(2a)^{8a+1}}{(8a+1)!} \\ &\leq e^{2a}(e/4)^{8a+1} \leq e^{-c_0 a} =: \epsilon_a. \end{aligned} \quad (\text{RBV5})$$

This bounds the full omitted even tail by a larger exponential tail. Impose the explicit eventual guard $\epsilon_a \leq r_0/4$. Since $|\sinh(r_0(2u-a))| \geq r_0$, write the actual value as $is + e$ with s real and $|e| \leq |s|/4$. Direct expansion gives

$$\Re(is + e)^2 \leq -s^2 + 2|s||e| + |e|^2 \leq -7s^2/16. \quad (\text{RBV6})$$

The elementary finite geometric sum gives exactly

$$\sum_{u=0}^a \cosh(2r_0(2u-a)) = \frac{\sinh(2r_0(a+1))}{\sinh(2r_0)},$$

$$\sum_{\lambda} \cosh(t_0(\lambda-c))^2 = -\frac{a+1}{2} \left[\frac{\sinh(2r_0(a+1))}{\sinh(2r_0)} - (a+1) \right]. \quad (\text{RBV7})$$

Consequently the incoming finite sign and its constants are correct:

$$\boxed{\mathcal{W}(P_a, P_a) \leq -N_a < 0, \quad N_a = \frac{7(a+1)}{32} \left[\frac{\sinh(2r_0(a+1))}{\sinh(2r_0)} - (a+1) \right]}. \quad (\text{RBV8})$$

The bracket is $2 \sum_u \sinh^2(r_0(2u-a)) > 0$. There is a stronger explicit finite lower bound. For real s and $0 \leq \epsilon \leq |s|/2$, maximizing $\Re(is + x + iy)^2$ over $x^2 + y^2 \leq \epsilon^2$ gives $-(|s| - \epsilon)^2$. For $s > 0$, substitute $x^2 \leq \epsilon^2 - y^2$: the remaining concave quadratic $-s^2 - 2sy + \epsilon^2 - 2y^2$ takes its maximum on $[-\epsilon, \epsilon]$ at $y = -\epsilon$ because its vertex is $-s/2 \leq -\epsilon$. The case $s < 0$ follows by reflection. Consequently RBV8 also holds with

$$N_a^{\text{sharp}} = (a+1) \sum_{u=0}^a (|\sinh(r_0(2u-a))| - \epsilon_a)^2$$

$$= (a+1) \left\{ \frac{1}{2} \left[\frac{\sinh(2r_0(a+1))}{\sinh(2r_0)} - (a+1) \right] - 2\epsilon_a \frac{\cosh(r_0(a+1)) - 1}{\sinh r_0} + (a+1)\epsilon_a^2 \right\}. \quad (\text{RBV8a})$$

The additional finite sum follows by summing the positive and negative odd-index hyperbolic sines separately. In particular the coefficient $7/32$ may be strengthened to $9/32$ on the same guard. The original smaller N_a is retained in the downstream displayed bounds, all of which also hold with N_a^{sharp} . The exact error from RBV7 is at most

$$2\epsilon_a(a+1) \sum_{u=0}^a |\sinh(r_0(2u-a))| + q\epsilon_a^2. \quad (\text{RBV9})$$

Since the sum is bounded by $(a+1)e^{r_0a}$, this error divided by the magnitude in RBV7 is $O_{\delta, \gamma}(ae^{-(c_0+r_0)a})$. Keeping also the lower exponential and $(a+1)$ term in RBV7 proves

$$\mathcal{W}(P_a, P_a) = -\frac{(a+1)e^{2r_0(a+1)}}{4\sinh(2r_0)} [1 + o(1)]. \quad (\text{RBV10})$$

22.2 Original arithmetic density and finite full-form constants

The source remains

$$v_h(s) = 2\xi(s)/h(s), \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \quad d\mu_a(y) = w_h^{*a}(y) dy,$$

$$M_h(\theta) = \int_{\mathbb{R}} e^{\theta y} w_h(y) dy, \quad \alpha = \pi/2, \quad d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi), \quad \sigma(\mathbb{R}) = \sqrt{2\pi}. \quad (\text{RBV11})$$

RTM1–3/GTM1 proves evenness, positivity almost everywhere and the finite transforms at $|\theta| \leq \alpha$, using the complete unit. For the simple quartet, the original global constants in GTM1 are

$$B = 50, \quad p = 7/2, \quad \vartheta_h = \int_{-1}^1 w_h(y) dy, \quad \mathfrak{M} = M_h(\alpha),$$

$$b_a = \frac{c_h \vartheta_h^{a-3} e^{-\alpha(a-3)} (a-2)^{-50}}{C_{\Gamma} 2^{25}}, \quad A_a = \frac{C_h^{\text{tilt}}}{C_{\Gamma}} a^{p+1} \mathfrak{M}^{a-1}, \quad (\text{RBV12})$$

$$b_a(1+y^2)^{-25} d\sigma \leq d\mu_a \leq A_a d\sigma, \quad c_{\Gamma} = \sqrt{2}e^{-7/6}, \quad C_{\Gamma} = \sqrt{2\sqrt{5}}e^{2/3}.$$

The constants c_h, C_h^{tilt} are exactly the original TW7/BT bounds retained in RTM2, including both half-lines. This proof does not posit a positive lower bound for the one-factor zeta numerator.

Here is the finite polynomial comparison needed for the whole boundary bands. In the original orthonormal Gamma frame, multiplication by y has adjacent entries $\sqrt{(n+1)(n+1/2)} \leq n+1$ (GTM7). On polynomials of degree at most D , its two shifts each have norm at most $D+1$. Thus $\|yF\|_\sigma \leq 2(D+1)\|F\|_\sigma$. For $F \neq 0$, divide the positive integral $|F|^2 d\sigma$ by its own total only to apply the following scalar inequality; the source measure itself is never changed. Convexity of $(1+x)^{-25}$ on $x \geq 0$, proved by its positive second derivative, gives its tangent lower bound at the mean of y^2 . Integration therefore gives

$$\|F\|_{\mu_a}^2 \geq b_a[1 + 4(D+1)^2]^{-25} \|F\|_\sigma^2. \quad (\text{RBV13})$$

For all degrees through $2q$ we may use the exact constants

$$L_a^- = b_a[1 + 4(2q+1)^2]^{-25}, \quad L_a^+ = A_a, \quad \log(L_a^+/L_a^-) = O_h(a + \log q). \quad (\text{RBV14})$$

The original Gamma recurrence and mass dictionary are the Meixner–Pollaczek formulas in DLMF (18.19.8)–(18.19.9), (18.22.8) [29], with the exact substitution and mass verified in LT2 and NTG3. This is a full coefficient-form comparison, so it applies before either original boundary minimization.

22.3 Concentration of the complete relation bands

Set $Y = 2^{-16}q$ and $R = a\sqrt{\delta^2 + \gamma^2}$, and retain the explicit eventual guard $R \leq Y/4$. For every polynomial P of degree at most q and $F(S) = \chi(S)P(S)$, the original roots give

$$|\chi(c+iy)| \leq (Y+R)^q \quad (|y| \leq Y), \quad |\chi(c+iy)| \geq (q-R)^q \quad (q \leq y \leq 2q). \quad (\text{RBV15})$$

Use the shifted Legendre basis $\phi_d(y) = \sqrt{(2d+1)/q} L_d(2y/q - 3)$ on $[q, 2q]$, $0 \leq d \leq q$. Its Rodrigues formula and squared norm are the Legendre entries of DLMF (18.5.5), Tables 18.5.1 and 18.3.1 [29]. The full finite identity

$$L_d(z) = 2^{-d} \sum_{j=0}^d \binom{d}{j}^2 (z-1)^{d-j} (z+1)^j$$

gives $|L_d(z)| \leq 10^d$ on $|z| \leq 4$, using $\sum_j \binom{d}{j}^2 = \binom{2d}{d} \leq 4^d$. For $|y| \leq Y$, expansion of the original polynomial $P(c+iy)$ in this basis and Cauchy–Schwarz prove

$$|P(c+iy)|^2 \leq \frac{(q+1)^2}{q} 100^q \int_q^{2q} |P(c+it)|^2 dt. \quad (\text{RBV16})$$

Indeed the basis-square sum is bounded by $q^{-1} 100^q \sum_{d=0}^q (2d+1) = q^{-1} 100^q (q+1)^2$. The lower Gamma density on $[q, 2q]$ is $c_\Gamma e^{-\pi q} (1+2q)^{-1/2}$. Multiplying RBV15–16, integrating the full central interval using mass $\sqrt{2\pi}$, and using RBV14 in the denominator proves

$$\begin{aligned} \int_{|y| \leq Y} |F(c+iy)|^2 d\mu_a &\leq \bar{\eta}_a \|F\|_{\mu_a}^2, \quad \bar{\eta}_a = \min(1, \eta_a), \\ \eta_a &= \frac{L_a^+}{L_a^-} \frac{\sqrt{2\pi}}{c_\Gamma} \frac{(q+1)^2 \sqrt{1+2q}}{q} e^{\pi q} 100^q \left(\frac{Y+R}{q-R} \right)^{2q}, \\ \log \eta_a &\leq -c_1 q + O_h(a + \log q), \quad c_1 = 30 \log 2 - \pi - \log 100 > 0. \end{aligned} \quad (\text{RBV17})$$

The last ratio is at most 2^{-15} on the stated guard. This bound applies to every vector of either complete relation band, including arbitrary linear combinations and all original cross terms.

22.4 Both boundary energies, including strict positivity

Let $e_j = \chi\pi_j/\sqrt{\nu_j}$, where π_j is the actual monic polynomial orthogonal for the full weight $|\chi(c+iy)|^2 d\mu_a$ in the original coordinate. Define

$$\mathcal{B}_0 = \text{span}(e_0, \dots, e_{q-1}), \quad \mathcal{B}_1 = \text{span}(e_1, \dots, e_q), \quad \Delta_i P = -\text{Proj}_{\mathcal{B}_i} P. \quad (\text{RBV18})$$

These are the original COB two windows. Each band consists of χ times polynomials of degree at most q , so RBV17 applies to every unit vector in it. Their exact link to the original minimum forms follows without a choice of diagonal trial metric. Let $s_N : E \rightarrow \mathcal{P}_N$ send a class to its unique minimum-norm representative in $d\mu_a$. For the original representative $P \in \mathcal{P}_{q-1}$, its entire affine fibre is $P + \chi\mathcal{P}_{N-q}$. Completing its orthogonal square gives $s_N P = P - \text{Proj}_{\chi\mathcal{P}_{N-q}} P$. Consequently $\Delta_0 = s_{2q-1} - s_{q-1}$ and $\Delta_1 = s_{2q} - s_q$, exactly as in RBV18. Writing $G_N = s_N^* s_N$ in the unchanged class frame, orthogonality of these two nested projection differences gives $G_{q-1} - G_{2q-1} = \Delta_0^* \Delta_0$ and $G_q - G_{2q} = \Delta_1^* \Delta_1$. Thus each coefficient matrix is exactly $\Gamma_{i,I} = L_I^* \Delta_i^* \Delta_i L_I$ on its stated original residue injection. This proves the map into both boundary energies, including all cross terms of the actual minimum. These are precisely COB.5–6 and COB.13 in `COUPLED_ORIGINAL_CONE_BOUNDARY.tex`. The general positive coefficient-square operation is the Schur minimum in Boyd and Vandenberghe [37, Appendix A.5.5, p. 650]; the preceding formulas prove its full original-fibre specialization.

In fact both Δ_i are injective on the entire original $\mathcal{P}_{q-1}$. Pair the roots with indices u and $a - u$. Since $(a+1)^2/2$ is even, their original factors give exactly

$$\chi(c+iy) = \prod_{b=0}^a \prod_{0 \leq u < a/2} [(y - (2b-a)\gamma)^2 + (2u-a)^2 \delta^2] > 0. \quad (\text{RBV19})$$

If $P \perp \mathcal{B}_0$, test its orthogonality against $\chi P \in \mathcal{B}_0$ to obtain $\int \chi(c+iy)|P(c+iy)|^2 d\mu_a = 0$, hence $P = 0$. If $P \perp \mathcal{B}_1$, choose the unique b such that $F = P - b\chi$ is orthogonal to e_0 . Since χ is a multiple of e_0 , F remains orthogonal to $e_1, \dots, e_q$. Its degree is at most q , so χF belongs to $\text{span}(e_0, \dots, e_q)$. Testing against it and using RBV19 gives $F = 0$. Then $P = b\chi$ and $\deg P < q$ force $P = b = 0$. This proves strict positivity without an assumed missing metric bound.

On the original source line, $|P_a(c+iy)| \leq \cosh(t_0|y|) \leq e^{t_0|y|}$. The exact convolution transform, with all masses retained, is

$$\int e^{\theta y} d\mu_a = M_h(\theta)^a.$$

Evenness and $2t_0 < \alpha$ therefore give, with $\theta_1 = (\alpha - 2t_0)/2 > 0$, $\theta_2 = 2t_0 + \theta_1 < \alpha$,

$$\|P_a\|_{\mu_a}^2 \leq 2M_h(2t_0)^a, \quad \|1_{|y|>Y} P_a\|_{\mu_a}^2 \leq 2e^{-\theta_1 Y} M_h(\theta_2)^a. \quad (\text{RBV20})$$

For a unit $f \in \mathcal{B}_i$, split its inner product with P_a over the two original integration regions. Cauchy–Schwarz and RBV17 give $|\langle f, P_a \rangle| \leq \sqrt{\eta_a} \|P_a\| + \|1_{|y|>Y} P_a\|$. Taking the supremum over the full unit sphere of the band, then $(x+y)^2 \leq 2x^2 + 2y^2$, proves

$$0 < \|\Delta_i P_a\|_{\mu_a}^2 \leq B_a, \quad B_a = 4\eta_a M_h(2t_0)^a + 4e^{-\theta_1 2^{-16}q} M_h(\theta_2)^a, \quad i = 0, 1. \quad (\text{RBV21})$$

No band-dimension multiplier enters this estimate.

22.5 Relative eigenvalue and evaluated discrepancy

Let G_i^E be the Gram of Δ_i on the actual monomial frame of $\mathcal{P}_{q-1}$; it is strictly positive by RBV19. Let W^E be the matrix of RBV3 in that same frame. Then

$$T_i^E = (G_i^E)^{-1/2} W^E (G_i^E)^{-1/2}$$

is Hermitian. The coefficient column of P_a gives the actual Rayleigh quotient, so

$$\lambda_{\min}(T_i^E) \leq -N_a/B_a < 0, \quad \|I - T_i^E\| \geq 1 + N_a/B_a. \quad (\text{RBV22})$$

This follows by diagonalizing the Hermitian matrix and expressing the Rayleigh quotient as a positive weighted average of its real eigenvalues. With the original two matrices, the discrepancy is

$$\begin{aligned} \|\Delta_i P_a\|_{\mu_a}^2 - \mathcal{W}(P_a, P_a) &= \frac{(a+1)e^{2r_0(a+1)}}{4 \sinh(2r_0)} [1 + o_h(1)], \\ \lim_{a \rightarrow \infty} \frac{\|\Delta_i P_a\|^2 - \mathcal{W}(P_a, P_a)}{ae^{\pi\delta a/\gamma}} &= \frac{e^{\pi\delta/\gamma}}{4 \sinh(\pi\delta/\gamma)}. \end{aligned} \quad (\text{RBV23})$$

Indeed RBV21 has logarithm at most $-c_*q + O_h(a + \log q)$, where

$$c_* = \min\{30 \log 2 - \pi - \log 100, 2^{-16}(\pi/2 - \pi/\gamma)/2\} > 0. \quad (\text{RBV24})$$

This is negligible compared with RBV10. Also $\log N_a = 2r_0a + \log a + O_h(1)$, so RBV22 proves

$$\liminf_{a \rightarrow \infty} q^{-1} \log[-\lambda_{\min}(T_i^E)] \geq c_*. \quad (\text{RBV25})$$

All limits run over the original $a \equiv 1 \pmod{4}$ family at fixed actual quartet/unit and eventually satisfy the explicitly stated guards. No actual-period numerical eigenvalue is invented.

22.6 Exact return through every original conductor chart

This paragraph records the precise map needed to receive these results on the supplied COB coefficient image, without identifying two different sections of the same quotient. Write $\chi(S) = \sum_{j=0}^q c_j S^j$, $c_q = 1$, and define

$$\mathbf{p}_n(S) = \sum_{j=n+1}^q c_j S^{j-n-1}, \quad \mathbb{V}_{n,\lambda} = \lambda^n, \quad \mathcal{L}\nu = \sum_{\lambda} \nu_{\lambda} \frac{\chi(S)}{S - \lambda}, \quad F = \mathcal{L}\mathbb{V}^{-1}. \quad (\text{RBV26})$$

NCB5–7 proves $Fe_n = \mathbf{p}_n$ by the coefficient of S^{-1} in $\mathbf{p}_n S^j / \chi$. Thus F is invertible and its inverse is the exact Laurent map $(F^{-1}P)_n = [S^{-n-1}]P(S)/\chi(S)$, $0 \leq n < q$. Let $C^{\top}$ be the original conductor multiplication in exponential coefficients, and write $\mathbb{V}C^{\top} = P_v J$ with the original v leading zeros. For every retained invertible q' -row minor J_I , define

$$\begin{aligned} H_I &= J_I^{-1} E_I^{\top}, \quad Q_I = E_{I^c}^{\top} - J_{I^c} H_I, \quad L_I = F P_v E_{I^c}, \quad C_E = \mathcal{L}C^{\top}, \\ \eta_I &= H_I P_v^{\top} F^{-1} P_a, \quad z_I = Q_I P_v^{\top} F^{-1} P_a, \quad \boxed{P_a = C_E \eta_I + L_I z_I}. \end{aligned} \quad (\text{RBV27})$$

There is no low component because $\deg P_a < m$ makes all Laurent coordinates below $q - m \geq v$ zero. To prove the displayed identity, check the I and I^c rows to get $JH_I + E_{I^c} Q_I = I$. Multiply by FP_v and use $FP_v J = C_E$. These are the exact original residue and conductor maps; no coefficient section is inserted in place of another.

In particular the consecutive minor $I_0 = \{0, \dots, q' - 1\}$ has determinant

$$\det J_{I_0} = \mu_v^{q'} \prod_{r=0}^{q'-1} \binom{v+r}{v} \det \mathbb{V}_{\text{lower}} \neq 0. \quad (\text{RBV28})$$

Indeed its row r is the lower-root evaluation of a degree- r polynomial with leading coefficient $\binom{v+r}{v} \mu_v$; triangular coefficient elimination gives exactly this determinant. Here $\eta_{I_0} = 0$ and $\text{im } L_{I_0} = \mathcal{P}_{m-1}$ by the distinct monic degrees $m-1, m-2, \dots, 0$ of its columns. The literal inverse is

$$(z_{I_0})_j = [S^{-(q-m+j+1)}]P_a(S)/\chi(S). \quad (\text{RBV29})$$

Therefore RBV8–25 apply to the Gram $\Gamma_{i,I_0} = L_{I_0}^* G_i^E L_{I_0}$ and reflected matrix $L_{I_0}^* W^E L_{I_0}$ with precisely the same bounds, because the Rayleigh vector is the explicit column RBV29. For any actual other chart I , RBV27 supplies the full original coefficient vector (η_I, z_I) instead. With $M_I = [C_E, L_I]$, the same calculation holds on its full image with the two exact matrices

$$M_I^* G_i^E M_I = \begin{pmatrix} C_E^* G_i^E C_E & C_E^* G_i^E L_I \\ L_I^* G_i^E C_E & L_I^* G_i^E L_I \end{pmatrix}, \quad M_I^* W^E M_I. \quad (\text{RBV30})$$

The columns of M_I form the original direct conductor/native decomposition in the high-moment space, hence this Gram is positive definite. All cross terms in RBV30 are essential. This gives a proved negative class and exponential relative bound through every complete original chart. Selecting only the native block in a nonconsecutive chart is a further, different restriction: RBV27 evaluates its omitted conductor component exactly instead of declaring it zero. The companion original-image derivation identifies which selector is the specified current COB map; no root ordering alone is used here as a proof of an ordering of row subsets. For the original four-copy maps use $I_4 \otimes L_{I_0}$ or $I_4 \otimes M_I$ and the coefficient column with the displayed one-copy preimage in its first block and zero in the other three. Its reflected value and each boundary energy are exactly the one-copy values. Hence RBV22–25 have the same constants on these original four-copy images, with no additional rank factor.

22.7 The complete tensor residue, its multiplier, and the residual

We now expand the full FTR/FRT comparison on the present original simple-quartet domain. For the four original roots $z_{e\eta} = 1/2 + (2\epsilon - 1)\delta + i(2\eta - 1)\gamma$, retain every original unit value in

$$w_{e\eta} = \frac{v_h(z_{e\eta})}{h'(z_{e\eta})} \neq 0, \quad p(X, Y) = \sum_{\epsilon, \eta=0}^1 w_{e\eta} X^\epsilon Y^\eta, \quad c_{u,b}^{\text{ten}} = [X^u Y^b] p(X, Y)^a. \quad (\text{RBV31})$$

There are no omitted jets: the one-factor multiplicity is exactly one here. The original full-unit residue formula FTR.8, by taking the simple residues in every ordered tensor factor, gives

$$\mathcal{B}(f, g) = \sum_{u,b=0}^a c_{u,b}^{\text{ten}} f^{\sharp a}(\lambda_{u,b}) g(\lambda_{u,b}). \quad (\text{RBV32})$$

Indeed each ordered root tuple contributes the product of its a unchanged $w_{e\eta}$ factors; the exponents of X, Y record the two original plus counts. Expansion of p^a counts every ordered tuple once. Thus the tensor pullback multiplier in the cyclic residue form is the actual class with values $b_{\text{ten}}(\lambda_{u,b}) = \chi'(\lambda_{u,b}) c_{u,b}^{\text{ten}}$. This is the complete simple-root specialization of FTR.6–12.

Put $\mathcal{D} = \{(u, b) : c_{u,b}^{\text{ten}} \neq 0\}$ and $\mathcal{Z} = \{0, \dots, a\}^2 \setminus \mathcal{D}$, and retain the original idempotents $e_{u,b} = \chi(S)/((S - \lambda_{u,b})\chi'(\lambda_{u,b}))$. The trace multiplier and complementary form are exactly

$$a_{\text{tr}} = \sum_{(u,b) \in \mathcal{D}} (c_{u,b}^{\text{ten}})^{-1} e_{u,b}, \quad \mathcal{W}_{\mathcal{Z}}(f, g) = \sum_{(u,b) \in \mathcal{Z}} f^{\sharp a}(\lambda_{u,b}) g(\lambda_{u,b}), \quad (\text{RBV33})$$

$$b_{\text{ten}} a_{\text{tr}} = e_{\mathcal{D}} \chi' \quad \text{in } E, \quad \boxed{\mathcal{W}(f, g) = \mathcal{B}(f, a_{\text{tr}} g) + \mathcal{W}_{\mathcal{Z}}(f, g)}.$$

All equations follow by evaluation at every simple root; the evaluation map is an isomorphism, so they prove the typed algebra map, not just matching dimensions. They are exactly FRT.2–5. The trace equality can also be verified by taking the residue of $\chi' f^{\sharp a} g / \chi$ at each root. The classical multiplication trace construction is recorded by the Stacks Project Authors, Section 49.3, tags 0BVH and 0BJF [18]; the displayed root computations prove the full specialization here.

In particular the retained residual has the exact finite value

$$\mathcal{W}_{\mathcal{Z}}(P_a, P_a) = \sum_{(u,b) \in \mathcal{Z}} P_a(\lambda_{u,b})^2. \quad (\text{RBV34})$$

Conjugation of the full unit and h gives $w_{\epsilon,1-\eta} = \overline{w_{\epsilon\eta}}$; their original reflection gives $w_{1-\epsilon,\eta} = -\overline{w_{\epsilon\eta}}$. Since a is odd, the coefficient sets $\mathcal{D}, \mathcal{Z}$ are closed under both grid symmetries. Thus both partial sums of $P_a(\lambda)^2$ are real. RBV6 proves that RBV34 is nonpositive, and strictly negative whenever $\mathcal{Z}$ is nonempty. This does not determine that set by a rank assumption: its exact definition retains all cancellations in the actual coefficients RBV31.

There is also a negative pairing directly in the retained tensor residue without needing to determine its interior zero coefficients. The four edge coefficients of p^a are binomial products of nonzero original w 's, so every perimeter point belongs to $\mathcal{D}$. In particular its two complete edges $u = 0, a$ contain $2(a+1)$ points with absolute comparison value $\sinh(r_0 a)$. Applying RBV8a at these points and RBV6 at all remaining points of $\mathcal{D}$ proves

$$\boxed{\mathcal{B}(P_a, a_{\text{tr}} P_a) \leq -2(a+1)(\sinh(r_0 a) - \epsilon_a)^2 < 0.} \quad (\text{RBV35})$$

The second argument is the explicitly computed full polynomial class $a_{\text{tr}} P_a \bmod \chi$; it is not identified with a native coefficient vector unless its Laurent coordinates satisfy that specific section's membership test.

Finally the original arithmetic injection is $\mathcal{I} : E \hookrightarrow Q^{\otimes a}$ with the complete $[v_h]_h^{\otimes a}$ factor. Its inverse on its named image is the original full Mellin-jet observation, division by that invertible Taylor factor and inverse sum substitution. Thus $\mathcal{I} M_{a_{\text{tr}}} \mathcal{I}^{-1}$ is the exact arithmetic map realizing RBV33–35 on $\mathcal{I}(e_{\mathcal{D}} E)$. In RBV35 the first argument may be written $e_{\mathcal{D}} P_a$, since the complementary first-input block pairs to zero under $\mathcal{B}$ by RBV32; the separate value RBV34 remains in the full reflected trace. It commutes with the original sum action because both act by multiplication before this injection. The proper boundary image uses instead the full COB.8 map $\Psi_2 : [P] \mapsto [P(D_1 + \dots + D_a) F_h^{\otimes a}]$ on $\mathbb{C}[S]/\chi^{a+1}$, whose one-factor Taylor unit is $[v_h]_{h^2}$; these two different original target spaces are not silently identified. The ordinary class is recovered on the named boundary image by $\mathcal{I} \Delta_i^{-1} \Psi_2^{-1}$ exactly as in COB.12. This retains the full tensor-residue residual and every original unit. The finite packet trace is not substituted for the full infinite Weil distribution, and none of these values is assigned to the native minimum metric.

The tensor calculation also has a relative-metric receiver, with the residual retained at matrix level. On any of the certified images above let L denote its displayed injection and set $\Gamma_i = L^* G_i^E L$. With $W_{\mathcal{D}}^E, W_{\mathcal{Z}}^E$ the two explicitly computed reflected matrices, define

$$\begin{aligned} T_{i,\mathcal{D}} &= \Gamma_i^{-1/2} L^* W_{\mathcal{D}}^E L \Gamma_i^{-1/2}, & T_{i,\mathcal{Z}} &= \Gamma_i^{-1/2} L^* W_{\mathcal{Z}}^E L \Gamma_i^{-1/2}, \\ T_i &= T_{i,\mathcal{D}} + T_{i,\mathcal{Z}}, & \boxed{\lambda_{\min}(T_{i,\mathcal{D}}) \leq -\frac{2(a+1)(\sinh(r_0 a) - \epsilon_a)^2}{B_a} < 0.} \end{aligned} \quad (\text{RBV36})$$

Both partial matrices are Hermitian by their proved symmetry sets. Use the very same preimage of P_a as in RBV22 and the strict boundary bound RBV21 to prove the displayed inequality. Its negative magnitude again has logarithmic lower rate c_* after division by q . This is a bound for the actual tensor-residue comparison $\mathcal{B}(f, a_{\text{tr}} g)$; the other matrix remains in the exact displayed sum. No equality of the smallest eigenvalues of those three matrices is asserted.

23 Uniform recovery on the original fixed-period observation image

This calculation derives U1–U8 of the supplied *A uniform inverse bound for the original period observation*, 19 September 2026. It corrects one finite-module degree statement and makes the linear division operator and its degree-uniform bound explicit. The actual observation, Fourier frame and inherited Gram are AKS7–10 in the existing cumulative source [19]. Every period constant below is at the same fixed original nonzero admitted period and its original logarithm branch. No numerical certificate for that period is claimed.

23.1 The unchanged coefficient-dual map and its exact kernel

Retain a simple stipulated quartet

$$\frac{1}{2} \pm \delta \pm i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2,$$

the original arithmetic unit U , primary idempotent matrix $\mathcal{E}_{\text{prim}}$, and period matrix Π . The precise one-factor matrix and Fourier data are

$$F_{jt} = \zeta^{jt} - 1, \quad (F^{-1})_{tj} = \zeta^{-tj}/5, \quad \zeta = e^{2\pi i/5}, \quad H = F^{-1}\Pi U \mathcal{E}_{\text{prim}}, \quad D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4). \quad (\text{UPR1})$$

The original provider establishes invertibility of H . In $\mathcal{A} = \mathbb{C}[x_1, x_2, x_3, x_4]$, put

$$Q(x) = Q_0(H^{-1}x), \quad Q_0(t) = t_1 t_4 - t_2 t_3.$$

For degree $k \geq 0$ let $\mathcal{B}_k = \text{span}\{Z^a W^b : 0 \leq a, b \leq k\}$. The original coefficient-dual map is

$$\Theta_k : \mathcal{A}_k \longrightarrow \mathcal{B}_k, \quad f(x) \longmapsto f(H(ZW, Z, W, 1)^\top). \quad (\text{UPR2})$$

The arithmetic receiver below uses the original subsequence $k \equiv 1 \pmod{4}$, $k \geq 5$; these finite polynomial constructions also define the intermediate graded maps at every degree.

Here is an exact proof of its kernel and an explicit full lift. For each pair $0 \leq a, b \leq k$, put $n = \max(0, a + b - k)$, and define

$$\ell_{a,b}(t) = t_1^n t_2^{a-n} t_3^{b-n} t_4^{k-a-b+n}. \quad (\text{UPR3})$$

All exponents are nonnegative, their sum is k , and substitution $t = (ZW, Z, W, 1)$ gives $Z^a W^b$. Every homogeneous monomial in the t_i can be reduced by $t_1 t_4 = t_2 t_3$ until either its first or its fourth exponent is zero. The two exponents $a = n_1 + n_2$, $b = n_1 + n_3$, and its total degree k then determine all four exponents uniquely, yielding (UPR3). Thus the remaining monomials have linearly independent images, and

$$\ker \Theta_k = Q \mathcal{A}_{k-2}, \quad \Theta_k \mathcal{A}_k = \mathcal{B}_k, \quad (\text{UPR4})$$

where a negative homogeneous degree means zero. This argument also proves the complete relation ideal; it is not a dimension-only identification.

Define the linear lift on the displayed biform frame by $L_k(Z^a W^b) = \ell_{a,b}(H^{-1}x)$. For the ordinary coefficient ℓ^1 norm put

$$h_- = \max \left(1, \max_i \sum_j |(H^{-1})_{ij}| \right).$$

The sum of absolute coefficients of a product is at most the product of those sums. Apply this to the k linear factors of (UPR3), and then to a linear combination of biform monomials:

$$\Theta_k L_k = I_{\mathcal{B}_k}, \quad \|L_k f\|_1 \leq h_-^k \|f\|_1. \quad (\text{UPR5})$$

The full arithmetic unit and period remain in H and h_- .

23.2 A fixed homogeneous module with the corrected degree bound

Put

$$\mathcal{S} = \mathbb{C}[X_1, X_2, X_3, X_4], \quad X_i = x_i^5, \quad \mathcal{R} = \{0, 1, 2, 3, 4\}^4.$$

The exact decomposition

$$\mathcal{A} = \bigoplus_{r \in \mathcal{R}} \mathcal{S} x^r \quad (\text{UPR6})$$

follows by Euclidean division of each monomial exponent by five. The invariant submodule for the original D -action is

$$\mathcal{A}^D = \bigoplus_{r \in \mathcal{R}_0} \mathcal{S}x^r, \quad \mathcal{R}_0 = \{r : \sum_{i=1}^4 ir_i = 0 \bmod 5\}. \quad (\text{UPR7})$$

There are 625 residue vectors and 125 invariant residue vectors: the fourth residue is uniquely determined by the first three in the displayed congruence.

Use the original grading $\deg X_i = 5$ and $\deg x^r = |r|$. For the second source summand below the residue basis has the additional degree-two shift:

$$\mathcal{M} : \left(\bigoplus_{r \in \mathcal{R}_0} \mathcal{S}x^r \right) \oplus \left(\bigoplus_{r \in \mathcal{R}} \mathcal{S}x^r[2] \right) \longrightarrow \bigoplus_{r \in \mathcal{R}} \mathcal{S}x^r, \quad (f, g) \longmapsto f + Qg. \quad (\text{UPR8})$$

It is an exact degree-preserving map, and

$$\text{im } \mathcal{M} = \mathcal{A}^D + Q\mathcal{A}. \quad (\text{UPR9})$$

The entries of the multiplication-by- Q block have total X -degree at most *two*, rather than the at-most-one assertion in the incoming U2. Indeed multiplication of x^r by a quadratic monomial can cause at most two carries in division of exponents by five. For distinct i, j , the term $x_i x_j$ and a residue with $r_i = r_j = 4$ produce the entry $X_i X_j$. The actual grading and the fact that this is one fixed finite polynomial matrix are unchanged by this correction.

23.3 Finite basis, homogeneous lifts and a specified linear division

We use Buchberger's polynomial-basis method [20], with its finiteness supplied by Dickson's lemma [21]. The needed module argument and norm estimates are supplied here.

Enumerate the 625 target residues once in lexicographic order, with integer position $b_r \in \{0, \dots, 624\}$. Order target monomials $X^\alpha x^r$ lexicographically by $(\alpha_1, \alpha_2, \alpha_3, \alpha_4, b_r)$. This is a well-order compatible with multiplication by every X^β .

Start with the nonzero columns of (UPR8), retaining their corresponding source coordinate vectors. Make each column monic by division by its actual nonzero leading coefficient. For pairs whose leading target position agrees, form the usual least-common-monomial cancellation, reduce it using the current monic list, and append each nonzero remainder with its simultaneously updated source lift. Order the finite pair lists lexicographically to specify the choices. A reduction of a homogeneous vector by a homogeneous column retains its full shifted degree, because the cancelled monomials occupy the same target position. Every resulting column g_j and retained lift h_j are therefore homogeneous and

$$\mathcal{M}h_j = g_j. \quad (\text{UPR10})$$

For completeness, an infinite sequence in $\mathbb{N}^d$ has an earlier term componentwise at most a later one. For $d = 1$, either some value occurs infinitely often, or an infinite strictly increasing subsequence can be selected. In either case there is an infinite nondecreasing subsequence. Apply this to the first coordinate, and apply the induction hypothesis to the other coordinates of that subsequence. This proves the assertion for every finite d , including $d = 4$. An infinite sequence of appended leading monomials would contain infinitely many in one target position and then an earlier one dividing a later one. That contradicts the complete remainder reduction before appending. Hence only finitely many columns are appended, and the pair algorithm terminates.

The output is a module Gröbner basis. To verify this criterion directly, take a nonzero element of $\text{im } \mathcal{M}$ and a representation by monomial multiples of the output columns whose largest occurring leading monomial is least possible; among these, minimize the number of terms at that largest monomial. If it exceeds the leading monomial of the represented vector, at least two terms at the same target position cancel. Their least-common-monomial combination is a multiple of one of

the processed pairs. That pair has a representation with all occurring leading monomials strictly smaller than its cancelled monomial, because its remainder is zero. Substitute this representation and combine the remaining terms at the largest monomial. This decreases their number, or the largest monomial itself, a contradiction. Thus some leading basis monomial divides the leading monomial of the given element. Division consequently has zero remainder exactly on the image module. A reduced basis may be obtained by discarding divisible leading monomials and reducing the tails, updating their lifts throughout; reducedness is not needed in the subsequent estimate.

We now specify positive integer scores that strictly decrease in every rewrite. Let D_* be the maximum of all absolute exponent differences, coordinate by coordinate, between a leading term and a tail term in the finite list g_j ; set $D_* = 0$ when all tails are empty. Set

$$T_* = 2(D_* + 625) + 2, \quad w_i = T_*^{5-i} \quad (1 \leq i \leq 4), \quad W_* = \max_i w_i, \quad B_* = 624. \quad (\text{UPR11})$$

Give $X^\alpha x^r$ score $\text{sc}(\alpha, r) = \sum_i w_i \alpha_i + b_r$. If a leading and tail term first differ at variable index i , their score difference is at least

$$T_*^{5-i} - D_* \sum_{j>i} T_*^{5-j} - 624 > 0.$$

The last inequality follows from $D_*/(T_* - 1) < 1/2$ and $T_*/2 > 624$. If all four exponents agree, the leading position has strictly larger b_r . Thus every such difference is an integer at least one. Multiplication by a monomial shifts both scores equally. All scores are nonnegative.

Set the following fixed constants from the finite basis and its actual lifts:

$$C_* = \max \left(2, 1 + \max_j \|\text{tail}(g_j)\|_1 \right), \quad H_* = \max \left(1, \max_j \|h_j\|_1 \right). \quad (\text{UPR12})$$

Here and below coefficient norms in the free module use its displayed residue basis; under (UPR6) these are precisely the ordinary polynomial coefficient norms.

Define a linear division operator by a recursion on individual monomials, rather than by choices depending on a whole input polynomial. If no leading g_j divides a target monomial t , set $\mathcal{R}(t) = 0$, $\text{NF}(t) = t$. Otherwise use the first such g_j , put $m = t / \text{LM } g_j$, and define

$$\begin{aligned} \mathcal{R}(t) &= mh_j - \mathcal{R}(m \text{ tail } g_j), \\ \text{NF}(t) &= -\text{NF}(m \text{ tail } g_j). \end{aligned} \quad (\text{UPR13})$$

Extend each right side linearly over its finitely many monomials. All recursive arguments have lower score, so this is a finite definition. Finally extend to arbitrary polynomials by their monomial expansion. The resulting maps are linear, and induction in the recursion proves

$$f = \mathcal{M}\mathcal{R}(f) + \text{NF}(f). \quad (\text{UPR14})$$

The normal form has only irreducible monomials; if f is in the image module, (UPR14) and the just-proved leading-monomial criterion imply that it is zero. No choice of a degree-dependent minor is involved.

If t has score s , another induction proves

$$\|\mathcal{R}(t)\|_1 \leq H_* C_*^{s+1}. \quad (\text{UPR15})$$

Indeed the first lift has norm at most H_* and the total absolute coefficient mass of its tails is at most $C_* - 1$. Their scores are at most $s - 1$, so the induction gives the bound

$$H_* + (C_* - 1)H_* C_*^s \leq H_* C_*^{s+1}.$$

At score zero a reducible term has no tail, and the same inequality holds directly. Monomial multiplication does not change coefficient norms. Homogeneity of (UPR10) shows that this recursion preserves the original shifted degree throughout.

On degree k , $5|\alpha| + |r| = k$, so the score is at most $W_*k/5 + B_*$. By linearity, for every degree- k element of the actual image module,

$$\mathcal{MR}(f) = f, \quad \|\mathcal{R}(f)\|_1 \leq H_* C_*^{W_*k/5+B_*+1} \|f\|_1. \quad (\text{UPR16})$$

These constants are defined by one finite calculation over the fixed field of actual coefficients. The proof establishes their finiteness and the uniform bound. It does not claim that this basis has been numerically computed for an unspecified hypothetical quartet or that exact zero tests on arbitrary supplied complex numbers are available to a numerical implementation.

23.4 A linear right inverse on the original invariant image

Let $f \in \Theta_k(\mathcal{A}_k^D)$. Its fixed lift $L_k f$ differs from an invariant lift by a multiple of Q , by (UPR4). Thus $L_k f \in \text{im } \mathcal{M}$. Write $\mathcal{R}(L_k f) = (p, g)$ in (UPR8). Then

$$p \in \mathcal{A}_k^D, \quad L_k f = p + Qg, \quad \Theta_k p = f.$$

The map $f \mapsto p$ is linear by (UPR5) and (UPR13). Since $\dim \mathcal{B}_k = (k+1)^2$,

$$\|p\|_2 \leq \|p\|_1 \leq \mathcal{R}_k \|f\|_2, \quad \mathcal{R}_k = H_* C_*^{W_*k/5+B_*+1} h_-^k (k+1). \quad (\text{UPR17})$$

In particular $\log \mathcal{R}_k = O_H(k + \log(k+1))$. Here k is the original tensor degree; the coefficient module and all its constants are fixed.

To compare with the actual matrix, expand $\prod_t (\sum_\alpha H_{t\alpha} t_\alpha)^{\nu_t}$ for each invariant ν . The coefficient with plus-real count a , plus-imaginary count b , is

$$B_{\nu; (a,b)} = \sum_{\substack{\sum_\alpha n_{t\alpha} = \nu_t \\ \text{original plus counts } a,b}} \left(\prod_t \frac{\nu_t!}{\prod_\alpha n_{t\alpha}!} \right) \prod_{t,\alpha} H_{t\alpha}^{n_{t\alpha}}, \quad (\text{UPR18})$$

exactly AKS8. Thus the matrix of the invariant restriction of (UPR2) is $B_k^\top$ in these unchanged coefficient frames. For any map T , its least-norm preimage has norm at most the preimage supplied by a right inverse. Equation (UPR17) therefore bounds the Moore–Penrose inverse of $B_k^\top$ by $\mathcal{R}_k$. A singular-value factorization $B_k = U\Sigma V^*$ gives $B_k^\top = \bar{V}\Sigma(\bar{U})^*$, with exactly the same nonzero singular values. Consequently

$$\|(B_k)^+\| \leq \mathcal{R}_k. \quad (\text{UPR19})$$

Zero singular values are kept as the actual observation kernel. If its whole image is zero, the pseudoinverse is zero and the same assertion holds.

For comparison the forward norm is also degree-exponential. With $h_+ = \max(1, \max_i \sum_j |H_{ij}|)$, expansion of the same linear factors has coefficient ℓ^1 norm at most h_+^k . Substitution $t = (ZW, Z, W, 1)$ only combines coefficients and cannot increase that norm. It follows that

$$\|B_k\| = \|B_k^\top\| \leq h_+^k \sqrt{\binom{k+3}{3}}. \quad (\text{UPR20})$$

Hence the condition number on its nonzero coefficient singular values is $\exp(O_H(k + \log(k+1)))$.

23.5 The exact original word Gram, with each multiplicity once

Direct summation of fifth roots of unity gives

$$(F^* F)_{tu} = \sum_{j=1}^4 (\zeta^{-jt} - 1)(\zeta^{ju} - 1) = 5(\delta_{tu} + 1). \quad (\text{UPR21})$$

Thus $F^*F = 5(I + \mathbf{1}\mathbf{1}^*)$, whose eigenvalues are five with multiplicity three and twenty-five with multiplicity one.

For $|\nu| = k$, let e_ν be the sum of all distinct ordered standard tensor words with multiplicity vector ν . These are the unscaled symmetric-word vectors of the original provider. Different e_ν are orthogonal and

$$\|e_\nu\|^2 = k!/\nu!, \quad 1 \leq k!/\nu! \leq 4^k. \quad (\text{UPR22})$$

The upper bound follows because the sum of all these multinomial integers is 4^k . If $\mathcal{F}_0$ has columns $F^{\otimes k}e_\nu$ for the original invariant index set, then its inherited Gram entries are

$$(\mathcal{F}_0^* \mathcal{F}_0)_{\nu\mu} = \sum_{\substack{\sum_u n_{tu} = \nu_t \\ \sum_t n_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} n_{tu}!} \prod_{t,u} [5(\delta_{tu} + 1)]^{n_{tu}}. \quad (\text{UPR23})$$

Indeed expansion of $(\bar{x}^\top F^* F y)^k$ gives the displayed coefficients; equivalently expand both ordered tensor-word sums. This is the exact AKS10 formula, including every row cross term.

The squared singular values of $F^{\otimes k}$ lie between 5^k and 25^k , since tensor eigenvalues multiply. For an arbitrary coefficient column $a = (a_\nu)$, (UPR22) gives

$$\sum_\nu |a_\nu|^2 \leq \left\| \sum_\nu a_\nu e_\nu \right\|^2 \leq 4^k \sum_\nu |a_\nu|^2.$$

Apply the tensor singular-value bounds to this same vector, restricted to the actual invariant indices:

$$\boxed{5^k I \leq \mathcal{G}_0 := \mathcal{F}_0^* \mathcal{F}_0 \leq 100^k I.} \quad (\text{UPR24})$$

The row multiplicities in (UPR18) and the word Gram (UPR23) have different exact roles. No additional multinomial is inserted into either of them.

The original AKS9 map in root-value coordinates is

$$j\Lambda_k v = \mathcal{F}_0 B_k v. \quad (\text{UPR25})$$

The map $\mathcal{F}_0$ is injective, with $\|\mathcal{F}_0^+\| \leq 5^{-k/2}$ by (UPR24). For y in the actual image of (UPR25), define

$$T_k y = B_k^+ \mathcal{F}_0^+ y.$$

Then $B_k T_k y = \mathcal{F}_0^+ y$ and $\mathcal{F}_0 B_k T_k y = y$, because the latter vector lies in $\text{im } B_k$. Furthermore

$$\|T_k y\|_2 \leq 5^{-k/2} \mathcal{R}_k \|y\|_{\text{original output}}. \quad (\text{UPR26})$$

This is a specified linear right inverse on the image. Its values may be passed to the quotient by $\ker B_k$; it does not recover components in that kernel.

23.6 Original arithmetic quotient metric at every cutoff

Now keep the original grid

$$q = (k+1)^2, \quad c = k/2, \quad \lambda_{a,b} = c + (2a-k)\delta + i(2b-k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{a,b}),$$

and let $R = k\sqrt{\delta^2 + \gamma^2}$, $d_* = 2\min(\delta, \gamma) > 0$. Distinct roots have separation at least d_* , because at least one of their two grid-coordinate differences is a nonzero integer. For an actual root-value column v , the degree- $(q-1)$ Lagrange representative is

$$P_v(S) = \sum_\lambda v_\lambda \frac{\chi(S)}{(S - \lambda)\chi'(\lambda)}. \quad (\text{UPR27})$$

It has exactly those root values. For $S = c + iy$, each numerator factor has modulus at most $|y| + R$, whereas $|\chi'(\lambda)| \geq d_*^{q-1}$. Cauchy–Schwarz in the actual q -term sum proves

$$|P_v(c + iy)|^2 \leq q d_*^{-2q+2} (|y| + R)^{2q-2} \|v\|_2^2. \quad (\text{UPR28})$$

Retain the arithmetic source

$$w_h(y) = |2\xi(1/2 + iy)/h(1/2 + iy)|^2 / (2\pi), \quad m_k = w_h^{*k}, \quad M_h(\eta) = \int_{\mathbb{R}} e^{\eta y} w_h(y) dy, \quad \eta = \pi/4.$$

The inherited exponential envelope gives a finite positive $M_h(\eta)$; no probability rescaling is made. The nonnegative exponential series proves

$$t^n \leq n! \eta^{-n} e^{\eta t} \quad (t \geq 0, n \in \mathbb{N}).$$

Tonelli’s theorem in the original convolution coordinates gives $\int e^{\eta y} m_k(y) dy = M_h(\eta)^k$. Evenness of the unchanged density yields

$$\int_{\mathbb{R}} e^{\eta|y|} m_k(y) dy = 2 \int_0^\infty e^{\eta y} m_k(y) dy \leq 2M_h(\eta)^k.$$

Apply these facts to (UPR28), including the factor $e^{\eta R}$:

$$\|P_v\|_{m_k}^2 \leq \mathcal{B}_{h,k} \|v\|_2^2, \quad \mathcal{B}_{h,k} = 2q(2q-2)! (\eta d_*)^{-2q+2} e^{\eta R} M_h(\eta)^k. \quad (\text{UPR29})$$

Both half-lines, every root and every original source mass remain.

For each original cutoff $N \geq q-1$, the minimum metric G_N on the class of P_v is bounded above by the norm of this particular admitted representative. Composing (UPR26) with (UPR27), and then with the actual class map, gives a linear right inverse $\mathcal{T}_{k,N}$ on the original observation image:

$$\boxed{\|\mathcal{T}_{k,N}\|_{\text{original output} \rightarrow G_N} \leq \sqrt{\mathcal{B}_{h,k}} 5^{-k/2} \mathcal{R}_k.} \quad (\text{UPR30})$$

Passing its value further to the original quotient by $\ker \Lambda_k$ can only decrease the quotient norm, so the same bound applies to that receiver. Since $\log(2q-2)! \leq (2q-2) \log(2q-2)$, $R = O_h(k)$, and (UPR17) has logarithm $O_{h,u}(k + \log(k+1))$,

$$\log \|\mathcal{T}_{k,N}\| \leq O_{h,u}(q \log(q+2)), \quad (\text{UPR31})$$

uniformly over every such N . All constants refer to the original fixed quartet, unit, period and branch. The construction is a large, finite recovery allowance on the actual image. It supplies no small native angular estimate, does not remove the observation kernel, and does not identify the arithmetic quotient metric with the independent proper-source metric.

The original Montgomery finite-part identity

The original Six Readers files are preserved unchanged. This note completes the scalar identity left open after `eq:poisson-pole-complete` in `proofs/B_guinand_weil_montgomery_interfaces.tex`, and in the Montgomery paragraph of `satellites/10_literature_foundations.tex`. It independently derives the new incoming note `03_MONTGOMERY_ARCHIMEDEAN_IDENTITY.md` from the literal test and finite-part convention in those complete sources. The distinct false local Guinand–Weil residual equality is corrected in the companion derivation; it is not used anywhere in this proof.

24 The original parameters, transform and domain

Throughout retain

$$\begin{aligned} 1 < \sigma_0 < 2, \quad t \in \mathbb{R}, \quad x \geq 1, \quad U = \log x, \quad a = \sigma_0 - \frac{1}{2}, \\ F(v) = e^{-a|v-U|} e^{-it(v-U)}, \quad C = F(0) = x^{1/2-\sigma_0+it}, \\ z = 1 - \sigma_0 + it, \quad w = \sigma_0 - it = 1 - z, \quad p = \sigma_0 + it, \quad A = (2\pi)^{-1}. \end{aligned} \quad (\text{MA1})$$

For every positive base, a complex power means $y^s = \exp(s \log y)$ with the real natural logarithm. The digamma function $\psi = \Gamma'/\Gamma$ is meromorphic, not a chosen branch of $\log \Gamma$. Weil’s real-place kernel and the retained finite part [32] are

$$\begin{aligned} K(v) &= \frac{e^{|v|/2}}{|e^v - e^{-v}|}, \\ \text{PF}(FK) &= \lim_{\Lambda \rightarrow \infty} \left\{ \int_{\mathbb{R}} (1 - e^{-\Lambda|v|}) F(v) K(v) dv - C \log \Lambda \right\}. \end{aligned} \quad (\text{MA2})$$

Indeed $|v|K(v) \rightarrow 1/2$ and $F(v) \rightarrow C$ as $v \rightarrow 0$, so the original subtraction $2\beta(0) \log \Lambda$ has $\beta(0) = C/2$. In particular its coefficient is C , not $2C$.

The test F is continuous. On the two intervals separated by U its derivative is $(a - it)F$ and $(-a - it)F$, respectively. The average derivative value at the single jump is $-it$, since $F(U) = 1$. These are exactly the first-kind jump data of Weil’s test convention. For every $0 < b < \sigma_0 - 1$,

$$|F(v)| + |F'(v)| \leq (1 + a + |t|) e^{aU} e^{-a|v|} = O(e^{-(1/2+b)|v|}).$$

Thus both literal conditions (A) and (B) of the recorded Weil source hold, including when $U = 0$. For $1 - \sigma_0 < \Re s < \sigma_0$, absolute convergence permits $v = U + u$:

$$\begin{aligned} \Phi(s) &:= \int_{\mathbb{R}} F(v) e^{(s-1/2)v} dv \\ &= x^{s-1/2} \left[\int_0^\infty e^{-[a-(s-1/2-it)]u} du + \int_0^\infty e^{-[a+(s-1/2-it)]u} du \right] \\ &= \frac{2ax^{s-1/2}}{a^2 - (s-1/2-it)^2}. \end{aligned} \quad (\text{MA3})$$

The strip contains $s = 0, 1$ and the entire closed critical strip. Factoring the two original denominators yields

$$\Phi(0) = -\frac{2ax^{-1/2}}{zp}, \quad \Phi(1) = \frac{2ax^{1/2}}{(\sigma_0 - 1 + it)(\sigma_0 - it)}. \quad (\text{MA4})$$

There is no zero denominator on (MA1): $-1 < \Re z < 0$, $1 < \Re w$, $\Re p < 2$.

25 The finite part as an ordinary, convergent regularized integral

On $v > 0$ write

$$J(v) = [F(v) + F(-v)]K(v).$$

The exact kernel is $K(v) = e^{-v/2}/(1 - e^{-2v})$. It is $1/(2v) + O(1)$ near zero, while $F(v) + F(-v) = 2C + O(v)$ by local Lipschitz continuity, including at $U = 0$. Consequently $J(v) = C/v + O(1)$. For $v \geq \max(1, U)$, the test has decay $O_{\sigma_0, t, x}(e^{-av})$ and $K(v) = O(e^{-v/2})$, giving $J(v) = O_{\sigma_0, t, x}(e^{-\sigma_0 v})$. Therefore the function

$$H(v) = J(v) - C \frac{e^{-v}}{v} \quad (\text{MA5})$$

is absolutely integrable on $(0, \infty)$. For finite $\Lambda > 0$, elementary differentiation in Λ and the value zero at $\Lambda = 0$ prove

$$\int_0^\infty \frac{e^{-v} - e^{-(1+\Lambda)v}}{v} dv = \log(1 + \Lambda).$$

It follows exactly that

$$\int_{\mathbb{R}} (1 - e^{-\Lambda|v|}) FK dv - C \log \Lambda = \int_0^\infty (1 - e^{-\Lambda v}) H(v) dv + C \log(1 + \Lambda^{-1}). \quad (\text{MA6})$$

Dominated convergence with the proved L^1 function H now gives

$$\boxed{\text{PF}(FK) = \int_0^\infty \left\{ [F(v) + F(-v)]K(v) - C \frac{e^{-v}}{v} \right\} dv.} \quad (\text{MA7})$$

This is an exact linear relation between the original finite-part functional and an ordinary regularized integral. The subtracted function and its constant have both been retained. The same estimates are uniform when (σ_0, t, x) varies over compact subsets of the domain (MA1), so this functional is continuous there, including the relative endpoint $x = 1$.

26 Direct integration on the two original intervals

The positive kernel has the geometric expansion

$$K(v) = \sum_{m=0}^{\infty} e^{-(2m+1/2)v}, \quad v > 0. \quad (\text{MA8})$$

For a fixed $m \geq 0$ put $\lambda_m = 2m + 1/2$. The actual test on the two intervals is

$$\begin{aligned} F(v) &= Ce^{(a-it)v} & (0 \leq v \leq U), \\ F(v) &= Cx^{2a}e^{-(a+it)v} & (v \geq U), \\ F(-v) &= Ce^{-(a-it)v} & (v \geq 0). \end{aligned} \quad (\text{MA9})$$

In particular the first interval is empty when $x = 1$, and remains finite when $m = 0$ and $\Re z < 0$. Direct integration gives

$$\begin{aligned} I_m &:= \int_0^\infty [F(v) + F(-v)]e^{-\lambda_m v} dv \\ &= C \frac{1 - e^{-(2m+z)U}}{2m+z} + Cx^{2a} \frac{e^{-(2m+p)U}}{2m+p} + \frac{C}{2m+w} \\ &= C \left(\frac{1}{2m+w} + \frac{1}{2m+z} \right) - \frac{2ax^{-2m-1/2}}{(2m+z)(2m+p)}. \end{aligned} \quad (\text{MA10})$$

Here $Cx^{-(2m+z)} = Cx^{2a-(2m+p)} = x^{-2m-1/2}$ and $p - z = 2a$. No integral of $e^{-(2m+z)v}$ over the infinite half-line was used.

For finite $\Lambda > 2$ all integrations and (MA8) may be exchanged in the regulated integral. To prove this for complex F , apply Tonelli to the absolute values: the dominating function is $(1 - e^{-\Lambda v})(|F(v)| + |F(-v)|)K(v)$, bounded near zero and exponentially integrable at infinity by the estimates preceding (MA5). The same integration as (MA10), with every $2m$ replaced by $2m + \Lambda$, gives

$$\begin{aligned} \int_{\mathbb{R}} (1 - e^{-\Lambda|v|})FK dv &= C \sum_{m \geq 0} \left[\frac{1}{2m+w} - \frac{1}{2m+w+\Lambda} + \frac{1}{2m+z} - \frac{1}{2m+z+\Lambda} \right] \\ &\quad - 2ax^{-1/2} \sum_{m \geq 0} \frac{x^{-2m}}{(2m+z)(2m+p)} \\ &\quad + 2ax^{-1/2-\Lambda} \sum_{m \geq 0} \frac{x^{-2m}}{(2m+z+\Lambda)(2m+p+\Lambda)}. \end{aligned} \quad (\text{MA11})$$

Each bracketed harmonic difference is $O_{\Lambda, \sigma_0, t}(m^{-2})$; the two rational series are separately absolutely convergent. Thus (MA11) splits only convergent series, not divergent harmonic sums.

27 The exact digamma difference and its complex limit

Euler's locally uniformly convergent product for $1/\Gamma$, as recorded in DLMF 5.8.2 [30], gives by local logarithmic differentiation

$$\psi(b) - \psi(a_0) = \sum_{m=0}^{\infty} \left(\frac{1}{m+a_0} - \frac{1}{m+b} \right), \quad (\text{MA12})$$

for a_0, b away from the nonpositive integers. The differentiated differences converge locally uniformly since they are $O(m^{-2})$. One can first differentiate on the positive real half-plane and then continue meromorphically; this yields the same expression without imposing a logarithm branch at $z/2$, whose real part lies in $(-1/2, 0)$. Applying (MA12) to (MA11) evaluates its first line exactly as

$$\frac{C}{2} [\psi((w+\Lambda)/2) - \psi(w/2) + \psi((z+\Lambda)/2) - \psi(z/2)]. \quad (\text{MA13})$$

For clarity, the limiting formula needed here follows directly from the same product. For real $r > 0$ it gives $\psi(r) = \lim_{N \rightarrow \infty} [\log N - \sum_{m=0}^N (m+r)^{-1}]$. The decreasing-integrand sum comparison gives

$$0 \leq \sum_{m=0}^N \frac{1}{m+r} - \int_0^{N+1} \frac{du}{u+r} \leq \frac{1}{r}, \quad |\psi(r) - \log r| \leq \frac{1}{r}.$$

For $r \geq \max(1, 2|c_0|)$, (MA12) also gives

$$|\psi(r+c_0) - \psi(r)| \leq 2|c_0| \sum_{m \geq 0} (m+r)^{-2} \leq \frac{4|c_0|}{r}.$$

Taking $r = \Lambda/2$, $c_0 = s/2$ proves the complex estimate

$$|\psi((\Lambda+s)/2) - \log \Lambda + \log 2| \leq \frac{2+4|s|}{\Lambda}, \quad \Lambda \geq \max(2, 2|s|). \quad (\text{MA14})$$

Thus no passage to real parts of the complex identity is involved. The shifted rational series in (MA11) satisfies, for $\Lambda > 2$,

$$\left| \sum_{m \geq 0} \frac{x^{-2m}}{(2m+z+\Lambda)(2m+p+\Lambda)} \right| \leq \frac{1}{(\Lambda-1)^2} + \frac{1}{2(\Lambda-1)}. \quad (\text{MA15})$$

Indeed the real parts of the two denominator factors exceed $2m + \Lambda - 1$, and the last bound is the sum-plus-integral bound for $(2u + \Lambda - 1)^{-2}$. Multiplication by $x^{-\Lambda}$ makes the error smaller when $x > 1$; (MA15) also proves its convergence to zero directly at $x = 1$.

Subtracting $C \log \Lambda$ in (MA11), then using (MA13)–(MA15), proves

$$\boxed{\text{PF}(FK) = -C \log 2 - \frac{C}{2} [\psi(w/2) + \psi(z/2)] - 2ax^{-1/2} \sum_{m=0}^{\infty} \frac{x^{-2m}}{(2m+z)(2m+p)}}. \quad (\text{MA16})$$

More quantitatively, for $\Lambda \geq \max(3, 2|w|, 2|z|)$ the difference between the regulated finite part in (MA2) and (MA16) is bounded by

$$\frac{|C|[2 + 2(|w| + |z|)]}{\Lambda} + 2ax^{-1/2-\Lambda} \left[\frac{1}{(\Lambda-1)^2} + \frac{1}{2(\Lambda-1)} \right]. \quad (\text{MA17})$$

This controls the actual regulator with its original subtraction. For $M \geq 1$, the tail of the unshifted rational series after $m = M$ satisfies

$$\left| \sum_{m>M} \frac{x^{-2m}}{(2m+z)(2m+p)} \right| \leq \frac{1}{2M}. \quad (\text{MA18})$$

For $x > 1$ it also satisfies the stronger bound $x^{-2M-2}/[2(M+1)^2(1-x^{-2})]$. To verify this, use $|2m+z| > 2m-1 \geq m$ and $|2m+p| > 2m$ for $m \geq 1$, then the integral or geometric bound. The separate $m = 0$ term is finite since $z, p \neq 0$.

28 Both prime orientations and the exact reflection identity

Let $\Lambda(n)$ be the von Mangoldt function, with $\Lambda(1) = 0$. The two orientations are, directly from (MA9),

$$n^{-1/2}F(\log n) = \begin{cases} Cn^{\sigma_0-1-it}, & n \leq x, \\ x^{\sigma_0-1/2+it}n^{-\sigma_0-it}, & n > x, \end{cases} \quad (\text{MA19})$$

$$n^{-1/2}F(-\log n) = Cn^{-w}.$$

At $n = x$ the two displayed positive-orientation expressions both equal $n^{-1/2}$. Thus the split does not create a jump or a half-weight at a prime power. At $x = 1$ the first sum contains only $n = 1$ and is zero. Absolute convergence of the infinite prime sums follows from $0 \leq \Lambda(n) \leq \log n$ and $\sigma_0 > 1$.

For completeness, the Euler product for $\zeta(s)$ on $\Re s > 1$ [31] is nonzero and has the locally absolutely convergent logarithm $\sum_{\ell \text{ prime}} \sum_{r \geq 1} \ell^{-rs}/r$. Absolute convergence follows from $\sum_{\ell} \sum_{r \geq 1} \ell^{-r\sigma_0}/r < \infty$; local termwise differentiation is permitted by the corresponding convergent $\log n n^{-\sigma_0}$ bound. Unique prime-power factorization therefore gives the exact signs

$$P_- := \sum_{n \geq 1} \Lambda(n) n^{-1/2} F(-\log n) = C \sum_{n \geq 1} \Lambda(n) n^{-w} = -C \frac{\zeta'(w)}{\zeta(w)}, \quad -P_- = +C \frac{\zeta'(w)}{\zeta(w)}. \quad (\text{MA20})$$

For example, for any integer N beyond e^{1/σ_0} , the absolute tail of the Dirichlet sum after N is at most

$$\int_N^{\infty} (\log u) u^{-\sigma_0} du = N^{1-\sigma_0} \left[\frac{\log N}{\sigma_0 - 1} + \frac{1}{(\sigma_0 - 1)^2} \right].$$

Its outside factor $|C|$ is retained for P_- .

Put $\mathcal{L}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$. The exact zeta functional equation is $\mathcal{L}(s) = \mathcal{L}(1-s)$, equivalently the ξ reflection with its $s(s-1)/2$ factor [31]. At w the Euler product and the nonvanishing

Gamma factor give $\mathcal{L}(w) \neq 0$. At $z = 1 - w$, which has $-1 < \Re z < 0$, $\Gamma(z/2)$ is finite and nonzero, so reflection also proves $\zeta(z) \neq 0$. Thus every logarithmic derivative below exists. Differentiating the equality as a meromorphic equality and dividing by its nonzero value gives

$$\begin{aligned} -\frac{1}{2} \log \pi + \frac{1}{2} \psi(z/2) + \frac{\zeta'(z)}{\zeta(z)} &= - \left[-\frac{1}{2} \log \pi + \frac{1}{2} \psi(w/2) + \frac{\zeta'(w)}{\zeta(w)} \right], \\ \frac{\zeta'(w)}{\zeta(w)} - \log \pi + \frac{1}{2} [\psi(w/2) + \psi(z/2)] &= - \frac{\zeta'(z)}{\zeta(z)}. \end{aligned} \quad (\text{MA21})$$

This derivation uses ratios, not an unproved global logarithm branch.

29 Pole cancellation and the completed identity

Substitute (MA16) and (MA20) into the literal residual expression. Its constant is $\log A = -\log(2\pi)$, so $-\log(2\pi) + \log 2 = -\log \pi$. Using (MA21) only after that cancellation yields

$$\begin{aligned} &\Phi(0) + C \log A - \text{PF}(FK) - P_- \\ &= \Phi(0) + C \left\{ \frac{\zeta'(w)}{\zeta(w)} - \log \pi + \frac{\psi(w/2) + \psi(z/2)}{2} \right\} + 2ax^{-1/2} \sum_{m \geq 0} \frac{x^{-2m}}{(2m+z)(2m+p)} \\ &= -C \frac{\zeta'(z)}{\zeta(z)} + \underbrace{\Phi(0) + \frac{2ax^{-1/2}}{zp}}_{=0 \text{ by (MA4)}} + 2ax^{-1/2} \sum_{m \geq 1} \frac{x^{-2m}}{(2m+z)(2m+p)}. \end{aligned} \quad (\text{MA22})$$

The two denominator presentations are exactly related by

$$2m + z = -(\sigma_0 - 1 - it - 2m), \quad 2m + p = \sigma_0 + it + 2m.$$

Therefore (MA22) is precisely the required complete scalar identity

Theorem 29.1. *For every original parameter tuple (MA1), with the literal finite part (MA2), all the terms below exist and*

$$\begin{aligned} &\Phi(0) + C \log((2\pi)^{-1}) - \text{PF}(FK) - \sum_{n \geq 1} \Lambda(n) n^{-1/2} F(-\log n) \\ &= -C \frac{\zeta'(1 - \sigma_0 + it)}{\zeta(1 - \sigma_0 + it)} - x^{-1/2} \sum_{m \geq 1} \frac{(2\sigma_0 - 1)x^{-2m}}{(\sigma_0 - 1 - it - 2m)(\sigma_0 + it + 2m)}. \end{aligned} \quad (\text{MA23})$$

The proof has not used a zero sum, a zero-location hypothesis, or the incorrect printed Guinand–Weil residual equality.

30 The endpoint and the receiving Montgomery formula

At $x = 1$, the actual test is $F(v) = e^{-a|v|} e^{-itv}$ and $C = 1$. The direct two-interval integration was already valid with $U = 0$. Independently, (MA12) and $p - z = 2a$ give

$$2a \sum_{m \geq 0} \frac{1}{(2m+z)(2m+p)} = \sum_{m \geq 0} \left(\frac{1}{2m+z} - \frac{1}{2m+p} \right) = \frac{1}{2} [\psi(p/2) - \psi(z/2)].$$

Substitution in (MA16) proves the endpoint value

$$\boxed{\text{PF}(FK) = -\log 2 - \frac{1}{2} [\psi((\sigma_0 - it)/2) + \psi((\sigma_0 + it)/2)] \quad (x = 1).} \quad (\text{MA24})$$

All series converge directly at this endpoint by (MA18); no omitted limiting or prime-power convention is required.

We now receive the established identity into the recorded original Weil formula, with its symmetric zero cutoff. Its rational trivial-character specialization is [32, formula (11)]

$$\lim_{T \rightarrow \infty} \sum_{|\Im \rho| < T} \Phi(\rho) = \Phi(0) + \Phi(1) + C \log A - P_+ - P_- - \text{PF}(FK), \quad (\text{MA25})$$

where P_+ is the positive-orientation sum from (MA19) and all nontrivial zeros are counted with multiplicity. We use this classical explicit formula at its original test domain, whose hypotheses were verified after (MA2). The full source provides its literal normalization and the two prime orientations. Substitution of the independently proved (MA23) gives the zero-side formula without changing any zero coordinate.

On the real-ordinate domain used in Montgomery’s 1973 paper, namely RH, $\rho = 1/2 + i\gamma$ with $\gamma \in \mathbb{R}$ and

$$\Phi(\rho) = \frac{(2\sigma_0 - 1)x^{i\gamma}}{(\sigma_0 - 1/2)^2 + (t - \gamma)^2}.$$

Equations (MA4), (MA19), (MA23) and (MA25) therefore produce exactly

$$\begin{aligned} & (2\sigma_0 - 1) \sum_{\gamma} \frac{x^{i\gamma}}{(\sigma_0 - 1/2)^2 + (t - \gamma)^2} \\ &= -x^{-1/2} \left[\sum_{n \leq x} \Lambda(n)(x/n)^{1-\sigma_0+it} + \sum_{n > x} \Lambda(n)(x/n)^{\sigma_0+it} \right] \\ & \quad - C \frac{\zeta'(1 - \sigma_0 + it)}{\zeta(1 - \sigma_0 + it)} + \frac{x^{1/2}(2\sigma_0 - 1)}{(\sigma_0 - 1 + it)(\sigma_0 - it)} \\ & \quad - x^{-1/2} \sum_{m \geq 1} \frac{(2\sigma_0 - 1)x^{-2m}}{(\sigma_0 - 1 - it - 2m)(\sigma_0 + it + 2m)}. \end{aligned} \quad (\text{MA26})$$

This is Montgomery’s complete retained equation [33, pp. 185–187]. The sum means at least the original symmetric cutoff in (MA25); its convergence is supplied by that cited theorem. No later real-ordinate estimate is declared valid for a complex ordinate by this receiving calculation. The scalar (MA23) itself is unconditional on (MA1); the RH qualification in (MA26) belongs only to the original choice of real zero coordinates.

31 Exact backward replacement

In the complete Original Six Readers provider, the proved transform, prime-range and pole formulas remain as printed. Replace the status sentence “The remaining gamma/trivial-zero commutator is not claimed proved” and the concluding paragraph beginning “Still to be proved is” in `proofs/B_guinand_weil_montgomery_interfaces.tex` by (MA23), with the proof (MA5)–(MA24). In `satellites/10_literature_foundations.tex`, replace the paragraph beginning “What is still open is one explicit commutator” by the same completed identity and its exact parameter domain. These are successor replacement instructions; no archived source bytes are changed here.

The separate local labels `eq:guinand-weil-arch-map` and `eq:guinand-weil-residual-complete` need the nonzero correction proved in the companion note, not (MA23) with a discarded term. On its explicit common test subspace the exact additional functional is

$$\mathcal{D}(f) = \int_0^\infty f(v)[2e^{-v/2} - \text{sech}(v/2)] dv.$$

The original right side must receive $+\mathcal{D}(f)$. In particular this change must be propagated to the asserted local subtraction proof and its status records; it is not a correction to Guinand’s published theorem. The companion proves the exact test-space inclusion with decay exponent strictly greater than $1/2$, all jump terms, and a nonzero admissible example.

Exact correction of the archived Guinand–Weil identity

This note concerns the equation labelled `eq:guinand-weil-arch-map` and its expanded version `eq:guinand-weil-residual-complete` in the programme's *Original Six Readers*. The local equations, cosine convention and finite-part subtraction are retained literally. The result is a direct calculation of their exact difference, with no use of RH or of a zero sum. The incoming note *Exact correction to the archived Guinand–Weil comparison*, 19 September 2026, identified this difference. Here every interchange and endpoint is proved, the test-domain assertion is corrected, and the entire exponential family is evaluated. The result corrects the local transcription/comparison; it makes no assertion against Guinand's published theorem [34] or Weil's published formula [35].

32 The original functions and the exact domain inclusion

For $c > 0$, define $\mathcal{D}_c$ to consist of complex continuous functions $f : [0, \infty) \rightarrow \mathbb{C}$ with a finite partition $0 = a_0 < a_1 < \cdots < a_m < \infty$ such that f is C^2 on each open piece, f'' is absolutely integrable on every piece, and f' has finite one-sided values at the finite endpoints. On the final unbounded piece require

$$|f(x)| + |f'(x)| + |f''(x)| \leq C_f e^{-cx} \quad (x \geq X_f). \quad (\text{A03GC.1})$$

At a derivative jump use its two one-sided values, and use their average as its assigned representative. This representative does not change an integral. Put

$$\mathcal{D}_0^{\text{exp}} = \bigcup_{c>0} \mathcal{D}_c, \quad \mathcal{D}_W^{\text{exp}} = \bigcup_{c>1/2} \mathcal{D}_c. \quad (\text{A03GC.2})$$

These are vector spaces: for addition take the common refinement of the two finite partitions and the smaller of the two decay exponents.

The original definitions are

$$g_f(t) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(tx) dx, \quad t \geq 0, \quad (\text{A03GC.3})$$

$$2H_f = \int_0^\infty f(x) \left(\frac{1}{x} - \frac{e^{-x/2}}{\sinh x} \right) dx, \quad (\text{A03GC.4})$$

$$K(x) = \frac{e^{|x|/2}}{|e^x - e^{-x}|}, \quad x \neq 0, \quad (\text{A03GC.5})$$

$$P_f = \lim_{\Lambda \rightarrow \infty} \left\{ \int_{\mathbb{R}} (1 - e^{-\Lambda|x|}) f(|x|) K(x) dx - f(0) \log \Lambda \right\}, \quad (\text{A03GC.6})$$

$$\mathcal{L}_f = \sqrt{\frac{2}{\pi}} \int_0^\infty g_f(t) \log \frac{t}{2\pi} dt, \quad (\text{A03GC.7})$$

$$W_f = 2 \int_0^\infty f(x) e^{-x/2} dx - P_f - f(0) \log(2\pi). \quad (\text{A03GC.8})$$

In (A03GC.6), $|x|f(|x|)K(x) \rightarrow f(0)/2$, so the original Weil subtraction $2\beta(0) \log \Lambda$ is exactly $f(0) \log \Lambda$; this factor is not adjusted. The archived equality was $W_f = 2H_f + \mathcal{L}_f$.

Proposition 32.1 (Exact inclusion in the recorded common domain). *For every $f \in \mathcal{D}_W^{\text{exp}}$ and $1 < p \leq 2$, the function f satisfies the Guinand test conditions recorded in the source: it is an indefinite integral of its locally integrable derivative, $f(x) \rightarrow 0$, and $xf'(x) \in L^p(0, \infty)$. The map*

$$E : \mathcal{D}_W^{\text{exp}} \longrightarrow \mathcal{W}_{\text{even}}^{G,p}, \quad E(f)(x) = \frac{f(|x|)}{\sqrt{2\pi}}, \quad f(t) = \sqrt{2\pi} E(f)(t) \quad (t \geq 0) \quad (\text{A03GC.9})$$

is an injective linear map into the recorded Guinand–Weil common test space, and the displayed restriction is its inverse on its image. Moreover

$$\int_{\mathbb{R}} E(f)(x) e^{itx} dx = g_f(t), \quad T^{1/2} f(T) \longrightarrow 0. \quad (\text{A03GC.10})$$

Proof. The integrability of f'' and finite derivative traces make f' bounded on each bounded piece. Continuity of f cancels the boundary values in the fundamental theorem on adjacent pieces. Thus $f(x) = f(0) + \int_0^x f'(u) du$ on every bounded interval. The exponential bound gives the limit at infinity and the L^p assertion; near zero and each finite break the derivative is bounded. The even extension is continuous and piecewise continuously differentiable. Its derivative at 0 has one-sided values $-f'(0+)/\sqrt{2\pi}$ and $f'(0+)/\sqrt{2\pi}$; at $\pm a_j$ it has precisely the corresponding reflected derivative jumps. Thus it satisfies the recorded Weil condition (A), including the averaged representatives. Choose a decay exponent $c > 1/2$ in (A03GC.1) and put $b = c - 1/2 > 0$. Then both $E(f)$ and its piecewise derivative are $O(e^{-(1/2+b)|x|})$, which is condition (B). The two expressions in (A03GC.9) compose to the identity. Splitting the integral at zero gives $2/\sqrt{2\pi} = \sqrt{2/\pi}$ times the cosine integral, proving (A03GC.10). Its last limit follows directly from (A03GC.1). These are the common-domain conditions recorded from Guinand’s Theorems 3 and 3’ and Weil’s (A), (B) [34, pp. 112–115][35, Collected Papers II, pp. 52–58]. $\square$

The same expression $E(f)$ is a defined even Fourier test for $f \in \mathcal{D}_0^{\text{exp}}$ and satisfies the same transform identity, but the inclusion into Weil’s (B) requires the faster exponent. For example $f(x) = e^{-x/4}$ belongs to $\mathcal{D}_0^{\text{exp}}$, whereas

$$e^{(1/2+b)x} |E(f)(x)| = \frac{e^{(1/4+b)x}}{\sqrt{2\pi}} \longrightarrow \infty \quad (b > 0). \quad (\text{A03GC.11})$$

Consequently the incoming note’s phrase “exponential decay” alone does not prove its claimed inclusion in the original common domain. Equations (A03GC.2)–(A03GC.10) give the exact inclusion and inverse. All functional identities below hold on $\mathcal{D}_0^{\text{exp}}$, and their application as a correction within the recorded common domain is restricted to $\mathcal{D}_W^{\text{exp}}$.

33 Cosine decay, inversion at zero, and the logarithmic functional

Lemma 33.1 (Jump-retaining decay and the endpoint inversion). *For $f \in \mathcal{D}_0^{\text{exp}}$ set $\Delta_j = f'(a_{j+}) - f'(a_{j-})$ and $B_f = |f'(0+)| + \sum_{j=1}^m |\Delta_j| + \int_0^\infty |f''(x)| dx$, where f'' is the piecewise derivative. For $t > 0$,*

$$g_f(t) = -\frac{\sqrt{2/\pi}}{t^2} \left(f'(0+) + \sum_{j=1}^m \Delta_j \cos(ta_j) + \int_0^\infty f''(x) \cos(tx) dx \right). \quad (\text{A03GC.12})$$

In particular g_f and $g_f(t) \log(t/(2\pi))$ are absolutely integrable, and

$$\sqrt{\frac{2}{\pi}} \int_0^\infty g_f(t) dt = f(0). \quad (\text{A03GC.13})$$

Proof. First integrate $f(x) \cos(tx)$ by parts on all pieces up to a finite endpoint R in the last piece. Continuity makes the internal f boundary terms cancel. The term at zero contains $\sin 0 = 0$, and the term at R tends to zero. This gives $\int f \cos = -t^{-1} \int f' \sin$. On the second integration, $\sin(tx) = -t^{-1}(\cos(tx))'$. The endpoint at zero contributes $f'(0+)/t$; the two values at a_j give $\Delta_j \cos(ta_j)/t$; the endpoint at infinity vanishes. Multiplication by the first factor $-1/t$ proves (A03GC.12). Thus $|g_f(t)| \leq \sqrt{2/\pi} \min(\|f\|_1, B_f/t^2)$. This proves integrability of the transform and of its logarithmic weight both at zero and infinity.

For $r > 0$, absolute convergence permits the exact Laplace–cosine calculation

$$\sqrt{\frac{2}{\pi}} \int_0^\infty e^{-rt} g_f(t) dt = \frac{2}{\pi} \int_0^\infty f(x) \frac{r}{r^2 + x^2} dx = \frac{2}{\pi} \int_0^\infty \frac{f(ru)}{1 + u^2} du. \quad (\text{A03GC.14})$$

Indeed $\int_0^\infty e^{-rt} \cos(tx) dt$ is the real part of $(r - ix)^{-1}$ and equals $r/(r^2 + x^2)$; the double integral is bounded by $\|f\|_1/r$. The function f is bounded. Dominated convergence in the right-hand integral as $r \downarrow 0$ gives $f(0)$, since $\int_0^\infty (1 + u^2)^{-1} du = \pi/2$. Dominated convergence by $|g_f|$ in the left-hand integral proves (A03GC.13) without an unproved endpoint inversion assertion. $\square$

Theorem 33.2 (The exact cosine-log functional). *For $f \in \mathcal{D}_0^{\text{exp}}$,*

$$\boxed{\mathcal{L}_f = - \int_0^\infty \frac{f(x) - f(0)e^{-x}}{x} dx - f(0) \log(2\pi).} \quad (\text{A03GC.15})$$

Proof. Set $h(x) = f(x) - f(0)e^{-x}$. It belongs to $\mathcal{D}_0^{\text{exp}}$, $h(0) = 0$, and $h(x) = O(x)$ near zero because its right derivative is bounded there. Its exponential decay gives

$$\int_0^\infty \frac{|h(x)|}{x} dx < \infty, \quad \int_0^\infty g_h(t) dt = 0. \quad (\text{A03GC.16})$$

For $t > 0$ the Frullani identity used here has an elementary proof:

$$\int_0^\infty \frac{e^{-r} - e^{-tr}}{r} dr = \int_1^t \left(\int_0^\infty e^{-ur} dr \right) du = \int_1^t \frac{du}{u} = \log t. \quad (\text{A03GC.17})$$

The interval $[1, t]$ is oriented if $t < 1$; absolute integrability on the bounded u interval justifies the exchange. For fixed t the integrand in r has constant sign. Therefore every truncated integral $\ell_{\epsilon, R}(t) = \int_\epsilon^R (e^{-r} - e^{-tr}) dr / r$ satisfies $|\ell_{\epsilon, R}(t)| \leq |\log t|$.

At $0 < \epsilon < R < \infty$, Fubini and (A03GC.16) give

$$\begin{aligned} \sqrt{\frac{2}{\pi}} \int_0^\infty g_h(t) \ell_{\epsilon, R}(t) dt &= -\sqrt{\frac{2}{\pi}} \int_\epsilon^R \frac{1}{r} \int_0^\infty e^{-rt} g_h(t) dt dr \\ &= -\frac{2}{\pi} \int_\epsilon^R \int_0^\infty \frac{h(x)}{r^2 + x^2} dx dr. \end{aligned} \quad (\text{A03GC.18})$$

For the first exchange, $2\|g_h\|_1 \log(R/\epsilon)$ is a finite majorant. For inserting the cosine transform in the second line use $\int_\epsilon^R r^{-1} \int_0^\infty e^{-rt} dt dr \|h\|_1 < \infty$. For the untruncated double integral, Tonelli applied to the absolute value gives exactly

$$\int_0^\infty \int_0^\infty \frac{|h(x)|}{r^2 + x^2} dx dr = \frac{\pi}{2} \int_0^\infty \frac{|h(x)|}{x} dx < \infty. \quad (\text{A03GC.19})$$

The left side of (A03GC.18) tends to the logarithmic integral by the majorant $|g_h(t) \log t|$ from the lemma. The right side tends by (A03GC.19) to $-\int h(x)/x dx$.

For the retained reference e^{-x} its transform is $\sqrt{2/\pi}/(1 + t^2)$. The substitution $t = 1/u$ in the absolutely convergent integral yields $\int_0^\infty \log t (1 + t^2)^{-1} dt = 0$. Hence its $\mathcal{L}$ value is $-\log(2\pi)$. Adding $f(0)$ times this reference to the result for h proves (A03GC.15), including its constant. $\square$

34 The original finite part and the defect map

Theorem 34.1 (Evaluation with the original regulator). *For $f \in \mathcal{D}_0^{\text{exp}}$ the limit in (A03GC.6) exists and*

$$\boxed{P_f = \int_0^\infty \left(\frac{f(x)e^{x/2}}{\sinh x} - \frac{f(0)e^{-x}}{x} \right) dx.} \quad (\text{A03GC.20})$$

The displayed combined integrand is absolutely integrable.

Proof. Write $A_f(x) = f(x)e^{x/2}/\sinh x - f(0)e^{-x}/x$ for $x > 0$. The expansions $e^{x/2}/\sinh x = x^{-1} + 1/2 + O(x)$ and $e^{-x}/x = x^{-1} - 1 + O(x)$, together with $f(x) = f(0) + f'(0+)x + o(x)$, show $A_f(x) \rightarrow f'(0+) + 3f(0)/2$ at zero. At infinity its two terms decay exponentially. Thus $A_f \in L^1$. Evenness changes the full-line regulated integral into $\int_0^\infty (1 - e^{-\Lambda x})f(x)e^{x/2}/\sinh x dx$, with no residual factor of two. Subtract and add the reference to obtain the exact finite- Λ identity

$$\begin{aligned} & \int_{\mathbb{R}} (1 - e^{-\Lambda|x|})f(|x|)K(x)dx - f(0)\log \Lambda \\ &= \int_0^\infty (1 - e^{-\Lambda x})A_f(x)dx + f(0)\log(1 + \Lambda^{-1}). \end{aligned} \quad (\text{A03GC.21})$$

Here the reference integral is $\int_0^\infty (e^{-x} - e^{-(1+\Lambda)x})dx/x = \log(1 + \Lambda)$ by (A03GC.17). Dominated convergence by $|A_f|$, and $\log(1 + \Lambda^{-1}) \rightarrow 0$, prove (A03GC.20). $\square$

Theorem 34.2 (Exact comparison and receiving map). *Define*

$$d(x) = 2e^{-x/2} - \operatorname{sech}(x/2) = \frac{2e^{-3x/2}}{1 + e^{-x}}, \quad D : \mathcal{D}_0^{\text{exp}} \rightarrow \mathbb{C}, \quad D(f) = \int_0^\infty f(x)d(x)dx. \quad (\text{A03GC.22})$$

Then the original expressions satisfy the exact identity

$$\boxed{W_f - (2H_f + \mathcal{L}_f) = D(f)}. \quad (\text{A03GC.23})$$

In particular the map

$$T : \mathcal{D}_0^{\text{exp}} \rightarrow \mathbb{C}^3, \quad T(f) = (W_f, H_f, \mathcal{L}_f), \quad (1, -2, -1)T = D \quad (\text{A03GC.24})$$

is the exact typed relation between the two presentations. Its restriction along the inclusion $\mathcal{D}_W^{\text{exp}} \hookrightarrow \mathcal{D}_0^{\text{exp}}$ is the corrected common-domain comparison. The defect is strictly positive on every nonzero real nonnegative test.

Proof. The kernel in (A03GC.4) tends to $1/2$ at zero and is $O(1/x)$ at infinity, proving absolute convergence of H_f . All other terms were proved convergent above. Substitute (A03GC.15) and (A03GC.20); to avoid separating divergent terms, first combine their integrands on $[\eta, R]$ and then pass to the limits. The $f(0)\log(2\pi)$ constants and the reference subtractions cancel exactly, as do the $f(x)/x$ terms. The remaining kernel is

$$2e^{-x/2} + \frac{e^{-x/2} - e^{x/2}}{\sinh x} = 2e^{-x/2} - \frac{1}{\cosh(x/2)} = d(x). \quad (\text{A03GC.25})$$

It is bounded at zero and bounded by $2e^{-3x/2}$ at infinity. Thus the limit is $D(f)$, proving (A03GC.23) and (A03GC.24). The second expression for d in (A03GC.22) is strictly positive. A continuous nonnegative nonzero f is positive on an interval, so its integral against d is strictly positive. $\square$

For an explicit receiving functional retaining all old terms, set

$$H_f^{\text{rec}} = H_f + \frac{1}{2}D(f), \quad (\text{A03GC.26})$$

$$2H_f^{\text{rec}} = \int_0^\infty f(x) \left(\frac{1}{x} - \frac{e^{x/2}}{\sinh x} + 2e^{-x/2} \right) dx, \quad W_f = 2H_f^{\text{rec}} + \mathcal{L}_f. \quad (\text{A03GC.27})$$

The kernel in this integral tends to $3/2$ at zero. Equations (A03GC.26)–(A03GC.27) follow by the already proved identity and (A03GC.25); they explicitly record the map from the archived H_f . They do not rename the archived H_f or identify H_f^{rec} with an uninspected historical transcription.

35 The full exponential family and the original counterexample

For $a > 0$ let $f_a(x) = e^{-ax}$. Its cosine transform, obtained by taking the real part of $(a - it)^{-1}$, is

$$g_{f_a}(t) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2 + t^2}. \quad (\text{A03GC.28})$$

In particular $f_a \in \mathcal{D}_0^{\text{exp}}$ for all $a > 0$ and $f_a \in \mathcal{D}_W^{\text{exp}}$ for $a > 1/2$. Put $z = a/2 + 1/4$ and $w = a/2 + 3/4$. We obtain

$$P_{f_a} = -\log 2 - \psi(z), \quad (\text{A03GC.29})$$

$$2H_{f_a} = \psi(w) + \log(2/a), \quad (\text{A03GC.30})$$

$$\mathcal{L}_{f_a} = \log(a/(2\pi)), \quad (\text{A03GC.31})$$

$$W_{f_a} = \frac{2}{a + 1/2} + \psi(z) - \log \pi, \quad (\text{A03GC.32})$$

$$2H_{f_a} + \mathcal{L}_{f_a} = \psi(w) - \log \pi, \quad (\text{A03GC.33})$$

$$D(f_a) = \frac{2}{a + 1/2} + \psi(z) - \psi(w) = \psi(z + 1) - \psi(w) > 0. \quad (\text{A03GC.34})$$

Here $\psi = \Gamma'/\Gamma$ has its usual unscaled argument.

To verify every constant in these identities, use the primary special function formula [36, 5.9.12]

$$\psi(v) = \int_0^\infty \left(\frac{e^{-t}}{t} - \frac{e^{-vt}}{1 - e^{-t}} \right) dt, \quad v > 0. \quad (\text{A03GC.35})$$

Its integrand is $O(1)$ at zero and exponentially integrable at infinity. In (A03GC.20), substitute $t = 2x$ to get the combined integrand $e^{-zt}/(1 - e^{-t}) - e^{-t/2}/t$. Adding and subtracting e^{-t}/t inside the combined integral gives $-\psi(z)$ and $\int_0^\infty (e^{-t} - e^{-t/2})dt/t = -\log 2$ by (A03GC.17). This proves (A03GC.29). The same substitution in (A03GC.4) gives $e^{-at/2}/t - e^{-wt}/(1 - e^{-t})$; the difference from (A03GC.35) integrates to $\log(2/a)$, proving (A03GC.30). In the transform integral put $t = au$. The constant term integrates to $\log(a/(2\pi))$, while the $\log u$ term is zero by $u \mapsto 1/u$. This proves (A03GC.31). The remaining identities follow by substitution, using the exact recurrence $\psi(z + 1) = \psi(z) + 1/z$ [36, 5.5.2]; here $1/z = 2/(a + 1/2)$. Positivity follows independently from (A03GC.22).

For the incoming test $a = 3/2$ one has $z = 1$, $w = 3/2$. The special values $\psi(1) = -\gamma_E$ and $\psi(1/2) = -\gamma_E - 2\log 2$, and the recurrence at $1/2$, give [36, 5.4.12–13 and 5.5.2]

$$\begin{aligned} P_{f_{3/2}} &= \gamma_E - \log 2, & 2H_{f_{3/2}} &= 2 - \gamma_E - \log 3, \\ \mathcal{L}_{f_{3/2}} &= \log \frac{3}{4\pi}, & W_{f_{3/2}} &= 1 - \gamma_E - \log \pi, \\ 2H_{f_{3/2}} + \mathcal{L}_{f_{3/2}} &= 2 - \gamma_E - \log \pi - 2\log 2. \end{aligned} \quad (\text{A03GC.36})$$

Consequently

$$\boxed{D(f_{3/2}) = 2\log 2 - 1 > 0.} \quad (\text{A03GC.37})$$

There is also a direct evaluation independent of digamma values:

$$D(f_{3/2}) = 2 \int_0^\infty \frac{e^{-3x}}{1 + e^{-x}} dx = 2 \int_0^1 \frac{u^2}{1 + u} du = 2 \left(\frac{1}{2} - 1 + \log 2 \right). \quad (\text{A03GC.38})$$

Strict positivity also follows from $\log 2 = \int_1^2 dt/t > 1/2$, without a numerical approximation. Thus the contradiction to the archived equality occurs inside its actual recorded common domain, and does not depend on the repaired inclusion for more slowly decaying functions.

36 Exact supersession and provenance

Equation (A03GC.23) supersedes the claimed zero-defect equality at the following two archived labels: `eq:guinand-weil-arch-map` and `eq:guinand-weil-residual-complete`. The proof that the latter follows from the two displayed local scalar formulas cannot validate both scalar transcriptions with the printed definitions: subtracting them gives the contradicted equality. The calculation here locates and evaluates the discrepancy exactly, while preserving both displayed functionals and all their original factors. It does not determine from this discrepancy alone which published formula was miscopied. That would require comparing the historical formulas themselves, rather than using the local assertion that they were literal.

The source snapshot is `06_Foundations_and_Original_Six_Readers.tex`, SHA-256

`e653a19b9bfc7b4788550e59de42367d52ed666598e46540d040e06d7362cad0.`

Its relevant original lines are 201784–201795 (Guinand domain and transform), 201798–201843 (extension), 201953–202030 (expanded false comparison), 202639–202660 (Weil conditions and regulator), and 202842–202906 (printed comparison). The incoming correction is `06_PRINTED_ARCHIMEDEAN_CORRECTION.md`, SHA-256

`6d0f28606fe98d8ab49f25b54d94377e1481ce83a511ce8189bfebfba629dcee.`

The source files are preserved unchanged. The three functional identities and the admissible exponential example are established here with complete proofs; no zero-location conclusion, mixed-support control, or arithmetic allocation follows merely from this correction.

Backward receivers and the original canonical primary-angle profile

37 The complete canonical scalar and its primary-angle profile

Keep the full packet and original density of AG1–3 and PT1–5. Thus $e = 1 + k(m_0 - 1)$, $q = e(k + 1)^2$, $k \equiv 1 \pmod{4}$, and A is multiplication by S on the original $E = \mathbb{C}[S]/\chi$. Let G_N be the original attained arithmetic quotient metric, $N \geq q - 1$. The canonical centered weight is

$$H_N = G_N^{-1/2}(A^*G_N + G_NA - kG_N)G_N^{-1/2}.$$

The exact two-boundary-row formula PT23 gives rank at most two; reflection in the original full polynomial gives the opposite signs, and PT7 makes both nonzero. Thus its nonzero eigenvalues are $+\epsilon_N, -\epsilon_N$. Let $u_{N,+}, u_{N,-}$ be unit eigenvectors and let Π_N be the Euclidean orthogonal projection onto $G_N^{1/2}E_+$, where E_+ is the complete positive-real-displacement primary subspace of A . All primary orders remain in this definition. PT7 and PT24 give

$$L_{h,k} = \frac{\delta q(k+1)}{2} = \text{Tr}(\Pi_N H_N) = \epsilon_N (\|\Pi_N u_{N,+}\|^2 - \|\Pi_N u_{N,-}\|^2). \quad (\text{A3R1})$$

Set

$$\mathfrak{d}_N = \|(I - \Pi_N)u_{N,+}\|^2 + \|\Pi_N u_{N,-}\|^2, \quad \mathfrak{t}_N = 1 - \mathfrak{d}_N = \frac{L_{h,k}}{\epsilon_N}. \quad (\text{A3R2})$$

These are the original primary alignment defects. The equality follows by subtracting the first squared projection from one in (A3R1). The positive-trace minimum proved in PT8–16 gives $0 < \mathfrak{t}_N \leq 1$ and $0 \leq \mathfrak{d}_N < 1$. Neither eigenvector phase changes them.

The original two-window action, before any asymptotic expansion, is

$$J_k^{\text{action}} = \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N^2, \quad w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \ (q \leq N \leq 2q-2). \quad (\text{A3R3})$$

There are exactly $2q$ weighted terms. This is CAI13–15, with the original adjacent windows $[q-1, 2q-2]$ and $[q, 2q-1]$ both present. Consequently its surplus over the positive-primary benchmark is exactly

$$J_k^{\text{action}} - 4q \log L_{h,k} = -2 \sum_N w_N \log(1 - \mathfrak{d}_N), \quad \left(\prod_N \mathfrak{t}_N^{w_N} \right)^{1/(2q)} = \exp \left[-\frac{J_k^{\text{action}} - 4q \log L_{h,k}}{4q} \right]. \quad (\text{A3R4})$$

This proves a metric and action morphism from the original full primary projection to the computed scalar, rather than equating it with the different original observation-kernel projection.

Use the exact constants of AG5, AG9, AG24 and AG26, and put $C_A = C_\sigma + 8 \log 2 - 4 - (4m_0 - 2)C_\Gamma - 4 \log(\delta/2)$. The newly evaluated AG31 now gives the original weighted geometric mean to inverse-logarithmic precision:

$$\begin{aligned} \frac{1}{2q} \sum_N w_N \log \mathfrak{t}_N = & -\frac{C_{Bq}}{4} + \log(k+1) - \frac{C_A}{4} - \frac{\log 2}{4\pi} I_h \frac{k^2}{q} + \frac{\eta_h M}{2} \frac{k(k+2)}{q} \\ & - \frac{C_\Gamma}{8(\log q + c_\zeta)} + o_h(1/\log q). \end{aligned} \quad (\text{A3R5})$$

This is direct division of (AG31) by $-4q$ in the exact identity (A3R4). The finite version retains the full SR remainder, the $O_h(\log q)$ Gamma bound, and the CAI upper/lower interval AG28. The individual $\mathfrak{t}_N$ and the original eight observation-flag allocations are not recovered by a geometric mean. Their maps remain those of CRP/CRS and JRE.

PT41 gives a further exact receiver. Here D_N is the off-diagonal action block between E_+ and its actual G_N -orthogonal complement. In the original primary coordinates the latter is the graph $(-K_N v, v)$ over E_- , with $K_N = P_N^{-1} X_N$ and the full positive metric blocks of PT36; this is not an assertion that E_+ and E_- are already orthogonal. Use the exact orthonormal frame of PT36; then, if $\|\cdot\|_1$ denotes the sum of its singular values,

$$\epsilon_N^2 \geq L_{h,k}^2 + \|D_N\|_1^2, \quad \sum_N w_N \log \left(1 + \frac{\|D_N\|_1^2}{L_{h,k}^2} \right) \leq J_k^{\text{action}} - 4q \log L_{h,k}. \quad (\text{A3R6})$$

Taking logarithms of each positive ratio in PT41 and summing the actual weights proves the second inequality. It supplies an upper budget for these couplings; a large total alone does not give a lower bound on any one coupling. The explicit Sylvester inverse in PT35–41 retains all root differences and nilpotent powers for the next such calculation.

The new angle notation $\mathfrak{d}_N$ is distinct from CAI's original volume contraction. Write the latter explicitly as $r_n = \det G_n / \det G_{n-1}$, and keep its original recurrence coefficient $b_n^{\text{rec}} = \omega_n / \omega_{n-1}$. CAI12 and (A3R2) give the exact connecting map

$$\mathfrak{t}_{q-1}^2 = \frac{L_{h,k}^2}{b_q^{\text{rec}}(r_q^{-1} - 1)}, \quad \mathfrak{t}_N^2 = \frac{L_{h,k}^2}{b_{N+1}^{\text{rec}}(1 - r_N)(r_{N+1}^{-1} - 1)} \quad (N \geq q). \quad (\text{A3R6a})$$

Every denominator is positive on the original full packet by the nonzero canonical weight and CAI12. This connects the computed angle product to the actual attained covariance increments without equating the two defects.

38 The original section and every retained metric cross block

For the coefficient programme retain its actual degree $a = k + 4$, simple quartet, $q = (a + 1)^2$, $q' = (a - 7)^2$, $m = q - q' - v$, and actual selected invertible conductor row set I . Let I_0 be the consecutive row set. The exact matrix identity COI9–11 is

$$E_{J_I} = J_v D_{0I} + E_{J_0} M_{0I}, \quad L_I = F_C D_{0I} + L_0 M_{0I}, \quad M_{0I} \text{ invertible}. \quad (\text{A3R7})$$

Here L_I is the original arithmetic residue injection, F_C is its conductor injection, and L_0 is the computed consecutive injection. For every positive arithmetic form G on the full packet, this gives

$$\begin{aligned} L_I^* G L_I &= M_{0I}^* L_0^* G L_0 M_{0I} + D_{0I}^* F_C^* G F_C D_{0I} \\ &\quad + M_{0I}^* L_0^* G F_C D_{0I} + D_{0I}^* F_C^* G L_0 M_{0I}. \end{aligned} \quad (\text{A3R8})$$

The proof is multiplication of (A3R7) and its adjoint. The same expansion holds for either original boundary difference and for the full action form. In contrast, division by the conductor translates the native minimizing fibre by an admitted lower exponential. That translation is a bijection, so its unchanged minimum gives exactly

$$D_{I,a,N-v}^{(s)} = M_{0I}^* D_{0,a,N-v}^{(s)} M_{0I}, \quad s = 1, a. \quad (\text{A3R9})$$

These two equations are the actual relation between the presentations. The absence of the conductor cross terms in (A3R9) is justified by its specific quotient, and does not remove them from (A3R8).

With four copies set $\mathcal{G}_N = (I_4 \otimes L_I)^* \text{diag}(G_N, 4)(I_4 \otimes L_I)$ and $\mathcal{D}_N = \text{diag}(D_{I,a,N-v}^{(s)}, 4)$. Let $\mathcal{R}$ have signs $(-, -, +, +)$ at $N = q - 1, q, 2q - 1, 2q$ and set $T_N = \mathcal{G}_N^{-1/2} \mathcal{D}_N \mathcal{G}_N^{-1/2}$. The full native total is $\mathcal{T} = \mathcal{R} \log \det D_{I,a,N-v}^{(s)}$. Taking determinants before their signed sum proves

$$\mathcal{R} \log \det T_N = 4\mathcal{T} + F_{\text{ar}}, \quad F_{\text{ar}} = -\mathcal{R} \log \det \mathcal{G}_N \geq 0. \quad (\text{A3R10})$$

Monotonicity follows by minimizing over the nested original relation spaces, and persists under the fixed injection. It proves the last sign separately for the two original endpoint links. For $I = I_0$, CAL1–18 bounds $F_{\text{ar}} \leq -8m \log(1 - \varepsilon_a)$ with its explicitly estimated exponentially small ε_a . For every other I , the exact full arithmetic value is retained in (A3R8) and NMR’s complete projection formula. No root-column ordering selects a row-subset ordering by itself.

If $b_- \mathcal{G}_N \preceq \mathcal{D}_N \preceq b_+ \mathcal{G}_N$ at all four endpoints, then all $4m$ eigenvalues of T_N lie in $[b_-, b_+]$. Thus each signed pair costs at most $4m \log(b_+/b_-)$, and (A3R10) gives

$$\log(b_+/b_-) \geq \frac{4\mathcal{T} + F_{\text{ar}}}{8m}. \quad (\text{A3R11})$$

This retains the full relative metric term at every original section.

39 Exact negative-trace and period receivers

RBV1–36 evaluates an explicit degree- $8a$ polynomial P_a in the original full packet. If the retained section is I_0 , its exact preimage is the Laurent coefficient vector of COI7. For any other I , solve the full square frame $[J_v, E_{J_I}]$ for the moment vector of P_a ; this returns both its conductor component and its native component. Substitution in (A3R8) gives the full energy with both cross terms. This is the receiver for both boundary windows; it does not assert that the native component alone represents P_a .

The complete tensor-residue comparison on the same full packet is

$$\mathcal{W}(f, g) = \mathcal{B}(f, a_{\text{tr}}g) + \mathcal{W}_Z(f, g), \quad a_{\text{tr}} = \sum_{c_\lambda \neq 0} \frac{e_\lambda}{c_\lambda}, \quad (\text{A3R12})$$

where c_λ is the actual tensor coefficient of the original four-term polynomial using $v_h(\zeta)/h'(\zeta)$, and e_λ is the original CRT idempotent. The residual is the sum over $c_\lambda = 0$. This identity follows site by site: a nonzero c_λ cancels its displayed reciprocal, and a zero coefficient is supplied by the residual. RBV31–36 retains the full unit, proves both terms nonpositive on P_a , and obtains a strict negative tensor value from the perimeter sites. The second argument is the displayed full-packet polynomial; (A3R7)–(A3R8) give its exact further receiver whenever a coefficient presentation is used.

UPR1–31 constructs the fixed homogeneous module

$$\mathbb{C}[X_1, X_2, X_3, X_4]^{125} \oplus \mathbb{C}[X_1, X_2, X_3, X_4]^{625} \longrightarrow \mathbb{C}[X_1, X_2, X_3, X_4]^{625},$$

with $(f, g) \mapsto f + Qg$, with $X_i = x_i^5$ and original shifted degrees. Multiplication by the quadratic may produce $X_i X_j$ from a residue having both exponents four, so the coefficient degree bound is two. The fixed decreasing rewrite score still gives a homogeneous image inverse of coefficient norm $\exp(O_{\text{actual}}(k))$. The original word Gram and Lagrange map then give the actual period-to-arithmetic image quotient bound $\exp(O_{h,u}(q \log(q+2)))$. The map is defined on the original image, with the original kernel left in its quotient, so it does not set an observation-kernel restriction to zero.

40 Correct the archived scalar identities in their original domains

The MA1–26 proof completes the original Montgomery finite part on $1 < \sigma_0 < 2$, $x \geq 1$, $t \in \mathbb{R}$, retaining its jump and both poles. Its exact replacement equation and full proof are incorporated alongside this receiver. GC proves that the two archived Guinand–Weil functionals are related by the additional linear functional

$$D(f) = \int_0^\infty f(x)[2e^{-x/2} - \text{sech}(x/2)] dx = \int_0^\infty f(x) \frac{2e^{-3x/2}}{1 + e^{-x}} dx. \quad (\text{A3R13})$$

The equality of kernels follows by writing $\operatorname{sech}(x/2) = 2e^{-x/2}/(1 + e^{-x})$. On $f = e^{-3x/2}$ the substitution $u = e^{-x}$ gives $D(f) = 2 \int_0^1 u^2/(1 + u) du = 2 \log 2 - 1 > 0$. Consequently the previous equality without $D(f)$ must be replaced at both named source equations; it cannot be used as a zero-defect premise. The corrected identity has the original common-domain inclusion for decay exponent greater than $1/2$, while GC proves the functional identity for positive exponential decay. The updated foundation derivatives preserve both original source files and record the exact replacement patches.

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